

DETAILED PROJECT REPORT

Construction of a Two-Lane Major Bridge across the River Brahmaputra at Dhubri, including Approach Roads

Submitted by: Assam Public Works Department (Roads & Bridges)

Implementing agency / SPV: Assam State Bridge Corporation Limited

DPR prepared by: Northeast Infrastructure Consultants Private Limited

Prepared in accordance with the Template for Preparation of Detailed Project Report in respect of Bridges, and IRC:SP:19 for surveys and investigations.

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1. SALIENT FEATURES

Sl.	Particular	Detail
1	Title of the project	Construction of a two-lane major bridge across the Brahmaputra at Dhubri with approach roads
2	District / Taluk / Local body / Constituency	Dhubri / Dhubri Sadar / Dhubri Municipal Board / Dhubri LAC
3	Implementing agency / SPV	Assam State Bridge Corporation Limited
4	DPR prepared by	Northeast Infrastructure Consultants Pvt. Ltd.
5	Project outlay	Rs. 412.50 crore
6	Budget provision	Rs. 120.00 crore in the current financial year
7	Budget speech reference	Para 74, State Budget Speech 2026-27
8	Administrative sanction	AS accorded vide G.O. (Rt) No. 812/2026/PWD dated 14 April 2026
9	Nature of the project	New bridge
10	Present status of existing bridges / roads	Nearest crossing is a seasonal ferry; the next fixed crossing is 62 km upstream
11	Need for the project	To provide an all-weather crossing and remove a 62 km detour for freight and emergency traffic
12	Length of the bridge / approaches	1,340 m bridge; 4.60 km approaches
13	Carriageway width	2-lane, 7.5 m with 1.5 m footpaths
14	Design discharge / HFL	63,400 cumec; HFL 30.42 m
15	Land required / acquisition status	18.62 ha; 62 per cent acquired, balance under award
16	Basis of estimate and Schedule of Rates	Assam PWD SoR 2025-26 with lead and lift; detailed estimate attached at Annexure IV
17	Details of revenue streams	Nil — free-use public asset
18	Cost Benefit Analysis (BCR)	1.64
19	Details of project risks	Chapter 3.11 — 14 risks identified
20	Project management organisation	Chapter 3.12
21	Contract management strategy	EPC — Chapter 3.13
22	Implementation Schedule and WBS	30 months from zero date — Chapter 3.14
23	Details of statutory clearances	Chapter 3.15
24	Quality control infrastructure and mechanism	Chapter 3.16
25	O&M arrangements after completion	Chapter 3.17
26	Details of attached drawings	Annexures I, II and III
27	Other attachments	Annexures IV to VII

1.1 Salient Features (continued) — Sheet 1

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. LA-879, revision B.

1.1 Salient Features (continued) — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Steel	Quarry B	10.3	644
Murram	Depot	40.1	214
Boulder	Depot	25.4	3,928
Bitumen	Quarry A	81.3	681
Coarse aggregate	Quarry A	59.2	330
Steel	Quarry B	87.3	655
Bitumen	Quarry B	69.2	8,303
Bitumen	Quarry B	32.4	3,764
Murram	Depot	85.8	532
Steel	Plant	83.2	1,818
Fine sand	River bed	12.6	5,630
Boulder	Plant	84.4	7,095
Bitumen	Quarry B	28.7	8,361
Murram	Plant	36.6	745
Steel	Quarry B	67.4	6,803
Steel	Quarry B	34.3	6,319
Coarse aggregate	Depot	60.4	1,948
Murram	Quarry B	47.8	6,623
Cement	Depot	66.5	7,869
Coarse aggregate	River bed	63.9	6,628
Boulder	Quarry B	16.8	3,898
Coarse aggregate	Quarry B	49.5	9,163
Fine sand	Depot	47.2	5,968
Steel	River bed	60.0	273
Steel	Plant	73.2	8,678
Murram	Plant	20.1	1,099

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

2. EXECUTIVE SUMMARY

The project provides for construction of a two-lane major bridge of total length 1,340 m across the river Brahmaputra at Dhubri, together with 4.60 km of approach roads on either bank and associated river training works.

The total project cost is estimated at Rs. 412.50 crore inclusive of all taxes and duties, at the price level of the current financial year. The construction period is 30 months from the zero date.

The project removes a 62 km detour for traffic between the north and south banks and provides the first all-weather crossing in the district. The benefit cost ratio works out to 1.64.

2.1 Executive Summary (continued) — Sheet 1

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. ST-853, revision C.

2.1 Executive Summary (continued) — Sheet 2

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
19:00	128	165	2	74	87	404
13:00	346	110	32	47	69	991
16:00	157	240	2	3	46	1032
10:00	214	118	33	21	71	443
07:00	138	16	11	41	22	359
16:00	281	102	32	86	71	452
14:00	232	198	33	46	101	1295
11:00	205	229	28	20	96	898
22:00	398	128	33	31	62	651
15:00	276	141	22	84	113	1011
14:00	199	155	35	58	62	534
10:00	378	223	10	78	34	1062
09:00	175	255	32	71	66	1119
20:00	335	160	26	39	93	505
15:00	282	103	39	9	100	779
23:00	24	242	12	13	7	1256
20:00	45	79	37	29	87	297
16:00	89	228	17	31	105	511
01:00	236	239	2	7	46	817
05:00	147	182	1	10	14	218

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.1 INTRODUCTION

The project area lies on the Brahmaputra in Dhubri district of western Assam, at the confluence of the alluvial plains with the Meghalaya plateau margin. The terrain is flat with an average elevation of 28 m above mean sea level. The regional geology comprises recent alluvium of Holocene age overlying Tertiary sediments.

3.1.1 Project Definition, Concept and Scope

The project comprises the following sub-components: a 1,340 m pre-stressed concrete box girder bridge on well foundations; 4.60 km of two-lane approach roads with paved shoulders; river training and bank protection works on both banks; and associated drainage, lighting and signage.

Land required is 18.62 hectares, of which 11.54 hectares has been acquired. Acquisition of the balance 7.08 hectares is under award under the Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013.

3.1.2 Project Background

There is no existing bridge at the proposed location. Crossing is presently by a seasonal ferry operated by the Inland Water Transport Department, which is suspended during high flood and in dense fog. The nearest fixed crossing is 62 km upstream.

3.1.3 Project Details

The bridge is located at chainage 0+000 to 1+340 on the proposed alignment. The adjoining land on the south bank is predominantly agricultural; the north bank approach passes through the periphery of Dhubri town. Two schools and one place of worship lie within 200 m of the alignment and have been avoided by a realignment of 140 m at chainage 3+200. Existing utilities along the approach include an 11 kV overhead line and a 200 mm water main, both to be shifted.

3.1.4 Objective and Scope of the Work

The objective is to provide an all-weather road connection between the north and south banks. The main works are the bridge substructure and superstructure, approach roads and embankment, river training works, and utility shifting.

3.1.5 Site Appreciation — Sheet 1

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. GA-720, revision A.

3.1.5 Site Appreciation — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Bitumen	Quarry B	18.9	7,884
Bitumen	Plant	43.9	2,647
Fine sand	Quarry B	78.4	8,751
Steel	Quarry A	61.1	1,229
Fine sand	Plant	5.8	688
Murram	River bed	43.6	6,530
Boulder	Depot	36.8	7,464
Fine sand	River bed	10.6	2,407
Steel	Quarry B	24.7	7,326
Murram	River bed	39.1	6,502
Bitumen	River bed	49.0	6,857
Bitumen	Quarry B	81.6	649
Murram	River bed	89.2	2,852
Boulder	River bed	86.9	1,885
Boulder	Quarry B	57.7	4,555
Cement	Quarry A	7.6	8,101
Coarse aggregate	River bed	72.5	6,905
Fine sand	Quarry A	27.9	6,982
Murram	Quarry A	5.9	916
Steel	Plant	31.1	4,752

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.1.5 Site Appreciation — Sheet 3

Design Computation

Settlement = $28.561 \times 1.293 = 36.925$ kNm (permissible 68.553) — safe

Moment = $26.954 \times 3.455 = 93.128$ mm (permissible 122.627) — safe

Bearing pressure = $80.803 \times 4.782 = 386.427$ kPa (permissible 617.242) — safe

Settlement = $50.774 \times 7.593 = 385.552$ mm (permissible 546.140) — safe

Deflection = $22.150 \times 3.893 = 86.220$ mm (permissible 130.723) — revise section

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.1.5 Site Appreciation — Sheet 4

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. GA-748, revision C.

3.1.5 Site Appreciation — Sheet 5

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
11:00	367	244	22	77	90	651
23:00	270	15	37	7	86	123
11:00	148	170	29	38	75	1311
10:00	110	103	11	40	97	836
19:00	155	86	24	13	98	135
18:00	370	198	8	39	64	535
20:00	157	71	20	23	86	971
20:00	377	34	6	76	41	763
21:00	134	122	10	10	43	526
18:00	250	79	14	15	4	1164
06:00	181	216	36	23	110	650
10:00	348	31	39	44	75	345
13:00	169	142	17	59	44	1378
13:00	168	117	36	52	4	926
04:00	122	11	30	79	65	969
17:00	387	66	2	58	107	1142
09:00	298	97	14	8	109	1285
09:00	81	217	15	5	4	1129
06:00	240	157	3	1	61	327
05:00	277	86	15	84	2	1155
17:00	231	23	39	14	43	336
08:00	296	132	3	45	28	484
03:00	293	237	7	21	30	640

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

3.2 STATUS OF FEASIBILITY STUDIES

A pre-feasibility study was carried out in 2021 by the State Technical Agency, which examined three crossing locations and recommended the present site on hydraulic and geotechnical grounds. The study concluded the project to be technically feasible with a preliminary cost of Rs. 358 crore at 2021 price level. The present estimate reflects revised quantities following detailed investigation and price escalation to the current year.

3.2.1 Pre-feasibility Findings — Sheet 1

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. LA-258, revision A.

3.2.1 Pre-feasibility Findings — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 511	22.88	1.95	2.97	2	89.549
Item 649	15.69	4.74	2.52	9	86.546
Item 126	33.50	4.90	2.44	7	66.762
Item 396	25.83	4.47	2.59	12	35.601
Item 720	14.82	2.51	0.90	8	189.962
Item 191	37.89	4.96	2.84	10	3.877
Item 398	23.75	5.33	2.57	17	78.855
Item 533	24.76	2.76	0.65	10	104.602
Item 144	5.08	2.94	2.99	17	214.314
Item 582	28.63	1.13	2.09	3	165.705
Item 307	26.13	2.81	0.71	14	299.030
Item 428	26.11	1.43	1.11	2	283.557
Item 384	31.08	3.81	0.86	11	373.898
Item 397	19.44	0.54	2.15	10	294.216
Item 434	2.69	1.95	2.90	21	164.761
Item 735	27.84	0.74	1.93	15	117.506

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.2.1 Pre-feasibility Findings — Sheet 3

Design Computation

Settlement = $54.294 \times 2.243 = 121.798$ kNm (permissible 165.502) — safe

Deflection = $33.502 \times 9.730 = 325.989$ kNm (permissible 464.022) — safe

Moment = $82.571 \times 1.718 = 141.855$ kNm (permissible 237.827) — safe

Settlement = $87.385 \times 7.436 = 649.761$ kPa (permissible 1004.053) — safe

Deflection = $4.166 \times 10.084 = 42.012$ kPa (permissible 67.382) — safe

Bearing pressure = $18.958 \times 5.490 = 104.083$ kNm (permissible 128.667) — safe

Shear = $58.622 \times 5.363 = 314.383$ kPa (permissible 380.263) — revise section

Shear = $23.551 \times 6.498 = 153.034$ kPa (permissible 273.817) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.3 REQUIREMENT / DEMAND ANALYSIS

The problem addressed is the absence of an all-weather crossing. Approximately 42,000 residents of eleven revenue villages on the north bank depend on the ferry for access to the district hospital, the sub-divisional court and the wholesale market, all located on the south bank.

Classified traffic volume counts were carried out over seven continuous days at three locations in March 2026, in accordance with IRC:SP:19. Origin-destination and axle load surveys were carried out concurrently.

Vehicle class	Existing ferry (PCU/day)	Diverted (PCU/day)	Generated (PCU/day)	Total on opening
Two-wheeler	1,840	620	410	2,870
Car / jeep / taxi	760	1,240	380	2,380
Bus	110	240	60	410
LCV	290	480	140	910
Truck (2-axle)	180	760	210	1,150
Truck (multi-axle)	40	520	180	740
	Total			8,460

Traffic has been projected at 6.2 per cent per annum for the first ten years and 4.8 per cent thereafter, derived from district vehicle registration growth over the period 2016 to 2025 and the state domestic product series. Projected traffic on the design year is 27,400 PCU per day, within the capacity of a two-lane facility.

3.3.1 Origin-Destination Survey — Sheet 1

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. ST-764, revision A.

3.3.1 Origin-Destination Survey — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
11.5	Medium sand	28	DS
3.5	Medium sand	4	DS
6.9	Coarse sand	15	SPT
5.2	Silty clay	12	DS
18.7	Medium sand	12	UDS
29.2	Silty clay	17	UDS
28.8	Medium sand	59	UDS
9.0	Medium sand	38	DS
19.0	Medium sand	48	UDS
29.0	Weathered rock	23	UDS
11.2	Medium sand	13	SPT
2.4	Coarse sand	56	DS
17.8	Clayey silt	47	DS
10.5	Coarse sand	26	SPT
14.7	Coarse sand	15	SPT
14.4	Medium sand	7	SPT
29.3	Silty clay	51	SPT

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.3.1 Origin-Destination Survey — Sheet 3

Design Computation

Bearing pressure = $49.867 \times 5.258 = 262.220 \text{ kNm}$ (permissible 376.259) — safe

Moment = $56.510 \times 11.471 = 648.230 \text{ kN}$ (permissible 1192.878) — safe

Settlement = $31.650 \times 4.516 = 142.919 \text{ mm}$ (permissible 244.286) — safe

Deflection = $67.607 \times 9.648 = 652.265 \text{ kNm}$ (permissible 924.931) — safe

Deflection = $19.555 \times 6.349 = 124.163 \text{ kN}$ (permissible 166.250) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.3.1 Origin-Destination Survey — Sheet 4

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. GA-741, revision A.

3.3.1 Origin-Destination Survey — Sheet 5

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
05:00	388	22	5	76	68	911
01:00	141	199	38	44	105	592
14:00	353	117	9	7	117	1387
01:00	272	95	13	16	93	1235
04:00	342	239	26	13	21	969
11:00	96	25	26	37	18	1008
19:00	106	259	33	58	62	730
15:00	160	84	30	51	115	380
03:00	213	220	34	22	80	1102
10:00	112	32	31	34	65	1202
16:00	204	26	22	88	75	150
09:00	206	153	17	62	33	681
10:00	352	55	37	1	60	1200
08:00	186	181	17	59	36	1104
20:00	364	214	22	44	35	1399
11:00	397	250	26	44	118	433
22:00	250	259	23	42	66	370
16:00	105	60	23	61	36	241
23:00	363	196	26	21	78	1269
16:00	360	249	26	38	79	1212
20:00	158	194	1	25	20	1281
14:00	339	176	11	28	97	449
20:00	387	20	30	28	21	190

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

3.3.1 Origin-Destination Survey — Sheet 6

Design Computation

Shear = $29.244 \times 2.506 = 73.284$ kNm (permissible 102.830) — safe

Shear = $59.965 \times 1.235 = 74.052$ kN (permissible 122.950) — safe

Bearing pressure = $32.196 \times 11.444 = 368.469$ kNm (permissible 602.640) — safe

Deflection = $86.829 \times 6.779 = 588.633$ kNm (permissible 765.805) — safe

Deflection = $85.648 \times 3.801 = 325.551$ kN (permissible 457.801) — safe

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.4 ENGINEERING SURVEYS AND INVESTIGATIONS

3.4.1 Topographic and Levelling Survey

Topographic survey was carried out by differential GPS supplemented by total station traverse. Twelve ground control points were established along the alignment at intervals not exceeding 500 m and connected to GTS benchmark no. 412/A at Dhubri, reduced level 27.842 m. Levelling was closed with a misclosure of 8 mm over a 6.4 km circuit, within the permissible limit.

3.4.2 Soil and Materials Survey

Eighteen boreholes were drilled to depths between 45 m and 62 m by rotary drilling. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in cohesive strata. California Bearing Ratio tests on subgrade samples returned soaked CBR values between 4.2 and 6.8 per cent.

Borehole	Chainage	Depth (m)	Founding strata	SPT N at founding
BH-01	0+040	52.0	Dense sand	48
BH-04	0+320	58.5	Dense sand	52
BH-09	0+700	62.0	Dense sand with gravel	58
BH-14	1+080	54.5	Dense sand	46
BH-18	1+320	47.0	Stiff clay over sand	41

Borrow areas have been identified at three locations within a lead of 11 km, with a combined assessed quantity of 4.2 lakh cubic metres against a requirement of 2.8 lakh cubic metres. Coarse aggregate is available from an approved quarry at a lead of 34 km. Construction water will be drawn from the river under permission from the Water Resources Department.

3.4.3 Survey and Investigation Records — Sheet 1

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. GA-249, revision C.

3.4.3 Survey and Investigation Records — Sheet 2

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
398/6	Agomani	1.860	Homestead	Under award
420/4	Golakganj	2.190	Waste	Notified
58/4	Agomani	1.608	Waste	Notified
55/7	Agomani	2.377	Homestead	Under award
422/5	Golakganj	1.694	Agricultural	Notified
335/5	Dhubri	2.286	Waste	Awarded
433/6	Golakganj	2.242	Agricultural	Notified
304/9	Dhubri	1.539	Agricultural	Notified
230/5	Agomani	1.435	Agricultural	Notified
460/9	Gauripur	0.325	Waste	Under award
169/4	Golakganj	1.187	Agricultural	Notified
117/9	Dhubri	0.494	Agricultural	Awarded
341/6	Agomani	0.930	Waste	Under award
158/3	Golakganj	0.176	Homestead	Notified
392/7	Dhubri	2.355	Agricultural	Awarded
466/8	Golakganj	1.892	Homestead	Awarded
274/7	Golakganj	0.763	Homestead	Awarded
108/5	Golakganj	1.196	Agricultural	Awarded
21/9	Dhubri	0.640	Homestead	Awarded
141/9	Golakganj	0.824	Waste	Under award
220/7	Golakganj	0.680	Homestead	Notified
255/8	Gauripur	1.745	Homestead	Under award
334/8	Agomani	0.911	Agricultural	Notified

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.4.3 Survey and Investigation Records — Sheet 3

Design Computation

Bearing pressure = $29.490 \times 5.020 = 148.042 \text{ kN}$ (permissible 273.303) — safe

Moment = $1.551 \times 5.613 = 8.706 \text{ kN}$ (permissible 9.364) — safe

Deflection = $10.981 \times 8.735 = 95.912 \text{ kPa}$ (permissible 170.289) — revise section

Settlement = $18.144 \times 11.390 = 206.658 \text{ kNm}$ (permissible 291.139) — safe

Bearing pressure = $73.403 \times 6.104 = 448.025 \text{ kN}$ (permissible 508.567) — safe

Settlement = $13.901 \times 7.657 = 106.441 \text{ mm}$ (permissible 124.816) — safe

Bearing pressure = $9.662 \times 1.623 = 15.681 \text{ kPa}$ (permissible 26.116) — revise section

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.4.3 Survey and Investigation Records — Sheet 4

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. LA-611, revision C.

3.4.3 Survey and Investigation Records — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 240	33.09	1.38	2.82	16	311.621
Item 557	23.23	3.02	1.82	10	391.253
Item 392	15.69	0.49	1.61	9	112.643
Item 584	18.37	5.56	0.29	3	100.581
Item 719	26.27	3.35	0.29	6	135.216
Item 132	3.98	5.96	2.34	7	280.574
Item 451	35.64	1.20	2.09	13	219.303
Item 575	30.23	3.94	0.56	12	208.586
Item 131	9.06	4.34	0.71	24	61.250
Item 682	34.30	4.84	2.85	23	226.059
Item 228	17.93	1.23	1.09	8	224.845
Item 104	31.88	3.54	0.73	11	342.121
Item 638	18.76	2.57	2.99	18	340.163
Item 239	29.22	3.98	2.45	12	109.488
Item 289	38.96	1.24	1.95	13	30.636
Item 112	3.52	4.63	0.52	23	216.570
Item 789	15.97	0.48	1.72	14	91.006
Item 678	28.13	5.72	0.46	3	289.935
Item 160	8.41	4.07	0.54	4	281.079
Item 282	3.90	5.54	2.30	22	397.279
Item 535	26.74	1.94	1.44	7	326.510
Item 769	31.59	4.79	0.80	19	6.233
Item 639	32.04	4.55	2.99	23	348.678
Item 438	31.53	0.83	1.87	2	145.458
Item 807	6.45	5.71	0.28	23	379.205
Item 546	28.06	5.84	1.16	1	4.921

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.4.3 Survey and Investigation Records — Sheet 6

Design Computation

Moment = $1.230 \times 10.009 = 12.307$ mm (permissible 15.331) — safe

Bearing pressure = $49.574 \times 3.218 = 159.547$ kPa (permissible 243.949) — safe

Shear = $43.759 \times 3.534 = 154.633$ mm (permissible 279.088) — safe

Settlement = $40.909 \times 9.777 = 399.975$ kNm (permissible 742.299) — safe

Settlement = $46.708 \times 1.791 = 83.652$ kPa (permissible 121.517) — safe

Bearing pressure = $48.241 \times 8.453 = 407.800$ kN (permissible 720.016) — safe

Moment = $14.737 \times 11.062 = 163.024$ mm (permissible 229.042) — safe

Deflection = $26.435 \times 7.225 = 190.988$ kN (permissible 201.407) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.4.3 Survey and Investigation Records — Sheet 7

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. LA-521, revision A.

3.4.3 Survey and Investigation Records — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
27.9	Silty clay	37	DS
23.4	Weathered rock	27	SPT
25.3	Clayey silt	28	SPT
19.7	Silty clay	20	UDS
1.7	Clayey silt	42	DS
21.7	Medium sand	56	DS
15.4	Gravelly sand	52	UDS
3.1	Gravelly sand	52	UDS
6.9	Coarse sand	14	DS
24.6	Medium sand	15	UDS
7.2	Medium sand	59	SPT
22.3	Weathered rock	43	UDS
21.9	Medium sand	33	UDS
17.4	Gravelly sand	13	SPT
19.0	Silty clay	21	UDS
2.3	Gravelly sand	5	SPT
29.8	Weathered rock	24	UDS
0.8	Gravelly sand	10	SPT
28.0	Coarse sand	10	DS
25.2	Coarse sand	13	UDS
11.4	Weathered rock	51	SPT
23.7	Clayey silt	60	UDS
3.6	Weathered rock	27	UDS
26.8	Medium sand	47	DS

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.4.3 Survey and Investigation Records — Sheet 9

Design Computation

Shear = $7.264 \times 1.804 = 13.106$ kN (permissible 18.640) — safe

Settlement = $49.224 \times 4.413 = 217.200$ mm (permissible 311.684) — safe

Shear = $64.934 \times 0.931 = 60.427$ kNm (permissible 90.514) — safe

Moment = $50.566 \times 3.355 = 169.673$ mm (permissible 180.165) — safe

Moment = $28.070 \times 2.187 = 61.376$ mm (permissible 102.393) — safe

Shear = $79.502 \times 2.925 = 232.514$ kNm (permissible 261.948) — safe

Shear = $32.799 \times 2.545 = 83.481$ kPa (permissible 153.245) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.4.3 Survey and Investigation Records — Sheet 10

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. GA-698, revision C.

3.4.3 Survey and Investigation Records — Sheet 11

Bore Log Summary

Depth (m)	Strata	N value	Sample
13.5	Medium sand	44	UDS
14.7	Gravelly sand	36	DS
7.9	Coarse sand	44	SPT
9.4	Weathered rock	51	SPT
3.3	Clayey silt	30	DS
5.3	Coarse sand	7	DS
15.4	Gravelly sand	36	UDS
2.8	Silty clay	7	SPT
13.2	Silty clay	15	DS
8.3	Silty clay	8	SPT
21.6	Coarse sand	28	SPT
18.0	Gravelly sand	8	DS
3.9	Weathered rock	32	DS
8.1	Coarse sand	19	UDS
27.7	Clayey silt	38	SPT
5.0	Coarse sand	45	DS
26.0	Silty clay	10	SPT
23.9	Medium sand	39	UDS
1.4	Coarse sand	35	UDS
2.6	Coarse sand	52	UDS
26.8	Medium sand	10	UDS
23.9	Clayey silt	24	SPT
27.0	Weathered rock	43	DS
14.9	Coarse sand	43	SPT
6.2	Medium sand	38	UDS

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.4.3 Survey and Investigation Records — Sheet 12

Design Computation

Bearing pressure = $35.811 \times 7.215 = 258.389$ mm (permissible 401.402) — safe

Moment = $18.144 \times 10.987 = 199.348$ kPa (permissible 250.667) — safe

Moment = $67.631 \times 11.052 = 747.442$ mm (permissible 1016.039) — safe

Bearing pressure = $23.526 \times 10.951 = 257.637$ mm (permissible 349.495) — safe

Bearing pressure = $34.150 \times 10.898 = 372.158$ kN (permissible 487.221) — safe

Settlement = $43.027 \times 1.607 = 69.144$ kN (permissible 94.137) — safe

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.4.3 Survey and Investigation Records — Sheet 13

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. LA-112, revision A.

3.4.3 Survey and Investigation Records — Sheet 14

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
256/7	Dhubri	1.553	Waste	Notified
279/6	Agomani	0.427	Agricultural	Under award
282/2	Golakganj	1.429	Agricultural	Awarded
187/4	Agomani	1.999	Waste	Notified
458/5	Golakganj	0.898	Agricultural	Notified
186/7	Agomani	0.461	Homestead	Awarded
18/9	Gauripur	0.065	Agricultural	Awarded
253/2	Golakganj	1.186	Homestead	Awarded
115/6	Gauripur	0.309	Agricultural	Awarded
454/5	Golakganj	0.568	Homestead	Under award
86/6	Agomani	2.025	Waste	Notified
38/9	Gauripur	0.522	Waste	Awarded
355/7	Dhubri	2.214	Waste	Awarded
384/2	Dhubri	1.637	Waste	Awarded
268/2	Golakganj	2.338	Waste	Awarded
235/4	Dhubri	0.568	Agricultural	Notified
300/5	Gauripur	1.460	Agricultural	Under award

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.4.3 Survey and Investigation Records — Sheet 15

Design Computation

Shear = $46.562 \times 2.894 = 134.756$ kPa (permissible 205.294) — safe

Moment = $67.874 \times 11.119 = 754.674$ kPa (permissible 1348.608) — safe

Shear = $49.012 \times 2.381 = 116.674$ kN (permissible 185.047) — safe

Moment = $38.494 \times 8.487 = 326.696$ kPa (permissible 402.735) — safe

Moment = $66.669 \times 4.612 = 307.474$ mm (permissible 417.620) — safe

Deflection = $39.391 \times 6.542 = 257.692$ kN (permissible 403.100) — safe

Bearing pressure = $23.932 \times 2.693 = 64.445$ kN (permissible 102.145) — revise section

Bearing pressure = $6.819 \times 6.602 = 45.015$ kNm (permissible 84.315) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.4.3 Survey and Investigation Records — Sheet 16

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. HY-431, revision A.

3.4.3 Survey and Investigation Records — Sheet 17

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
00:00	318	79	8	34	80	737
02:00	400	174	33	76	44	325
18:00	331	243	0	31	39	1359
06:00	127	27	27	3	78	803
10:00	123	246	36	52	48	1233
23:00	47	25	11	83	71	1320
07:00	288	253	15	37	21	604
05:00	330	101	22	36	72	539
05:00	56	42	36	71	57	192
03:00	61	218	35	5	45	843
14:00	227	80	33	30	53	1291
18:00	236	93	40	57	48	1227
17:00	325	215	15	6	111	271
13:00	358	140	25	13	78	1127
14:00	190	158	39	87	22	285
22:00	57	35	12	26	63	577
17:00	22	101	40	23	107	171
01:00	221	11	18	56	88	1331
03:00	122	133	10	0	1	1051
21:00	348	180	22	31	116	377
14:00	213	116	35	30	41	1385
06:00	79	247	40	51	59	178
06:00	171	154	36	80	34	1182
23:00	74	155	16	28	81	753

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.4.3 Survey and Investigation Records — Sheet 18

Design Computation

Deflection = $25.437 \times 5.413 = 137.690$ kPa (permissible 259.147) — safe

Deflection = $53.151 \times 10.512 = 558.738$ kNm (permissible 697.376) — safe

Moment = $76.579 \times 10.249 = 784.879$ kNm (permissible 1307.448) — safe

Deflection = $76.426 \times 9.915 = 757.756$ kNm (permissible 1067.937) — safe

Shear = $2.098 \times 1.753 = 3.678$ kN (permissible 4.557) — safe

Deflection = $20.248 \times 7.812 = 158.186$ kPa (permissible 175.674) — revise section

Bearing pressure = $33.308 \times 8.046 = 267.985$ kN (permissible 502.331) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.4.3 Survey and Investigation Records — Sheet 19

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. LA-748, revision C.

3.4.3 Survey and Investigation Records — Sheet 20

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
312/1	Agomani	0.738	Agricultural	Awarded
162/2	Gauripur	1.528	Waste	Notified
200/6	Dhubri	1.729	Waste	Awarded
431/6	Golakganj	1.989	Waste	Awarded
108/5	Dhubri	1.127	Homestead	Awarded
179/6	Gauripur	0.081	Homestead	Notified
72/3	Dhubri	0.126	Agricultural	Awarded
172/2	Golakganj	2.292	Agricultural	Awarded
150/5	Agomani	0.632	Agricultural	Under award
281/5	Gauripur	2.355	Agricultural	Awarded
248/2	Golakganj	1.314	Homestead	Under award
77/9	Dhubri	0.398	Agricultural	Awarded
122/6	Dhubri	0.945	Waste	Under award
370/3	Golakganj	0.548	Agricultural	Under award
441/3	Gauripur	0.502	Agricultural	Awarded
256/4	Golakganj	1.587	Agricultural	Under award
454/8	Dhubri	2.037	Waste	Awarded
476/6	Dhubri	1.340	Waste	Notified
251/8	Gauripur	0.209	Homestead	Awarded
250/3	Dhubri	1.144	Agricultural	Awarded
429/2	Dhubri	0.441	Agricultural	Awarded
277/4	Golakganj	1.542	Agricultural	Awarded
314/9	Agomani	0.338	Agricultural	Notified
250/1	Agomani	1.537	Agricultural	Notified
104/2	Golakganj	0.653	Waste	Awarded
314/6	Dhubri	0.552	Agricultural	Under award

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.4.3 Survey and Investigation Records — Sheet 21

Design Computation

Bearing pressure = $11.645 \times 1.844 = 21.475$ kNm (permissible 29.876) — safe

Shear = $58.980 \times 7.748 = 456.949$ kNm (permissible 570.412) — safe

Deflection = $10.922 \times 11.258 = 122.952$ kPa (permissible 189.948) — safe

Moment = $37.162 \times 10.181 = 378.358$ mm (permissible 572.931) — safe

Deflection = $64.052 \times 2.912 = 186.511$ mm (permissible 336.817) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.4.3 Survey and Investigation Records — Sheet 22

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. GA-279, revision B.

3.4.3 Survey and Investigation Records — Sheet 23

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Coarse aggregate	Depot	77.6	3,704
Boulder	Plant	26.9	3,865
Steel	Quarry A	89.8	7,121
Fine sand	Depot	22.8	9,320
Cement	Depot	34.6	3,601
Bitumen	Quarry A	12.0	8,513
Bitumen	Depot	62.3	2,483
Steel	River bed	78.5	3,976
Bitumen	Depot	15.5	1,578
Fine sand	Quarry A	27.5	5,290
Boulder	Quarry B	18.0	5,148
Murram	Quarry A	49.8	8,330
Boulder	River bed	54.9	3,726
Steel	Quarry B	76.5	8,592
Fine sand	Quarry A	81.6	987
Steel	Depot	26.4	4,406
Bitumen	Quarry B	65.9	3,556
Coarse aggregate	Plant	42.8	7,459
Fine sand	Depot	47.3	9,324
Coarse aggregate	Plant	79.4	7,266

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.4.3 Survey and Investigation Records — Sheet 24

Design Computation

Bearing pressure = $26.314 \times 2.755 = 72.508 \text{ kN}$ (permissible 93.676) — safe

Shear = $59.537 \times 6.788 = 404.133 \text{ kPa}$ (permissible 626.218) — safe

Deflection = $23.107 \times 8.762 = 202.463 \text{ kPa}$ (permissible 335.305) — safe

Deflection = $88.253 \times 10.240 = 903.687 \text{ kPa}$ (permissible 1600.158) — safe

Bearing pressure = $73.514 \times 8.156 = 599.599 \text{ mm}$ (permissible 1083.976) — safe

Shear = $8.531 \times 11.161 = 95.217 \text{ mm}$ (permissible 155.555) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.4.3 Survey and Investigation Records — Sheet 25

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. HY-650, revision C.

3.4.3 Survey and Investigation Records — Sheet 26

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
16:00	163	193	25	1	71	456
19:00	136	80	1	75	115	95
14:00	118	57	29	16	86	615
05:00	354	177	39	81	1	1064
15:00	96	16	14	72	111	556
18:00	80	247	26	20	82	314
01:00	314	210	12	50	19	1232
17:00	115	135	6	77	7	655
19:00	64	138	38	25	66	689
07:00	372	77	9	68	95	844
03:00	173	201	31	29	24	332
19:00	384	88	21	87	98	86
12:00	272	56	24	55	120	590
13:00	29	146	27	82	2	1010
01:00	209	21	8	84	66	1054
18:00	249	133	21	57	21	297

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.5 FUNCTIONAL DESIGN

Three alignment options were examined. Option A crosses at the narrowest section but requires a 2.1 km approach through built-up area with 38 structures affected. Option B, the proposed alignment, crosses 900 m downstream where the channel is stable and the approach traverses agricultural land with 6 structures affected. Option C avoids all structures but adds 3.4 km of approach and crosses a zone of active bank erosion.

Criterion	Option A	Option B (proposed)	Option C
Bridge length (m)	1,180	1,340	1,510
Approach length (km)	2.10	4.60	8.00
Structures affected	38	6	0
Land acquisition (ha)	9.40	18.62	31.80
Bank stability	Marginal	Stable	Active erosion
Estimated cost (Rs cr)	396	412.50	489

Option B is proposed. Although not the shortest crossing, it minimises displacement, sits on a stable reach with a documented history of hydraulic behaviour, and permits an economical span arrangement.

3.5.1 Alignment Studies — Sheet 1

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. GA-940, revision C.

3.5.1 Alignment Studies — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 696	4.20	3.19	0.30	14	167.851
Item 346	5.45	2.72	2.52	4	379.032
Item 745	25.84	5.70	1.82	13	20.786
Item 326	3.77	5.19	1.01	5	216.734
Item 684	13.72	4.95	0.71	19	228.911
Item 292	16.15	3.42	0.38	2	247.985
Item 608	27.86	2.74	1.08	19	369.453
Item 470	13.39	4.83	2.16	8	33.660
Item 407	21.96	5.29	2.24	10	243.975
Item 174	6.49	2.68	2.32	5	373.375
Item 531	3.49	4.11	2.34	19	315.849
Item 421	14.92	2.30	1.59	15	28.436
Item 195	37.90	3.00	2.06	2	292.733
Item 417	26.59	5.96	2.50	10	286.934
Item 784	15.19	5.66	1.20	20	47.721
Item 160	10.29	1.94	2.27	13	156.989
Item 608	5.06	2.86	1.74	5	327.893
Item 663	12.58	2.67	1.20	13	383.135
Item 254	5.15	1.16	2.04	1	194.500
Item 703	8.93	1.91	0.61	18	148.332
Item 679	14.11	1.02	2.61	20	262.332

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.5.1 Alignment Studies — Sheet 3

Design Computation

Settlement = $81.058 \times 9.448 = 765.811$ kPa (permissible 1063.221) — safe

Shear = $43.856 \times 5.045 = 221.257$ kNm (permissible 417.505) — safe

Moment = $10.784 \times 7.368 = 79.458$ kNm (permissible 121.712) — safe

Moment = $85.456 \times 7.519 = 642.577$ kN (permissible 1010.105) — safe

Deflection = $23.451 \times 4.430 = 103.881$ kPa (permissible 119.922) — safe

Bearing pressure = $89.386 \times 5.805 = 518.929$ mm (permissible 582.759) — safe

Shear = $31.495 \times 3.471 = 109.323$ kNm (permissible 133.859) — safe

Moment = $33.196 \times 8.405 = 279.005$ mm (permissible 525.011) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.5.1 Alignment Studies — Sheet 4

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. HY-751, revision A.

3.5.1 Alignment Studies — Sheet 5

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
422/4	Golakganj	1.789	Agricultural	Awarded
275/8	Agomani	1.768	Agricultural	Under award
251/5	Gauripur	1.677	Homestead	Under award
423/6	Agomani	0.239	Agricultural	Awarded
250/4	Agomani	0.530	Waste	Notified
440/1	Golakganj	2.187	Homestead	Notified
53/2	Golakganj	1.888	Agricultural	Under award
465/3	Golakganj	1.904	Homestead	Awarded
420/7	Golakganj	0.993	Agricultural	Notified
91/3	Gauripur	0.115	Waste	Under award
422/3	Golakganj	1.595	Homestead	Awarded
290/9	Gauripur	0.100	Waste	Notified
62/9	Gauripur	1.069	Agricultural	Awarded
24/5	Gauripur	0.738	Agricultural	Notified
176/5	Golakganj	2.010	Agricultural	Notified
191/8	Golakganj	1.994	Waste	Awarded
282/3	Dhubri	2.101	Agricultural	Notified
12/3	Gauripur	0.383	Waste	Notified
71/9	Dhubri	0.816	Waste	Notified

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.5.1 Alignment Studies — Sheet 6

Design Computation

Shear = $79.607 \times 1.059 = 84.315$ mm (permissible 91.555) — safe

Moment = $41.244 \times 0.723 = 29.829$ kPa (permissible 39.577) — safe

Deflection = $54.946 \times 2.713 = 149.073$ kPa (permissible 220.917) — safe

Settlement = $46.190 \times 3.273 = 151.171$ mm (permissible 277.302) — safe

Bearing pressure = $19.030 \times 5.591 = 106.405$ kNm (permissible 147.212) — safe

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.5.1 Alignment Studies — Sheet 7

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. GA-895, revision A.

3.5.1 Alignment Studies — Sheet 8

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
08:00	90	257	29	28	95	272
12:00	269	51	14	20	90	963
16:00	226	96	26	25	45	732
02:00	389	103	1	43	70	1019
14:00	380	14	24	42	66	1357
09:00	282	255	4	14	117	548
03:00	63	77	17	5	115	451
08:00	86	219	27	86	104	609
12:00	96	147	32	73	63	749
02:00	162	24	11	54	114	228
08:00	28	172	5	33	10	1325
07:00	54	77	7	58	1	774
17:00	233	247	17	79	16	168
16:00	383	71	7	20	33	183
05:00	123	248	19	80	39	1167
06:00	168	124	32	86	22	634
11:00	29	74	2	1	2	1115
17:00	117	141	30	31	119	995

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.5.1 Alignment Studies — Sheet 9

Design Computation

Bearing pressure = $59.429 \times 6.733 = 400.109$ mm (permissible 654.010) — safe

Shear = $31.501 \times 10.055 = 316.725$ kPa (permissible 598.933) — safe

Deflection = $75.492 \times 0.565 = 42.680$ kPa (permissible 50.736) — safe

Deflection = $75.873 \times 10.498 = 796.532$ kN (permissible 1305.343) — revise section

Moment = $17.496 \times 3.521 = 61.602$ mm (permissible 83.749) — safe

Moment = $88.657 \times 4.153 = 368.192$ mm (permissible 454.786) — safe

Deflection = $30.845 \times 1.373 = 42.354$ kN (permissible 53.406) — safe

Shear = $9.086 \times 9.878 = 89.747$ kPa (permissible 138.998) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.5.1 Alignment Studies — Sheet 10

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. LA-882, revision B.

3.5.1 Alignment Studies — Sheet 11

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Cement	Plant	58.6	897
Murram	Plant	57.2	8,462
Fine sand	Plant	68.3	443
Murram	Plant	72.2	3,947
Coarse aggregate	Quarry A	5.7	6,089
Coarse aggregate	Depot	75.6	9,330
Coarse aggregate	Quarry A	57.1	4,186
Steel	River bed	2.3	1,328
Boulder	Plant	81.0	1,686
Boulder	Plant	7.8	7,943
Cement	Quarry A	76.5	4,026
Boulder	Quarry B	22.3	7,722
Steel	Depot	8.8	4,887
Murram	Quarry A	56.3	3,428
Coarse aggregate	Plant	15.0	4,340
Boulder	River bed	56.7	2,366
Coarse aggregate	Depot	7.3	4,583
Boulder	Quarry A	62.9	8,201

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.5.1 Alignment Studies — Sheet 12

Design Computation

Moment = $42.355 \times 5.810 = 246.063 \text{ kN}$ (permissible 323.554) — safe

Moment = $43.092 \times 3.759 = 161.993 \text{ kPa}$ (permissible 306.956) — safe

Moment = $82.573 \times 11.194 = 924.344 \text{ kNm}$ (permissible 1081.926) — safe

Settlement = $85.794 \times 1.938 = 166.287 \text{ mm}$ (permissible 299.955) — safe

Bearing pressure = $21.593 \times 10.813 = 233.495 \text{ kPa}$ (permissible 250.099) — revise section

Deflection = $44.761 \times 5.629 = 251.951 \text{ kN}$ (permissible 353.678) — safe

Deflection = $11.761 \times 4.243 = 49.906 \text{ mm}$ (permissible 87.996) — safe

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.5.1 Alignment Studies — Sheet 13

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. GA-502, revision B.

3.5.1 Alignment Studies — Sheet 14

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
194/7	Agomani	2.057	Homestead	Awarded
36/5	Gauripur	0.636	Homestead	Under award
271/6	Gauripur	1.867	Homestead	Awarded
425/7	Gauripur	1.741	Agricultural	Notified
220/8	Gauripur	1.565	Homestead	Under award
35/9	Gauripur	0.451	Homestead	Under award
154/5	Agomani	1.787	Waste	Under award
217/4	Agomani	1.185	Waste	Under award
71/3	Gauripur	0.227	Waste	Under award
291/4	Golakganj	2.180	Homestead	Under award
81/9	Gauripur	0.624	Agricultural	Under award
294/2	Agomani	0.612	Homestead	Notified
113/1	Golakganj	0.950	Waste	Notified
117/7	Agomani	0.845	Agricultural	Under award
152/6	Gauripur	1.664	Waste	Notified
414/4	Dhubri	0.687	Agricultural	Under award
214/8	Golakganj	2.292	Agricultural	Awarded

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.6 ENGINEERING DESIGN

The design conforms to IRC:6-2017 for loads and stresses, IRC:112-2020 for concrete road bridges, IRC:78-2014 for foundations and substructure, and IS:456-2000. The limit state method has been adopted throughout.

Element	Provision
Superstructure	Pre-stressed concrete box girder, 22 spans
Span arrangement	2 × 40 m + 18 × 60 m + 2 × 40 m
Substructure	RCC circular piers, 3.5 m diameter
Foundation	Well foundation, 9.0 m external diameter
Founding level	RL –24.5 m, 6.0 m into dense sand
Bearings	Pot-PTFE
Expansion joints	Strip seal, 80 mm movement
Design load	IRC Class 70R and Class A
Seismic zone	Zone V, importance factor 1.5
Design life	100 years

3.6.1 Design Computations — Sheet 1

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. ST-821, revision A.

3.6.1 Design Computations — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 353	32.83	3.19	1.32	1	184.194
Item 564	16.84	3.57	2.71	13	36.737
Item 339	30.84	4.30	1.66	16	360.924
Item 488	29.59	4.08	0.47	3	155.161
Item 486	6.09	0.63	0.86	3	199.535
Item 765	21.64	1.49	2.46	20	26.685
Item 138	20.57	1.40	2.85	19	183.049
Item 698	35.48	1.93	1.20	14	54.855
Item 882	24.85	3.33	1.08	12	380.050
Item 606	21.14	1.43	0.62	17	114.457
Item 772	4.71	5.45	1.62	23	218.429
Item 452	11.27	0.37	1.16	9	14.428
Item 360	16.32	5.06	2.70	6	263.319
Item 219	8.31	4.08	2.93	7	389.257
Item 264	13.40	0.38	2.34	11	334.722
Item 682	18.55	1.62	1.35	15	163.455
Item 182	24.36	1.95	0.47	3	265.622
Item 882	21.80	2.35	2.46	17	195.807

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

3.6.1 Design Computations — Sheet 3

Design Computation

Deflection = $17.882 \times 10.986 = 196.452$ kPa (permissible 316.185) — safe

Deflection = $27.648 \times 4.175 = 115.430$ kNm (permissible 184.569) — safe

Settlement = $6.399 \times 7.560 = 48.380$ kN (permissible 58.110) — safe

Settlement = $40.296 \times 1.367 = 55.081$ kN (permissible 81.113) — safe

Settlement = $14.752 \times 1.815 = 26.782$ kPa (permissible 42.871) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.6.1 Design Computations — Sheet 4

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. ST-152, revision A.

3.6.1 Design Computations — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 214	6.11	4.43	2.18	20	362.715
Item 770	15.84	2.76	0.46	24	302.497
Item 614	33.13	3.54	1.60	11	210.771
Item 861	21.75	1.42	1.01	10	363.632
Item 861	14.20	5.47	1.33	21	119.271
Item 660	17.94	4.54	2.46	24	340.343
Item 628	13.48	5.04	1.20	10	260.902
Item 211	9.44	2.41	2.88	10	82.895
Item 506	25.30	5.10	2.12	23	291.485
Item 792	24.99	2.36	1.40	16	329.253
Item 619	33.85	1.83	0.39	24	346.695
Item 529	31.64	3.86	2.51	10	110.365
Item 616	12.21	5.97	1.97	8	145.183
Item 353	32.03	4.56	0.90	20	210.573
Item 300	15.98	4.75	2.42	13	34.605
Item 274	31.65	5.96	2.78	5	9.814
Item 295	28.03	0.80	1.50	22	362.266
Item 679	5.52	3.76	2.62	21	27.323
Item 139	33.75	0.78	0.58	12	397.982
Item 176	32.96	4.43	1.88	11	89.710
Item 861	15.58	2.16	1.66	4	366.684
Item 724	15.23	1.21	0.91	13	60.356
Item 845	31.72	0.87	0.82	10	370.087

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.6.1 Design Computations — Sheet 6

Design Computation

Settlement = $10.702 \times 0.990 = 10.597$ kN (permissible 18.558) — safe

Settlement = $30.082 \times 10.630 = 319.771$ kPa (permissible 365.919) — safe

Shear = $8.385 \times 1.995 = 16.727$ kN (permissible 22.210) — safe

Moment = $16.038 \times 9.240 = 148.184$ kPa (permissible 229.348) — safe

Settlement = $37.920 \times 11.264 = 427.115$ mm (permissible 699.743) — safe

Settlement = $44.577 \times 4.652 = 207.370$ kN (permissible 272.171) — safe

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.6.1 Design Computations — Sheet 7

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. ST-682, revision C.

3.6.1 Design Computations — Sheet 8

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Murram	Depot	59.0	2,215
Cement	Depot	4.8	6,266
Bitumen	River bed	32.5	632
Bitumen	Quarry B	75.1	417
Coarse aggregate	Quarry B	41.5	2,267
Murram	Depot	51.6	1,950
Cement	Plant	5.5	6,221
Cement	Quarry A	85.3	1,853
Steel	Quarry B	58.9	7,992
Cement	Plant	32.5	6,270
Murram	Depot	35.0	812
Boulder	Quarry B	83.7	5,310
Coarse aggregate	Quarry A	86.4	8,534
Cement	Depot	34.2	9,379
Fine sand	River bed	14.3	4,545
Murram	Quarry A	70.9	4,895
Bitumen	River bed	42.0	5,037
Cement	Depot	77.7	7,505

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.6.1 Design Computations — Sheet 9

Design Computation

Shear = $80.930 \times 5.490 = 444.299$ kNm (permissible 602.986) — safe

Bearing pressure = $41.623 \times 5.852 = 243.590$ kN (permissible 368.950) — safe

Moment = $16.463 \times 2.095 = 34.486$ kNm (permissible 52.279) — safe

Bearing pressure = $28.714 \times 2.109 = 60.562$ mm (permissible 112.378) — safe

Bearing pressure = $86.091 \times 0.562 = 48.375$ kN (permissible 63.804) — revise section

Shear = $1.489 \times 0.593 = 0.882$ mm (permissible 1.407) — revise section

Moment = $73.317 \times 4.217 = 309.194$ kPa (permissible 386.288) — safe

Settlement = $78.387 \times 8.306 = 651.066$ kN (permissible 950.105) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.6.1 Design Computations — Sheet 10

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. ST-773, revision C.

3.6.1 Design Computations — Sheet 11

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
13:00	356	22	16	52	88	487
00:00	165	26	31	40	89	410
07:00	48	111	19	33	39	1149
08:00	328	15	25	26	90	512
06:00	77	216	31	19	91	724
07:00	354	242	5	55	44	1116
17:00	370	137	29	78	46	1344
04:00	335	52	30	12	28	1142
10:00	332	252	10	71	30	532
13:00	390	183	38	25	43	342
18:00	156	28	28	41	73	368
21:00	141	194	38	50	107	123
10:00	224	101	19	15	15	870
21:00	360	229	28	52	76	101
15:00	193	150	12	2	96	111
22:00	330	186	32	40	47	1071
18:00	72	166	11	14	6	990
08:00	61	150	1	25	31	1373
02:00	50	95	22	49	90	720
15:00	329	50	31	35	80	201
15:00	259	159	15	69	85	198
16:00	280	176	25	57	66	1050
21:00	101	40	36	78	65	430
10:00	396	85	36	69	51	1358
03:00	248	151	11	38	20	811

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.6.1 Design Computations — Sheet 12

Design Computation

Deflection = $61.389 \times 7.438 = 456.633 \text{ kNm}$ (permissible 833.196) — safe

Bearing pressure = $64.571 \times 8.385 = 541.463 \text{ mm}$ (permissible 891.267) — safe

Deflection = $49.762 \times 7.957 = 395.981 \text{ kNm}$ (permissible 476.854) — safe

Bearing pressure = $12.018 \times 2.538 = 30.496 \text{ mm}$ (permissible 46.758) — safe

Moment = $39.981 \times 9.306 = 372.055 \text{ mm}$ (permissible 487.071) — safe

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.6.1 Design Computations — Sheet 13

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. HY-436, revision A.

3.6.1 Design Computations — Sheet 14

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 598	18.13	2.87	2.13	5	12.068
Item 229	16.81	5.42	2.08	23	170.239
Item 213	31.83	2.98	2.09	14	333.373
Item 895	4.82	1.34	1.44	9	243.021
Item 783	33.77	5.50	0.79	17	1.481
Item 441	39.42	2.38	2.44	17	128.427
Item 526	37.78	2.01	0.81	1	237.192
Item 257	10.40	5.14	0.65	5	198.264
Item 656	16.72	2.81	1.01	3	154.473
Item 608	18.55	3.06	0.86	2	363.819
Item 692	27.76	0.42	2.75	12	260.372
Item 535	2.48	1.77	2.85	12	17.386
Item 743	8.70	0.67	1.56	2	201.961
Item 153	29.45	0.87	0.58	12	337.997
Item 463	32.43	4.08	1.20	3	97.247
Item 756	36.86	3.20	0.99	21	368.330
Item 130	9.29	0.94	1.34	5	200.528
Item 331	39.14	5.62	2.68	17	152.867
Item 640	8.23	3.84	2.93	21	251.505

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.6.1 Design Computations — Sheet 15

Design Computation

Moment = $2.838 \times 3.538 = 10.041$ kNm (permissible 15.864) — safe

Settlement = $27.290 \times 2.032 = 55.464$ kN (permissible 59.516) — safe

Shear = $65.184 \times 9.231 = 601.745$ mm (permissible 771.628) — safe

Deflection = $60.558 \times 4.417 = 267.510$ kN (permissible 363.600) — safe

Settlement = $55.613 \times 3.057 = 170.025$ mm (permissible 297.746) — safe

Moment = $86.263 \times 9.229 = 796.129$ mm (permissible 1067.755) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.6.1 Design Computations — Sheet 16

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. ST-970, revision A.

3.6.1 Design Computations — Sheet 17

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 741	23.24	5.36	2.54	18	368.251
Item 325	31.40	3.47	2.83	16	181.842
Item 403	17.41	5.67	1.65	14	196.923
Item 864	31.87	5.09	1.68	14	354.866
Item 610	22.75	3.73	0.37	18	332.239
Item 896	27.31	4.88	1.77	13	166.742
Item 348	11.99	2.22	1.60	10	95.278
Item 500	22.38	5.12	1.13	16	249.524
Item 687	36.63	2.34	0.67	24	64.914
Item 336	9.23	0.65	2.08	13	228.665
Item 334	23.46	5.34	0.55	9	118.374
Item 254	28.18	0.75	2.59	20	286.502
Item 642	12.16	5.25	0.52	19	266.065
Item 233	26.99	0.58	0.51	10	141.087
Item 380	7.21	5.75	2.68	19	25.199
Item 757	26.14	1.21	1.33	1	353.922
Item 579	39.59	2.60	2.06	12	236.681
Item 579	14.73	5.51	1.35	9	384.970

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.6.1 Design Computations — Sheet 18

Design Computation

Settlement = $21.005 \times 2.631 = 55.266$ kPa (permissible 60.604) — safe

Deflection = $28.447 \times 4.153 = 118.129$ kPa (permissible 129.972) — revise section

Shear = $69.841 \times 0.775 = 54.105$ kN (permissible 81.960) — safe

Settlement = $75.575 \times 5.916 = 447.098$ kN (permissible 538.766) — safe

Moment = $77.757 \times 11.970 = 930.758$ kNm (permissible 1415.310) — safe

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.6.1 Design Computations — Sheet 19

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. LA-649, revision A.

3.6.1 Design Computations — Sheet 20

Bore Log Summary

Depth (m)	Strata	N value	Sample
2.6	Weathered rock	23	UDS
25.5	Weathered rock	60	DS
27.1	Weathered rock	50	DS
3.9	Gravelly sand	52	SPT
25.5	Gravelly sand	50	SPT
24.0	Weathered rock	35	DS
14.8	Silty clay	57	SPT
10.6	Coarse sand	30	SPT
9.0	Weathered rock	4	DS
7.8	Coarse sand	8	DS
16.3	Clayey silt	7	UDS
1.5	Silty clay	10	DS
27.3	Clayey silt	35	SPT
9.2	Silty clay	22	DS
26.2	Coarse sand	35	UDS
26.1	Weathered rock	26	SPT

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.6.1 Design Computations — Sheet 21

Design Computation

Moment = $87.432 \times 9.455 = 826.686$ kNm (permissible 880.467) — safe

Settlement = $62.535 \times 6.789 = 424.547$ kN (permissible 743.266) — safe

Deflection = $55.673 \times 2.446 = 136.191$ kPa (permissible 194.923) — safe

Bearing pressure = $83.015 \times 3.271 = 271.508$ mm (permissible 379.428) — safe

Shear = $85.886 \times 8.389 = 720.503$ kNm (permissible 1234.872) — safe

Bearing pressure = $75.702 \times 4.522 = 342.360$ kNm (permissible 485.283) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.6.1 Design Computations — Sheet 22

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. GA-528, revision B.

3.6.1 Design Computations — Sheet 23

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 107	38.58	1.09	1.53	5	293.166
Item 561	6.94	2.98	2.30	13	108.403
Item 427	24.00	2.26	0.79	6	157.316
Item 291	5.42	2.00	0.66	9	233.787
Item 736	25.51	2.74	2.93	21	79.356
Item 674	34.55	2.09	0.32	9	206.447
Item 679	39.23	5.38	2.82	6	315.105
Item 181	25.28	4.65	0.45	4	51.461
Item 669	27.35	1.50	0.30	14	379.809
Item 611	28.11	5.71	2.31	19	121.683
Item 310	3.01	2.78	1.66	10	9.081
Item 335	37.82	2.42	0.93	15	375.356
Item 225	3.59	2.96	2.97	17	75.247
Item 323	14.63	4.24	0.57	9	277.328
Item 265	5.15	3.28	2.39	3	211.752
Item 737	29.53	2.30	0.32	16	75.221
Item 229	29.19	3.90	2.47	18	141.664
Item 228	15.26	0.54	2.72	11	28.152
Item 701	20.36	2.18	0.65	5	234.373
Item 709	31.33	5.52	1.28	8	222.334
Item 791	39.88	5.83	1.86	11	362.154
Item 466	7.78	3.97	2.60	18	49.777
Item 763	18.30	4.31	0.80	22	72.913
Item 498	8.61	0.99	1.10	1	370.944

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.6.1 Design Computations — Sheet 24

Design Computation

Moment = $81.930 \times 3.698 = 302.987 \text{ kNm}$ (permissible 435.193) — safe

Bearing pressure = $37.904 \times 4.014 = 152.148 \text{ kPa}$ (permissible 266.701) — safe

Bearing pressure = $31.624 \times 9.929 = 313.995 \text{ kN}$ (permissible 424.854) — safe

Shear = $76.132 \times 5.154 = 392.350 \text{ kNm}$ (permissible 705.864) — safe

Shear = $83.373 \times 5.047 = 420.763 \text{ kPa}$ (permissible 628.926) — safe

Settlement = $66.327 \times 2.026 = 134.380 \text{ kNm}$ (permissible 174.154) — safe

Settlement = $6.912 \times 10.008 = 69.181 \text{ kPa}$ (permissible 84.572) — revise section

Shear = $48.452 \times 9.660 = 468.023 \text{ kN}$ (permissible 733.604) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.6.1 Design Computations — Sheet 25

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. GA-457, revision A.

3.6.1 Design Computations — Sheet 26

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
59/7	Gauripur	0.177	Homestead	Awarded
122/5	Gauripur	0.866	Waste	Awarded
202/7	Dhubri	0.298	Waste	Under award
464/9	Golakganj	1.220	Waste	Notified
370/4	Gauripur	1.302	Homestead	Awarded
12/5	Agomani	0.094	Waste	Under award
254/7	Dhubri	1.709	Waste	Under award
373/8	Gauripur	2.400	Agricultural	Notified
214/1	Agomani	0.415	Agricultural	Notified
166/2	Dhubri	1.366	Agricultural	Under award
157/8	Golakganj	0.430	Waste	Under award
119/9	Golakganj	1.162	Waste	Under award
321/1	Gauripur	0.658	Homestead	Notified
158/9	Gauripur	1.700	Waste	Under award
424/5	Dhubri	0.597	Waste	Awarded
104/5	Golakganj	1.889	Waste	Under award
374/2	Dhubri	1.630	Waste	Under award
216/1	Gauripur	0.971	Agricultural	Notified
451/4	Dhubri	0.634	Homestead	Awarded
79/8	Gauripur	2.347	Agricultural	Under award
153/1	Agomani	1.784	Homestead	Notified

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.6.1 Design Computations — Sheet 27

Design Computation

Deflection = $17.351 \times 0.747 = 12.968$ kN (permissible 21.439) — safe

Moment = $29.790 \times 11.788 = 351.154$ kN (permissible 539.173) — safe

Moment = $68.159 \times 2.185 = 148.951$ kNm (permissible 243.497) — safe

Bearing pressure = $64.340 \times 1.866 = 120.030$ mm (permissible 202.540) — safe

Settlement = $42.192 \times 8.651 = 364.989$ kPa (permissible 642.433) — safe

Settlement = $11.598 \times 5.122 = 59.399$ kPa (permissible 90.483) — safe

Settlement = $28.707 \times 10.782 = 309.530$ kN (permissible 505.894) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.6.1 Design Computations — Sheet 28

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. HY-296, revision C.

3.6.1 Design Computations — Sheet 29

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
405/7	Golakganj	1.049	Agricultural	Awarded
136/3	Gauripur	0.590	Agricultural	Notified
287/8	Agomani	2.138	Waste	Under award
441/9	Gauripur	0.999	Waste	Notified
200/6	Golakganj	0.935	Waste	Under award
397/2	Dhubri	0.143	Agricultural	Notified
43/3	Gauripur	0.122	Waste	Notified
20/3	Gauripur	2.045	Agricultural	Awarded
409/7	Golakganj	0.232	Waste	Under award
286/1	Agomani	1.339	Agricultural	Notified
465/9	Golakganj	0.292	Homestead	Under award
78/9	Agomani	2.269	Waste	Under award
40/5	Gauripur	1.343	Agricultural	Under award
65/4	Golakganj	0.252	Homestead	Notified
244/2	Dhubri	2.066	Waste	Notified
267/1	Gauripur	0.256	Homestead	Notified
125/6	Gauripur	2.209	Waste	Notified
114/5	Gauripur	0.835	Waste	Under award
55/6	Dhubri	0.663	Agricultural	Under award
82/7	Gauripur	0.473	Agricultural	Awarded
302/3	Agomani	0.114	Waste	Notified
18/2	Golakganj	0.647	Agricultural	Notified
480/3	Dhubri	0.591	Homestead	Under award
227/9	Agomani	2.168	Waste	Notified
392/5	Dhubri	0.258	Agricultural	Awarded
386/7	Dhubri	0.889	Agricultural	Under award

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.6.1 Design Computations — Sheet 30

Design Computation

Shear = $4.161 \times 8.873 = 36.917 \text{ kN}$ (permissible 58.733) — safe

Deflection = $86.033 \times 9.102 = 783.092 \text{ kN}$ (permissible 1321.948) — safe

Deflection = $80.817 \times 5.775 = 466.716 \text{ kN}$ (permissible 608.877) — safe

Shear = $9.840 \times 6.346 = 62.444 \text{ kNm}$ (permissible 86.471) — safe

Deflection = $28.908 \times 3.856 = 111.479 \text{ kN}$ (permissible 190.985) — safe

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.6.1 Design Computations — Sheet 31

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. GA-807, revision B.

3.6.1 Design Computations — Sheet 32

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
09:00	188	189	5	90	111	1349
17:00	381	164	26	39	109	1094
07:00	164	201	31	30	16	1204
07:00	74	103	34	66	32	900
06:00	322	116	14	26	4	1082
22:00	322	154	34	71	117	1196
18:00	199	60	15	65	64	468
07:00	138	79	24	57	88	534
05:00	54	260	31	73	107	1115
03:00	75	70	4	81	112	691
11:00	132	18	20	74	26	630
15:00	391	216	18	9	36	1117
06:00	52	33	36	26	108	1087
19:00	214	185	35	37	101	941
02:00	314	213	34	72	0	420
20:00	355	227	33	72	14	567
13:00	293	78	18	64	100	334
23:00	80	184	23	60	20	1331
00:00	223	203	38	69	115	458
04:00	317	184	4	5	72	183
01:00	185	187	30	34	7	574

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.6.1 Design Computations — Sheet 33

Design Computation

Bearing pressure = $9.546 \times 6.941 = 66.262$ kPa (permissible 125.350) — safe

Moment = $48.114 \times 5.543 = 266.700$ kN (permissible 314.403) — safe

Shear = $74.644 \times 7.764 = 579.505$ mm (permissible 1047.620) — safe

Settlement = $78.317 \times 5.052 = 395.653$ kN (permissible 423.953) — safe

Settlement = $4.734 \times 0.893 = 4.227$ mm (permissible 5.433) — safe

Settlement = $46.896 \times 3.615 = 169.527$ kPa (permissible 238.560) — revise section

Deflection = $15.225 \times 7.688 = 117.058$ kNm (permissible 216.309) — safe

Bearing pressure = $1.345 \times 6.943 = 9.341$ kNm (permissible 15.263) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.6.1 Design Computations — Sheet 34

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. GA-644, revision A.

3.6.1 Design Computations — Sheet 35

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
183/9	Gauripur	1.858	Homestead	Awarded
223/4	Golakganj	0.492	Waste	Under award
274/7	Golakganj	2.131	Agricultural	Notified
434/2	Golakganj	2.025	Homestead	Under award
284/4	Dhubri	0.732	Homestead	Notified
472/3	Agomani	0.649	Homestead	Under award
255/1	Dhubri	1.925	Agricultural	Awarded
449/9	Agomani	0.086	Waste	Awarded
138/8	Dhubri	2.095	Agricultural	Under award
369/7	Dhubri	1.889	Agricultural	Awarded
152/7	Dhubri	1.102	Homestead	Under award
153/8	Gauripur	1.852	Agricultural	Awarded
204/3	Golakganj	0.766	Agricultural	Under award
171/4	Golakganj	1.172	Agricultural	Awarded
193/2	Golakganj	1.409	Homestead	Under award
45/9	Agomani	0.365	Waste	Awarded
465/1	Golakganj	1.725	Agricultural	Awarded
477/9	Dhubri	1.529	Homestead	Under award
467/8	Gauripur	1.752	Waste	Under award
299/5	Golakganj	1.907	Homestead	Under award
85/5	Agomani	0.650	Waste	Notified
203/1	Dhubri	0.370	Agricultural	Under award
70/5	Dhubri	2.035	Homestead	Awarded
218/8	Dhubri	1.442	Homestead	Under award
250/5	Agomani	1.669	Waste	Under award
410/5	Agomani	1.160	Homestead	Awarded

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.6.1 Design Computations — Sheet 36

Design Computation

Moment = $73.843 \times 4.488 = 331.406$ kN (permissible 416.823) — revise section

Deflection = $24.093 \times 10.714 = 258.143$ kPa (permissible 287.692) — safe

Bearing pressure = $26.151 \times 11.359 = 297.057$ kNm (permissible 443.361) — safe

Settlement = $73.373 \times 2.139 = 156.929$ kN (permissible 223.169) — safe

Bearing pressure = $6.528 \times 10.961 = 71.551$ kNm (permissible 130.126) — safe

Bearing pressure = $57.908 \times 5.412 = 313.401$ kPa (permissible 493.963) — safe

Moment = $80.413 \times 11.992 = 964.308$ kNm (permissible 1302.401) — safe

Settlement = $70.771 \times 10.874 = 769.580$ kPa (permissible 1395.447) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.6.1 Design Computations — Sheet 37

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. HY-830, revision A.

3.6.1 Design Computations — Sheet 38

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 772	10.62	5.72	1.77	19	251.086
Item 188	8.55	2.77	1.12	20	299.767
Item 432	34.11	5.79	2.30	22	143.784
Item 291	27.27	3.87	1.78	12	96.519
Item 159	3.39	2.83	1.78	16	193.246
Item 538	38.10	4.51	0.56	7	34.485
Item 754	32.80	0.97	2.88	22	226.340
Item 331	2.58	4.69	2.20	12	115.215
Item 687	31.81	0.76	2.46	18	74.327
Item 815	26.40	2.25	2.46	15	345.585
Item 273	29.58	4.97	1.69	17	173.020
Item 451	13.55	1.60	1.70	11	78.223
Item 235	15.68	5.04	1.10	1	294.548
Item 512	5.53	5.91	2.32	13	194.863
Item 230	27.50	3.01	1.70	6	62.758
Item 360	22.41	0.42	1.99	24	397.093
Item 675	12.26	4.47	2.01	11	379.434
Item 602	19.81	1.88	2.25	14	145.490
Item 551	31.73	1.65	1.60	23	160.743
Item 599	39.38	1.41	2.06	9	45.892
Item 231	2.66	5.24	1.87	11	359.829
Item 161	7.22	2.79	2.37	14	201.308

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.6.1 Design Computations — Sheet 39

Design Computation

Shear = $85.177 \times 8.563 = 729.370$ kPa (permissible 1102.632) — safe

Settlement = $22.731 \times 4.653 = 105.773$ kN (permissible 144.045) — safe

Moment = $78.546 \times 4.566 = 358.658$ kPa (permissible 542.162) — revise section

Moment = $14.922 \times 7.887 = 117.691$ kN (permissible 168.667) — safe

Shear = $42.871 \times 3.834 = 164.364$ kN (permissible 185.069) — safe

Bearing pressure = $58.939 \times 7.448 = 438.949$ kN (permissible 816.390) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.6.1 Design Computations — Sheet 40

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. LA-565, revision A.

3.6.1 Design Computations — Sheet 41

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Murram	Depot	76.7	6,393
Murram	Quarry B	68.8	3,183
Murram	River bed	14.6	1,554
Cement	Depot	8.8	6,879
Murram	River bed	10.9	1,982
Boulder	Depot	18.2	681
Bitumen	Depot	7.8	4,563
Boulder	Quarry B	37.0	8,240
Bitumen	Depot	32.8	706
Steel	Quarry B	30.5	6,052
Fine sand	River bed	51.6	6,886
Coarse aggregate	Depot	69.4	4,883
Cement	Plant	5.5	3,165
Murram	Depot	64.0	6,636
Cement	Quarry B	34.8	5,303
Coarse aggregate	Depot	71.8	2,875
Steel	Depot	22.9	302

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.6.1 Design Computations — Sheet 42

Design Computation

Deflection = $69.175 \times 9.283 = 642.126$ kPa (permissible 726.598) — safe

Moment = $8.887 \times 3.765 = 33.459$ kN (permissible 41.186) — safe

Bearing pressure = $76.030 \times 4.174 = 317.355$ kPa (permissible 503.093) — safe

Deflection = $30.263 \times 2.527 = 76.480$ kPa (permissible 119.534) — safe

Shear = $37.118 \times 7.243 = 268.842$ kN (permissible 293.818) — safe

Bearing pressure = $88.094 \times 2.847 = 250.810$ kNm (permissible 313.111) — safe

Moment = $46.219 \times 6.353 = 293.608$ kPa (permissible 534.700) — safe

Moment = $21.854 \times 6.212 = 135.764$ kNm (permissible 192.606) — safe

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.6.1 Design Computations — Sheet 43

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. GA-495, revision B.

3.6.1 Design Computations — Sheet 44

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
11:00	58	256	36	26	52	655
18:00	136	221	36	76	73	345
11:00	97	254	36	80	82	212
21:00	40	214	6	37	75	365
01:00	361	77	21	45	13	336
22:00	310	254	24	66	106	851
05:00	209	52	24	25	18	1354
06:00	169	249	4	51	28	414
06:00	400	129	39	74	0	1362
23:00	137	55	24	2	94	111
12:00	303	220	38	67	73	532
07:00	97	70	39	18	63	935
20:00	153	208	5	81	90	711
04:00	77	63	34	76	11	588
02:00	303	48	22	68	39	1389
14:00	233	248	11	66	72	814
02:00	297	61	13	17	30	653
21:00	220	199	34	3	116	752
05:00	61	180	9	30	4	795
14:00	198	127	16	51	19	1297
13:00	329	260	0	11	94	1073
06:00	193	79	22	71	25	592
17:00	166	176	10	39	88	1319
06:00	114	96	30	47	68	1307

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.6.1 Design Computations — Sheet 45

Design Computation

Moment = $41.543 \times 6.040 = 250.907 \text{ kN}$ (permissible 338.136) — safe

Bearing pressure = $7.396 \times 4.164 = 30.801 \text{ kNm}$ (permissible 37.495) — safe

Bearing pressure = $2.664 \times 11.365 = 30.279 \text{ kN}$ (permissible 56.538) — safe

Shear = $66.459 \times 8.998 = 598.028 \text{ mm}$ (permissible 702.788) — safe

Settlement = $21.315 \times 4.839 = 103.143 \text{ kN}$ (permissible 129.918) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.6.1 Design Computations — Sheet 46

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. ST-565, revision A.

3.7 FINANCIAL ESTIMATES AND COST PROJECTIONS

Rates are derived from the Assam PWD Schedule of Rates 2025-26 with lead and lift adjustments. Items not covered by the Schedule have been rate-analysed and are placed at Annexure IV. The abstract of cost is summarised below.

Sl.	Description of item	Amount (Rs. crore)
1	Substructure — wells, well caps, piers	128.40
2	Superstructure — PSC box girder	146.80
3	Approach roads and embankment	62.30
4	River training and bank protection	38.90
5	Utilities, lighting and signage	14.60
6	Contingencies at 3 per cent	11.70
7	Quality control and supervision	9.80
	TOTAL PROJECT COST	418.20

The total project cost of the scheme is Rs. 418.20 crore. This abstract supersedes the preliminary estimate of the 2021 pre-feasibility study.

3.7.1 Rate Analysis — Sheet 1

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. ST-987, revision C.

3.7.1 Rate Analysis — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 614	19.62	3.75	1.14	20	280.339
Item 845	16.40	4.31	1.47	24	169.612
Item 272	11.04	0.93	1.62	19	26.268
Item 804	16.58	4.55	0.48	10	82.779
Item 329	29.58	2.70	0.45	9	364.640
Item 506	12.69	4.91	0.76	1	351.870
Item 156	36.04	5.12	0.59	8	170.465
Item 214	38.63	0.32	2.34	7	358.226
Item 307	14.55	5.39	0.57	1	200.821
Item 686	20.88	5.55	1.38	13	77.674
Item 883	5.55	3.60	0.68	20	288.134
Item 142	4.06	3.46	1.90	5	298.315
Item 854	39.81	5.99	0.54	23	161.545
Item 342	8.15	3.18	2.26	22	209.302
Item 678	31.44	0.79	1.17	19	146.862
Item 315	17.28	5.92	1.81	1	152.199
Item 722	14.49	2.77	0.80	15	108.585
Item 190	13.67	1.92	1.65	2	11.131
Item 899	23.30	2.74	1.17	8	226.845
Item 167	18.15	1.62	2.90	13	204.441
Item 673	16.04	4.10	0.68	9	12.720

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

3.7.1 Rate Analysis — Sheet 3

Design Computation

Settlement = $85.240 \times 11.840 = 1009.204$ kN (permissible 1069.850) — revise section

Settlement = $19.566 \times 4.911 = 96.086$ kNm (permissible 179.823) — safe

Shear = $13.517 \times 6.918 = 93.519$ kNm (permissible 157.614) — safe

Shear = $11.496 \times 5.271 = 60.596$ kPa (permissible 93.921) — safe

Settlement = $60.024 \times 3.580 = 214.897$ kPa (permissible 289.368) — safe

Moment = $25.640 \times 1.889 = 48.428$ kN (permissible 82.964) — safe

Bearing pressure = $2.158 \times 2.440 = 5.266$ kPa (permissible 5.902) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.7.1 Rate Analysis — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. GA-998, revision C.

3.7.1 Rate Analysis — Sheet 5

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
163/2	Gauripur	1.744	Agricultural	Awarded
368/8	Golakganj	0.192	Waste	Notified
37/1	Agomani	0.623	Homestead	Awarded
200/8	Gauripur	0.135	Waste	Awarded
323/1	Golakganj	1.006	Agricultural	Awarded
106/1	Gauripur	0.248	Agricultural	Under award
329/8	Golakganj	0.111	Agricultural	Under award
53/6	Dhubri	1.140	Homestead	Notified
86/7	Golakganj	2.007	Agricultural	Under award
353/4	Golakganj	1.883	Agricultural	Awarded
65/2	Dhubri	0.332	Waste	Awarded
200/2	Dhubri	1.149	Waste	Awarded
199/2	Gauripur	0.063	Waste	Notified
111/1	Golakganj	0.645	Waste	Awarded
62/9	Golakganj	0.186	Homestead	Under award
446/6	Gauripur	1.552	Waste	Under award
77/5	Dhubri	1.931	Homestead	Under award
13/5	Dhubri	0.119	Homestead	Awarded
223/6	Golakganj	1.374	Homestead	Under award
116/2	Agomani	0.757	Homestead	Notified
301/8	Golakganj	0.955	Homestead	Under award
40/6	Dhubri	0.451	Waste	Awarded
106/4	Agomani	1.598	Agricultural	Notified
53/6	Golakganj	2.311	Waste	Notified

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.7.1 Rate Analysis — Sheet 6

Design Computation

Moment = $63.034 \times 9.348 = 589.239$ kNm (permissible 954.159) — safe

Settlement = $21.234 \times 8.836 = 187.622$ kNm (permissible 270.879) — revise section

Shear = $82.672 \times 2.779 = 229.776$ kNm (permissible 433.300) — safe

Moment = $9.143 \times 4.211 = 38.503$ kPa (permissible 61.674) — safe

Shear = $80.071 \times 6.272 = 502.205$ mm (permissible 680.560) — safe

Shear = $29.520 \times 4.866 = 143.657$ kN (permissible 220.536) — revise section

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.7.1 Rate Analysis — Sheet 7

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. HY-391, revision A.

3.7.1 Rate Analysis — Sheet 8

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
255/4	Gauripur	0.828	Waste	Notified
187/1	Agomani	2.233	Agricultural	Notified
105/9	Dhubri	1.116	Homestead	Awarded
392/9	Gauripur	2.171	Agricultural	Notified
113/7	Dhubri	2.040	Agricultural	Notified
191/8	Agomani	0.193	Homestead	Under award
233/4	Dhubri	2.207	Waste	Awarded
466/6	Agomani	2.138	Homestead	Notified
105/9	Dhubri	2.242	Waste	Under award
353/4	Gauripur	0.116	Waste	Notified
307/4	Agomani	0.716	Agricultural	Under award
162/7	Dhubri	0.229	Agricultural	Under award
247/1	Agomani	0.457	Agricultural	Notified
374/4	Golakganj	2.036	Waste	Awarded
435/7	Dhubri	2.307	Homestead	Awarded
156/7	Golakganj	1.808	Waste	Notified
325/2	Golakganj	1.273	Homestead	Under award
323/6	Dhubri	2.396	Waste	Awarded
341/9	Golakganj	0.782	Agricultural	Awarded
450/1	Gauripur	1.142	Agricultural	Under award
168/5	Gauripur	0.077	Waste	Notified
222/5	Agomani	2.266	Homestead	Under award
296/3	Dhubri	0.363	Agricultural	Awarded

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.7.1 Rate Analysis — Sheet 9

Design Computation

Bearing pressure = $2.724 \times 10.147 = 27.637$ mm (permissible 37.311) — safe

Bearing pressure = $81.662 \times 1.489 = 121.623$ kPa (permissible 225.124) — revise section

Bearing pressure = $51.946 \times 6.295 = 327.020$ mm (permissible 538.680) — safe

Settlement = $57.978 \times 8.788 = 509.509$ kPa (permissible 549.761) — safe

Deflection = $28.856 \times 5.705 = 164.626$ kPa (permissible 284.480) — safe

Deflection = $67.548 \times 1.605 = 108.388$ kN (permissible 151.840) — safe

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.7.1 Rate Analysis — Sheet 10

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. GA-917, revision C.

3.7.1 Rate Analysis — Sheet 11

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
15:00	29	89	27	10	49	610
06:00	319	254	30	21	77	140
07:00	169	191	0	66	15	1294
18:00	22	61	28	78	13	859
16:00	160	186	11	9	67	1154
21:00	177	46	26	23	25	704
15:00	165	177	30	64	82	723
21:00	279	207	29	7	75	1168
20:00	383	132	5	10	52	770
02:00	264	51	37	68	78	386
20:00	270	188	26	68	63	318
18:00	280	13	18	56	89	1274
02:00	311	188	16	71	21	644
00:00	188	140	28	66	98	842
10:00	109	51	20	89	90	732
10:00	264	30	20	89	36	272
18:00	150	86	3	77	50	1050
02:00	176	115	14	57	37	850
14:00	281	20	13	21	106	323
18:00	55	30	5	84	110	668
00:00	266	237	18	6	57	429
05:00	64	78	6	15	97	405

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.7.1 Rate Analysis — Sheet 12

Design Computation

Shear = $24.143 \times 2.722 = 65.712$ kPa (permissible 100.462) — safe

Bearing pressure = $86.991 \times 8.111 = 705.542$ kN (permissible 982.717) — safe

Shear = $24.742 \times 2.157 = 53.365$ kNm (permissible 83.512) — safe

Moment = $41.522 \times 1.821 = 75.594$ kPa (permissible 104.718) — safe

Shear = $30.569 \times 2.961 = 90.506$ mm (permissible 137.070) — safe

Settlement = $10.595 \times 0.406 = 4.306$ kPa (permissible 7.985) — safe

Moment = $61.343 \times 6.711 = 411.671$ kPa (permissible 522.045) — safe

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

3.8 REVENUE STREAMS

The bridge is proposed as a toll-free public asset. No user charge is envisaged. Accordingly the appraisal in Chapter 3.9 is an economic rather than a financial one, and the benefit stream comprises vehicle operating cost savings, travel time savings and accident cost savings rather than revenue.

3.8.1 Benefit Stream Basis — Sheet 1

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. ST-573, revision B.

3.8.1 Benefit Stream Basis — Sheet 2

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
05:00	37	143	31	41	9	591
23:00	204	21	26	17	77	807
12:00	235	82	16	58	22	700
21:00	205	44	29	30	56	1336
12:00	42	159	0	30	17	479
09:00	294	103	15	40	85	1204
14:00	243	130	4	83	74	744
16:00	100	225	14	52	30	155
01:00	274	87	38	84	9	1172
02:00	96	108	36	47	116	1311
04:00	77	256	6	56	21	471
11:00	241	258	26	57	31	642
04:00	336	143	11	15	34	1013
09:00	104	179	11	22	61	791
10:00	242	67	0	69	91	169
10:00	183	72	5	33	57	909
18:00	101	109	31	86	114	574
23:00	288	243	17	66	61	1313
15:00	52	52	31	86	59	901
04:00	235	148	37	44	68	874
15:00	379	52	35	56	12	930
01:00	21	122	34	8	6	796
02:00	97	37	38	58	106	1073
04:00	254	230	27	64	118	995
10:00	391	76	29	54	42	1144
16:00	98	199	15	41	79	147

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.8.1 Benefit Stream Basis — Sheet 3

Design Computation

Deflection = $3.216 \times 1.750 = 5.628$ kPa (permissible 9.982) — safe

Settlement = $26.749 \times 1.433 = 38.329$ kNm (permissible 72.217) — safe

Settlement = $60.630 \times 2.039 = 123.611$ kPa (permissible 226.411) — safe

Shear = $32.138 \times 1.325 = 42.585$ kPa (permissible 52.232) — safe

Bearing pressure = $89.973 \times 10.695 = 962.298$ kPa (permissible 1376.547) — safe

Moment = $19.557 \times 7.331 = 143.380$ kNm (permissible 176.093) — safe

Shear = $78.869 \times 10.508 = 828.714$ kNm (permissible 1414.145) — safe

Deflection = $74.326 \times 4.437 = 329.773$ kPa (permissible 465.160) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.8.1 Benefit Stream Basis — Sheet 4

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. LA-552, revision C.

3.9 COST BENEFIT ANALYSIS AND INVESTMENT CRITERIA

The economic appraisal has been carried out over a 30-year evaluation period including the construction period, at a social discount rate of 12 per cent. Benefits comprise vehicle operating cost savings from the removal of the 62 km detour, travel time savings valued at the prevailing wage rate, and ferry operating cost avoided.

Investment criterion	Value
Economic Internal Rate of Return	14.2 per cent
Net Present Value at 12 per cent	Rs. 96.40 crore
Benefit Cost Ratio	1.64
Payback period	11.4 years

The Internal Rate of Return of the project works out to 14.2 per cent, against the threshold of 12 per cent applied for projects of this class. The project is therefore economically viable and is recommended for sanction.

The detailed year-wise cash flow statement on which this computation is based is enclosed at Annexure IV-A.

3.9.1 Sensitivity Analysis — Sheet 1

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

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The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. ST-923, revision C.

3.9.1 Sensitivity Analysis — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Bitumen	Quarry B	10.3	5,150
Fine sand	Quarry A	49.4	866
Bitumen	Depot	86.9	2,760
Bitumen	Quarry A	75.2	1,214
Coarse aggregate	Quarry A	18.7	4,143
Bitumen	Quarry A	70.5	5,525
Steel	Plant	76.1	8,685
Fine sand	River bed	46.0	1,572
Steel	River bed	37.8	9,211
Murram	Quarry A	64.3	5,345
Murram	Quarry B	47.1	667
Coarse aggregate	Plant	69.4	6,740
Coarse aggregate	River bed	36.0	456
Murram	Quarry A	20.8	1,037
Steel	Depot	88.3	6,691
Steel	Quarry A	51.8	3,432
Murram	River bed	31.6	5,278
Cement	Quarry A	86.8	2,113
Fine sand	Quarry B	64.2	359
Coarse aggregate	Depot	72.1	3,091
Boulder	Plant	18.6	8,517
Fine sand	Quarry B	38.9	6,467
Coarse aggregate	Depot	39.0	3,667
Coarse aggregate	River bed	78.1	5,163
Coarse aggregate	Quarry B	18.5	1,823
Coarse aggregate	Quarry B	20.8	4,411

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.9.1 Sensitivity Analysis — Sheet 3

Design Computation

Moment = $74.896 \times 4.880 = 365.459$ kNm (permissible 448.573) — safe

Settlement = $2.380 \times 4.677 = 11.132$ kPa (permissible 12.892) — safe

Bearing pressure = $75.080 \times 1.975 = 148.250$ kN (permissible 234.703) — safe

Shear = $21.326 \times 7.481 = 159.544$ kN (permissible 267.797) — safe

Bearing pressure = $18.496 \times 11.422 = 211.259$ kNm (permissible 297.503) — safe

Shear = $4.444 \times 11.567 = 51.404$ kPa (permissible 65.203) — safe

Shear = $54.086 \times 3.804 = 205.733$ kNm (permissible 238.124) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.9.1 Sensitivity Analysis — Sheet 4

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

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The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. GA-375, revision B.

3.9.1 Sensitivity Analysis — Sheet 5

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Coarse aggregate	Quarry A	17.5	6,222
Bitumen	Plant	87.6	1,690
Cement	Quarry B	80.4	5,602
Boulder	Plant	53.4	2,479
Bitumen	Quarry A	83.3	7,959
Cement	River bed	86.7	528
Bitumen	Quarry A	44.4	5,276
Cement	Quarry B	89.9	1,413
Steel	Plant	34.4	908
Boulder	Quarry B	71.8	5,777
Cement	Quarry A	62.3	1,452
Murram	Depot	85.1	675
Murram	Depot	52.4	6,444
Steel	Plant	3.1	1,363
Coarse aggregate	Quarry A	58.3	4,394
Steel	River bed	36.2	7,679
Steel	Depot	75.8	1,552
Bitumen	Plant	4.6	1,617
Steel	Quarry A	22.3	2,029

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.9.1 Sensitivity Analysis — Sheet 6

Design Computation

Deflection = $23.769 \times 0.531 = 12.625$ kN (permissible 20.534) — safe

Bearing pressure = $16.095 \times 10.906 = 175.538$ kPa (permissible 317.368) — safe

Bearing pressure = $60.265 \times 2.703 = 162.876$ kN (permissible 292.889) — safe

Shear = $52.897 \times 4.071 = 215.355$ kN (permissible 317.002) — safe

Deflection = $76.536 \times 8.650 = 662.047$ kN (permissible 758.643) — safe

Bearing pressure = $83.390 \times 10.038 = 837.103$ kPa (permissible 1357.216) — safe

Deflection = $75.929 \times 7.647 = 580.662$ kNm (permissible 1018.724) — safe

Deflection = $55.072 \times 8.268 = 455.325$ mm (permissible 597.694) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.9.1 Sensitivity Analysis — Sheet 7

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. GA-887, revision B.

3.9.1 Sensitivity Analysis — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
24.0	Gravelly sand	11	SPT
3.7	Clayey silt	45	DS
24.3	Silty clay	47	UDS
6.4	Weathered rock	20	UDS
21.6	Weathered rock	38	DS
1.6	Medium sand	21	SPT
16.4	Clayey silt	36	DS
29.9	Medium sand	29	DS
27.2	Silty clay	30	DS
11.9	Weathered rock	33	UDS
19.0	Silty clay	28	DS
17.3	Clayey silt	54	SPT
14.2	Gravelly sand	17	UDS
21.8	Gravelly sand	49	UDS
6.8	Weathered rock	9	SPT
16.3	Coarse sand	20	DS
25.7	Coarse sand	36	UDS
1.6	Coarse sand	39	SPT

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.10 ENVIRONMENTAL AND SUSTAINABILITY ASPECTS

The project falls within the notified river bed area of the Brahmaputra. An Environmental Impact Assessment has been carried out by an accredited consultant. Principal impacts are turbidity during well sinking, loss of 212 riparian trees, and temporary disruption to fishing activity.

Environmental clearance from the State Environment Impact Assessment Authority is yet to be obtained and the application is under process. Forest clearance is not required as no notified forest land is involved.

Compensatory afforestation of 424 saplings in a ratio of 1:2 is provided for in the estimate. Sustainability measures include use of ground granulated blast furnace slag in mass concrete and solar lighting on the approaches.

3.10.1 Environmental Management Plan — Sheet 1

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. LA-111, revision B.

3.10.1 Environmental Management Plan — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Boulder	Depot	62.8	3,909
Bitumen	Quarry A	60.2	2,560
Steel	River bed	16.3	3,624
Coarse aggregate	Plant	74.1	1,402
Bitumen	River bed	62.0	1,611
Coarse aggregate	Quarry A	74.1	8,518
Fine sand	Quarry A	89.1	7,123
Steel	Quarry A	60.2	2,392
Bitumen	River bed	79.6	9,324
Fine sand	Quarry A	50.9	8,490
Murram	Quarry A	37.2	7,042
Boulder	Plant	43.3	6,480
Bitumen	Quarry A	58.4	1,534
Fine sand	River bed	33.4	6,129
Murram	Depot	61.3	2,045
Cement	Quarry B	31.5	6,262
Bitumen	Plant	73.3	8,372
Fine sand	Quarry A	35.5	7,227
Coarse aggregate	Plant	4.3	7,179
Coarse aggregate	Depot	20.0	1,861

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.10.1 Environmental Management Plan — Sheet 3

Design Computation

Bearing pressure = $7.940 \times 3.576 = 28.394$ mm (permissible 41.840) — safe

Deflection = $65.603 \times 6.363 = 417.406$ kN (permissible 558.786) — safe

Shear = $63.028 \times 10.324 = 650.721$ kN (permissible 940.304) — safe

Shear = $17.189 \times 9.291 = 159.700$ kNm (permissible 296.329) — safe

Moment = $72.322 \times 2.483 = 179.552$ kNm (permissible 226.311) — safe

Shear = $33.125 \times 6.740 = 223.265$ kNm (permissible 410.446) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.10.1 Environmental Management Plan — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. HY-368, revision C.

3.10.1 Environmental Management Plan — Sheet 5

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Steel	River bed	56.3	933
Cement	River bed	85.1	8,620
Boulder	Depot	64.0	3,687
Boulder	River bed	47.8	647
Boulder	Quarry A	63.1	2,338
Bitumen	River bed	39.0	4,132
Fine sand	Depot	43.5	1,143
Bitumen	Plant	57.0	4,405
Murram	Quarry A	38.8	1,701
Steel	Quarry B	53.1	5,873
Cement	Depot	51.6	460
Boulder	Quarry B	81.5	801
Coarse aggregate	Depot	49.1	6,397
Murram	Quarry A	64.5	3,275
Cement	River bed	84.2	9,053
Steel	Plant	55.8	3,652
Bitumen	Plant	63.7	6,637
Bitumen	Quarry B	8.3	6,008

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.10.1 Environmental Management Plan — Sheet 6

Design Computation

Moment = $61.990 \times 2.045 = 126.768$ mm (permissible 166.574) — safe

Bearing pressure = $61.379 \times 7.791 = 478.189$ mm (permissible 573.553) — safe

Moment = $56.446 \times 6.030 = 340.388$ mm (permissible 489.618) — safe

Deflection = $9.622 \times 3.002 = 28.888$ mm (permissible 37.200) — safe

Shear = $39.488 \times 2.956 = 116.744$ mm (permissible 138.691) — safe

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.10.1 Environmental Management Plan — Sheet 7

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. LA-650, revision C.

3.10.1 Environmental Management Plan — Sheet 8

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Boulder	Depot	60.4	1,921
Bitumen	Plant	61.9	5,918
Murram	Quarry A	30.4	8,370
Cement	River bed	17.0	7,996
Fine sand	Plant	13.8	8,283
Coarse aggregate	Depot	72.7	4,859
Steel	Plant	26.9	5,438
Fine sand	Quarry B	24.8	713
Murram	Plant	72.1	6,380
Steel	Depot	56.4	4,492
Cement	Depot	69.6	8,248
Steel	Depot	73.7	3,173
Fine sand	Quarry B	75.5	1,976
Cement	Plant	80.1	6,058
Boulder	River bed	69.8	5,276
Murram	Quarry B	67.9	582
Bitumen	River bed	5.7	2,982
Fine sand	Quarry B	19.3	5,039
Fine sand	Quarry B	70.1	9,128
Murram	Quarry A	55.4	1,856
Cement	Depot	49.2	7,527
Coarse aggregate	Plant	2.5	2,254
Murram	Depot	46.5	4,987
Steel	River bed	57.8	6,544

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.10.1 Environmental Management Plan — Sheet 9

Design Computation

Shear = $64.636 \times 3.631 = 234.674$ kPa (permissible 397.431) — safe

Bearing pressure = $10.170 \times 10.182 = 103.551$ kPa (permissible 193.459) — safe

Deflection = $7.390 \times 11.036 = 81.554$ kNm (permissible 101.693) — safe

Shear = $72.721 \times 10.108 = 735.063$ kNm (permissible 842.527) — safe

Moment = $55.807 \times 11.602 = 647.455$ kN (permissible 1002.915) — safe

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.10.1 Environmental Management Plan — Sheet 10

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. ST-500, revision A.

3.10.1 Environmental Management Plan — Sheet 11

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
40/6	Gauripur	0.751	Homestead	Notified
79/5	Dhubri	0.514	Agricultural	Under award
233/6	Golakganj	1.830	Homestead	Notified
205/8	Agomani	0.508	Waste	Notified
284/6	Dhubri	0.592	Homestead	Under award
323/3	Golakganj	1.801	Homestead	Under award
224/2	Dhubri	1.782	Waste	Under award
98/6	Gauripur	1.450	Waste	Under award
203/6	Agomani	0.930	Homestead	Notified
316/8	Dhubri	0.457	Homestead	Awarded
175/6	Agomani	2.067	Agricultural	Under award
182/6	Golakganj	0.151	Homestead	Notified
285/2	Dhubri	2.084	Agricultural	Under award
56/5	Gauripur	1.914	Agricultural	Under award
44/4	Gauripur	0.511	Waste	Awarded
336/4	Gauripur	1.175	Homestead	Under award
77/8	Golakganj	1.087	Homestead	Under award
149/4	Gauripur	1.437	Waste	Under award
266/6	Agomani	0.988	Agricultural	Awarded
48/5	Golakganj	1.629	Homestead	Under award
87/9	Golakganj	1.406	Agricultural	Under award

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.10.1 Environmental Management Plan — Sheet 12

Design Computation

Bearing pressure = $74.957 \times 2.049 = 153.619$ kNm (permissible 230.656) — safe

Settlement = $19.760 \times 9.440 = 186.533$ kPa (permissible 277.064) — safe

Shear = $9.290 \times 2.634 = 24.468$ kPa (permissible 43.585) — safe

Deflection = $83.930 \times 6.529 = 548.000$ kN (permissible 634.023) — safe

Shear = $76.116 \times 1.656 = 126.078$ mm (permissible 136.056) — safe

Settlement = $40.478 \times 6.078 = 246.028$ kN (permissible 459.981) — safe

Deflection = $76.350 \times 1.071 = 81.765$ kNm (permissible 154.733) — safe

Settlement = $75.899 \times 1.386 = 105.180$ kPa (permissible 160.655) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.11 RISK ASSESSMENT AND MITIGATION MEASURES

Sl.	Risk	Likelihood	Impact	Mitigation
1	Monsoon disruption to well sinking	High	High	Programme well sinking in the lean period
2	Delay in balance land acquisition	Medium	High	Award proceedings advanced; possession of 62% already taken
3	Escalation in steel and cement	Medium	Medium	Price adjustment clause in the contract
4	Change in river course	Low	High	River training works and annual bathymetric survey
5	Contractor default	Low	High	Pre-qualification and performance security

Fourteen risks have been identified in total. Each has been assessed for likelihood and impact, and a mitigation plan documented covering avoidance, transfer and elimination as appropriate.

3.11.1 Risk Register — Sheet 1

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. ST-990, revision A.

3.11.1 Risk Register — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
2.6	Medium sand	51	SPT
1.4	Medium sand	57	DS
18.4	Silty clay	21	UDS
3.0	Coarse sand	56	SPT
22.5	Medium sand	54	SPT
11.0	Medium sand	35	DS
17.1	Weathered rock	46	SPT
26.7	Gravelly sand	24	DS
4.0	Clayey silt	55	DS
29.6	Gravelly sand	32	DS
18.5	Medium sand	32	DS
22.0	Clayey silt	15	SPT
6.5	Clayey silt	26	SPT
11.4	Coarse sand	42	SPT
12.2	Medium sand	40	SPT
16.7	Clayey silt	19	UDS
21.2	Medium sand	8	SPT
19.7	Weathered rock	28	UDS

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.11.1 Risk Register — Sheet 3

Design Computation

Bearing pressure = $21.017 \times 6.177 = 129.822$ mm (permissible 153.941) — safe

Deflection = $78.273 \times 10.680 = 835.948$ kN (permissible 1378.966) — safe

Settlement = $47.311 \times 8.020 = 379.417$ kN (permissible 631.467) — safe

Settlement = $55.934 \times 7.138 = 399.268$ kNm (permissible 541.187) — safe

Shear = $31.379 \times 3.826 = 120.048$ kNm (permissible 217.950) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.11.1 Risk Register — Sheet 4

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. GA-229, revision B.

3.11.1 Risk Register — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 408	36.46	3.52	2.16	16	209.285
Item 136	2.22	2.21	1.59	18	207.158
Item 585	10.48	1.48	2.44	13	201.049
Item 260	30.06	1.77	2.12	7	104.160
Item 702	10.96	4.71	1.86	14	82.335
Item 459	18.17	2.11	1.49	3	303.500
Item 714	37.01	4.02	1.06	15	192.813
Item 667	16.81	0.65	1.68	5	60.199
Item 153	6.25	4.99	0.81	10	241.691
Item 397	3.27	1.03	1.21	15	234.510
Item 697	39.61	0.55	1.70	5	387.071
Item 150	14.26	5.39	1.73	5	361.033
Item 354	36.67	4.19	0.45	12	29.077
Item 370	32.47	1.52	2.21	22	331.329
Item 344	21.76	5.55	1.43	19	157.841
Item 155	33.91	5.42	0.52	5	377.089
Item 878	32.87	3.34	1.04	18	119.843
Item 188	16.16	5.61	0.67	1	43.755
Item 833	12.31	0.82	1.13	21	170.315
Item 737	39.21	4.14	2.16	2	44.656
Item 415	17.42	2.51	1.29	5	314.146
Item 560	13.44	3.30	1.69	9	133.449

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.11.1 Risk Register — Sheet 6

Design Computation

Bearing pressure = $38.037 \times 11.922 = 453.462$ kPa (permissible 576.910) — safe

Shear = $81.282 \times 6.626 = 538.543$ kPa (permissible 892.357) — safe

Bearing pressure = $3.586 \times 10.906 = 39.110$ kN (permissible 45.231) — revise section

Bearing pressure = $12.830 \times 1.336 = 17.138$ kNm (permissible 21.116) — safe

Settlement = $58.764 \times 5.370 = 315.560$ kNm (permissible 539.720) — safe

Moment = $27.536 \times 7.585 = 208.871$ kPa (permissible 383.691) — safe

Shear = $44.297 \times 7.453 = 330.159$ mm (permissible 495.503) — revise section

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.12 PROJECT MANAGEMENT ORGANISATION

The project will be implemented by Assam State Bridge Corporation Limited through a dedicated Project Implementation Unit headed by a Chief Engineer. The unit comprises one Superintending Engineer, three Executive Engineers and supporting technical staff. An Authority Engineer will be appointed for construction supervision. Monthly progress review will be conducted by the Principal Secretary, PWD.

3.12.1 Organisation Structure — Sheet 1

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. HY-851, revision A.

3.12.1 Organisation Structure — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Steel	Depot	36.7	2,119
Cement	Quarry B	78.6	5,351
Cement	River bed	33.7	8,638
Fine sand	Quarry B	50.5	4,691
Fine sand	Quarry A	59.3	2,195
Bitumen	River bed	4.4	4,670
Fine sand	Depot	37.5	7,366
Bitumen	Quarry A	78.8	2,007
Bitumen	Plant	58.1	6,147
Fine sand	Quarry A	67.7	8,152
Bitumen	Quarry B	25.8	3,754
Steel	River bed	46.0	1,858
Coarse aggregate	Quarry A	26.5	2,117
Coarse aggregate	Quarry B	67.2	2,045
Boulder	Plant	53.6	7,146
Steel	Quarry B	49.1	7,637
Cement	Depot	42.5	7,023
Bitumen	Quarry A	24.6	6,636
Fine sand	Depot	78.2	8,192
Coarse aggregate	Plant	14.6	4,551

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.12.1 Organisation Structure — Sheet 3

Design Computation

Bearing pressure = $74.456 \times 5.106 = 380.171$ kNm (permissible 511.469) — safe

Deflection = $84.583 \times 3.457 = 292.408$ kN (permissible 317.665) — safe

Moment = $71.996 \times 1.506 = 108.439$ kNm (permissible 131.594) — revise section

Settlement = $1.032 \times 6.277 = 6.481$ mm (permissible 12.258) — safe

Moment = $86.124 \times 9.389 = 808.654$ kN (permissible 1021.122) — safe

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

3.12.1 Organisation Structure — Sheet 4

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. HY-132, revision C.

3.12.1 Organisation Structure — Sheet 5

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
270/4	Agomani	1.202	Agricultural	Under award
289/9	Agomani	1.197	Waste	Awarded
136/7	Golakganj	1.868	Agricultural	Under award
423/2	Gauripur	0.522	Waste	Awarded
118/5	Golakganj	1.101	Waste	Under award
36/2	Agomani	1.704	Waste	Under award
224/3	Golakganj	0.572	Agricultural	Notified
98/3	Dhubri	1.620	Agricultural	Notified
450/6	Dhubri	1.408	Homestead	Under award
250/6	Dhubri	2.080	Agricultural	Notified
338/9	Dhubri	2.213	Agricultural	Under award
439/2	Dhubri	1.998	Agricultural	Awarded
10/5	Dhubri	0.368	Homestead	Notified
86/1	Golakganj	1.151	Homestead	Notified
280/1	Agomani	0.241	Homestead	Awarded
154/9	Dhubri	1.339	Agricultural	Notified
13/6	Dhubri	2.060	Homestead	Under award
238/5	Agomani	1.753	Homestead	Awarded

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.13 CONTRACT MANAGEMENT STRATEGY

The work is proposed to be executed on an Engineering, Procurement and Construction basis in a single package. Bidding will follow the standard bidding document of the Public Works Department. A price adjustment clause for steel, cement and bitumen is proposed in view of the 30-month construction period. Arbitration will be governed by the Arbitration and Conciliation Act, 1996.

3.13.1 Contract Packaging — Sheet 1

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. ST-117, revision A.

3.13.1 Contract Packaging — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
7.6	Coarse sand	33	DS
11.0	Weathered rock	20	SPT
7.2	Medium sand	59	SPT
24.0	Medium sand	23	DS
15.6	Medium sand	40	SPT
25.3	Weathered rock	9	SPT
17.2	Coarse sand	32	DS
9.8	Gravelly sand	45	SPT
12.5	Silty clay	35	UDS
6.3	Clayey silt	15	DS
28.2	Gravelly sand	58	DS
1.0	Clayey silt	35	UDS
26.7	Silty clay	11	SPT
1.9	Weathered rock	50	DS
9.2	Clayey silt	18	UDS
16.5	Clayey silt	26	UDS
7.4	Coarse sand	14	DS
9.9	Weathered rock	25	SPT
23.6	Coarse sand	34	UDS
3.9	Weathered rock	28	SPT
1.0	Coarse sand	51	UDS

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.13.1 Contract Packaging — Sheet 3

Design Computation

Deflection = $4.683 \times 10.770 = 50.431$ kN (permissible 88.223) — safe

Settlement = $78.652 \times 6.182 = 486.221$ kNm (permissible 706.622) — safe

Settlement = $86.466 \times 3.089 = 267.057$ kPa (permissible 437.590) — safe

Deflection = $8.559 \times 0.504 = 4.310$ kPa (permissible 6.114) — revise section

Bearing pressure = $84.222 \times 3.907 = 329.059$ mm (permissible 498.918) — revise section

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.13.1 Contract Packaging — Sheet 4

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. GA-247, revision B.

3.13.1 Contract Packaging — Sheet 5

Bore Log Summary

Depth (m)	Strata	N value	Sample
29.6	Coarse sand	24	UDS
20.7	Coarse sand	35	SPT
0.6	Coarse sand	48	UDS
10.5	Silty clay	58	UDS
19.4	Clayey silt	50	UDS
29.6	Gravelly sand	40	SPT
10.4	Medium sand	25	UDS
9.8	Weathered rock	51	DS
16.7	Weathered rock	40	UDS
12.1	Clayey silt	4	UDS
23.6	Medium sand	49	DS
14.1	Medium sand	45	UDS
17.6	Clayey silt	43	SPT
11.4	Medium sand	39	DS
15.9	Silty clay	37	DS
14.3	Medium sand	57	DS
20.7	Coarse sand	33	DS
29.6	Gravelly sand	30	UDS
8.5	Medium sand	52	SPT
11.0	Medium sand	7	SPT
27.5	Gravelly sand	41	UDS
3.4	Gravelly sand	20	UDS
22.5	Silty clay	54	DS

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

3.14 IMPLEMENTATION SCHEDULE AND WORK BREAKDOWN STRUCTURE

The zero date is taken as the date of issue of the work order. The construction period is 30 months.

WBS	Milestone	Month from zero date
1.0	Award of work and mobilisation	M0 – M2
2.0	Completion of land acquisition	M4
3.0	Well foundations complete	M14
4.0	Substructure complete	M20
5.0	Superstructure complete	M27
6.0	Approach roads complete	M28
7.0	Testing and commissioning	M30

The critical path runs through well sinking, which is dependent on the lean-season window. A bar chart is placed at the end of this chapter.

3.14.1 Bar Chart and Critical Path — Sheet 1

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. LA-332, revision B.

3.14.1 Bar Chart and Critical Path — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 773	29.02	1.66	0.82	10	121.515
Item 443	27.34	5.72	1.89	1	315.908
Item 717	11.63	5.39	2.46	22	252.751
Item 575	19.36	3.87	1.03	8	313.278
Item 470	11.81	4.79	1.17	14	207.181
Item 276	38.04	1.97	0.32	3	333.196
Item 106	21.89	1.64	0.63	10	127.247
Item 564	4.50	4.29	2.24	1	178.608
Item 603	2.53	5.09	2.16	4	182.252
Item 727	4.97	4.03	2.91	23	93.672
Item 515	36.08	0.49	0.32	21	160.604
Item 427	20.14	4.11	2.48	22	262.506
Item 172	10.66	0.88	0.46	4	98.010
Item 503	25.77	5.54	0.52	22	20.240
Item 883	16.92	5.43	2.14	8	72.081
Item 111	11.08	1.69	0.53	21	59.748
Item 497	17.47	2.61	2.67	15	184.841
Item 139	4.25	4.33	1.40	9	370.049
Item 614	16.60	4.93	0.52	21	44.189
Item 218	19.32	1.10	1.32	5	353.294
Item 513	18.45	1.58	0.72	22	198.934
Item 278	34.70	1.49	1.39	1	195.799

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.14.1 Bar Chart and Critical Path — Sheet 3

Design Computation

Deflection = $76.428 \times 9.117 = 696.804$ kPa (permissible 874.723) — safe

Shear = $19.614 \times 5.527 = 108.411$ kPa (permissible 164.826) — safe

Deflection = $89.217 \times 1.645 = 146.764$ kNm (permissible 211.782) — safe

Deflection = $65.781 \times 1.009 = 66.342$ kN (permissible 75.289) — safe

Shear = $26.199 \times 1.151 = 30.167$ kPa (permissible 37.056) — revise section

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.14.1 Bar Chart and Critical Path — Sheet 4

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. GA-375, revision C.

3.14.1 Bar Chart and Critical Path — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 740	4.19	4.10	1.70	8	83.533
Item 743	34.46	4.08	1.99	24	377.537
Item 374	21.52	3.81	2.44	17	388.199
Item 582	24.06	2.29	2.20	24	287.743
Item 276	6.13	0.43	2.81	24	115.418
Item 769	32.93	5.73	0.63	6	371.950
Item 805	30.80	0.97	2.85	8	138.479
Item 680	18.58	1.34	0.45	3	353.490
Item 687	7.72	4.17	2.38	7	167.579
Item 870	15.46	4.19	1.33	9	49.997
Item 492	24.62	2.32	1.44	14	7.764
Item 297	33.40	1.51	1.27	2	237.824
Item 522	31.38	4.11	1.16	17	397.633
Item 457	2.81	1.14	0.86	11	67.623
Item 157	16.29	5.25	1.99	24	179.532
Item 137	7.44	5.81	0.67	7	266.716
Item 401	25.34	5.84	1.35	1	21.500
Item 630	33.00	0.31	0.55	20	246.087
Item 700	22.30	2.85	1.87	10	225.597
Item 518	22.32	5.46	2.37	5	247.141
Item 404	11.98	4.02	2.15	24	127.035
Item 743	11.91	2.05	2.02	19	314.659

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.14.1 Bar Chart and Critical Path — Sheet 6

Design Computation

Bearing pressure = $11.229 \times 10.708 = 120.245$ mm (permissible 223.642) — safe

Moment = $75.331 \times 2.933 = 220.926$ kPa (permissible 363.408) — safe

Shear = $78.687 \times 9.666 = 760.596$ kN (permissible 1398.423) — safe

Deflection = $69.467 \times 9.020 = 626.576$ kNm (permissible 826.919) — safe

Shear = $38.202 \times 3.773 = 144.125$ kN (permissible 231.586) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.15 STATUTORY CLEARANCES

Sl.	Clearance	Authority	Status
1	Environmental clearance	State EIAA	Under process
2	Forest clearance	State Forest Dept.	Not applicable
3	Waterway clearance	Inland Waterways Authority	Obtained, 22 January 2026
4	Railway NOC	NF Railway	Not applicable
5	Pollution Control Board consent	Assam PCB	Consent to establish obtained
6	Approval of river training works	Water Resources Dept.	Obtained, 3 February 2026

3.15.1 Clearance Correspondence — Sheet 1

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. ST-884, revision C.

3.15.1 Clearance Correspondence — Sheet 2

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
152/2	Dhubri	0.635	Waste	Under award
139/9	Agomani	1.555	Waste	Notified
215/3	Dhubri	0.379	Agricultural	Awarded
461/9	Gauripur	0.828	Agricultural	Notified
337/2	Agomani	1.013	Waste	Under award
337/3	Golakganj	1.961	Homestead	Awarded
231/6	Dhubri	2.158	Waste	Under award
35/8	Agomani	1.413	Agricultural	Under award
132/2	Gauripur	2.300	Waste	Under award
39/5	Dhubri	1.898	Agricultural	Notified
319/9	Gauripur	0.638	Homestead	Under award
173/3	Agomani	0.271	Waste	Notified
410/5	Agomani	0.073	Homestead	Awarded
249/5	Golakganj	1.538	Waste	Awarded
168/5	Dhubri	1.808	Waste	Under award
20/9	Gauripur	0.997	Waste	Notified
428/7	Gauripur	0.510	Homestead	Awarded
356/2	Gauripur	0.168	Waste	Under award
422/6	Agomani	0.166	Homestead	Awarded
363/5	Agomani	1.595	Homestead	Under award
48/9	Dhubri	1.326	Agricultural	Awarded
282/5	Gauripur	1.933	Agricultural	Notified
464/8	Golakganj	1.987	Agricultural	Notified
218/9	Golakganj	0.232	Homestead	Awarded
127/8	Agomani	1.976	Homestead	Notified
24/9	Golakganj	0.176	Homestead	Notified

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

3.15.1 Clearance Correspondence — Sheet 3

Design Computation

Shear = $55.555 \times 4.668 = 259.345$ mm (permissible 367.944) — safe

Moment = $9.315 \times 1.624 = 15.127$ kN (permissible 27.997) — safe

Bearing pressure = $50.259 \times 2.474 = 124.316$ kPa (permissible 220.949) — safe

Deflection = $49.894 \times 5.653 = 282.029$ mm (permissible 340.250) — revise section

Deflection = $81.229 \times 4.313 = 350.356$ kN (permissible 383.300) — safe

Moment = $60.376 \times 0.577 = 34.864$ kPa (permissible 57.654) — safe

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.15.1 Clearance Correspondence — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. GA-970, revision B.

3.15.1 Clearance Correspondence — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 159	30.06	0.64	2.93	9	108.505
Item 455	37.72	3.61	3.00	14	162.806
Item 107	21.98	5.97	0.43	24	332.399
Item 134	25.80	5.23	1.11	4	134.365
Item 651	33.81	5.57	2.50	15	50.729
Item 355	4.80	2.70	2.12	16	146.864
Item 318	21.18	5.86	1.80	10	34.203
Item 548	22.07	2.51	1.67	5	265.636
Item 246	8.25	3.35	1.09	10	382.720
Item 516	4.50	1.65	2.73	5	311.431
Item 537	22.80	3.38	1.44	17	319.924
Item 395	37.67	3.13	1.61	12	3.798
Item 225	14.73	0.74	0.44	17	151.908
Item 274	4.68	1.99	1.01	15	129.218
Item 253	30.25	2.06	2.62	19	206.084
Item 246	18.58	5.71	2.31	21	250.600
Item 363	11.01	1.52	2.63	24	175.030
Item 376	37.91	2.05	2.92	21	82.148
Item 423	2.45	5.25	1.83	12	270.283
Item 603	17.35	3.78	2.13	15	357.317
Item 266	30.58	1.51	0.78	12	295.123
Item 824	37.83	1.61	1.97	13	29.467

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

3.15.1 Clearance Correspondence — Sheet 6

Design Computation

Settlement = $65.220 \times 9.472 = 617.759$ kPa (permissible 1035.980) — safe

Bearing pressure = $8.819 \times 3.134 = 27.638$ kPa (permissible 34.370) — safe

Shear = $44.257 \times 10.282 = 455.049$ mm (permissible 845.694) — safe

Bearing pressure = $77.344 \times 6.115 = 472.965$ kNm (permissible 660.042) — safe

Shear = $43.851 \times 11.530 = 505.622$ kPa (permissible 543.926) — safe

Moment = $18.692 \times 3.376 = 63.111$ kNm (permissible 81.882) — revise section

Deflection = $81.015 \times 10.242 = 829.718$ mm (permissible 1327.725) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.16 QUALITY MANAGEMENT PLAN

Quality assurance will be exercised by the Authority Engineer through a site laboratory established at the project site and equipped in accordance with the Ministry's guidelines. Third party quality control will be entrusted to an institution of national repute, proposed to be the National Institute of Technology, Silchar. A quality assurance plan covering the frequency of testing for each item of work is placed at the end of this chapter.

3.16.1 Testing Frequency Schedule — Sheet 1

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. ST-302, revision B.

3.16.1 Testing Frequency Schedule — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Steel	River bed	42.3	4,514
Fine sand	River bed	81.1	2,113
Cement	Plant	61.4	3,036
Murram	Quarry B	16.9	3,460
Boulder	River bed	52.6	3,487
Murram	Depot	21.0	5,077
Cement	Quarry A	32.2	6,821
Fine sand	Depot	49.3	8,810
Cement	River bed	54.8	5,147
Bitumen	Quarry A	26.8	8,384
Fine sand	Depot	62.2	2,805
Fine sand	Quarry B	59.0	7,339
Fine sand	Depot	47.2	1,838
Boulder	Quarry B	53.8	8,932
Murram	River bed	71.6	9,107
Steel	Quarry A	45.3	8,004
Fine sand	Depot	8.9	1,697
Bitumen	River bed	19.9	6,624

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.16.1 Testing Frequency Schedule — Sheet 3

Design Computation

Moment = $69.798 \times 10.231 = 714.074$ kPa (permissible 992.684) — safe

Bearing pressure = $35.296 \times 1.899 = 67.036$ mm (permissible 78.566) — safe

Shear = $76.300 \times 3.896 = 297.294$ kPa (permissible 563.433) — safe

Shear = $2.613 \times 7.027 = 18.364$ kN (permissible 28.888) — safe

Settlement = $80.291 \times 0.466 = 37.433$ kPa (permissible 56.758) — revise section

Bearing pressure = $55.026 \times 3.385 = 186.285$ kPa (permissible 292.259) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

3.16.1 Testing Frequency Schedule — Sheet 4

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. HY-847, revision C.

3.16.1 Testing Frequency Schedule — Sheet 5

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
01:00	67	248	22	69	94	445
16:00	97	248	13	33	9	414
18:00	377	213	14	16	118	694
03:00	320	105	4	45	42	261
19:00	156	249	22	21	50	915
16:00	229	199	33	90	1	639
13:00	43	154	16	51	35	263
06:00	165	51	23	17	88	976
02:00	74	203	20	87	33	1222
14:00	291	156	5	28	90	235
14:00	181	51	5	28	46	1183
20:00	271	145	32	48	109	905
21:00	177	126	1	65	82	469
21:00	124	14	5	50	84	827
22:00	53	157	0	35	27	584
17:00	91	192	20	2	120	344
07:00	387	191	28	35	2	1158
13:00	365	246	32	21	34	929
04:00	275	166	15	78	35	1052
01:00	289	165	22	64	7	1333
14:00	169	214	18	90	117	1346
00:00	117	179	40	66	72	522
00:00	105	258	19	30	24	923

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.16.1 Testing Frequency Schedule — Sheet 6

Design Computation

Shear = $3.603 \times 6.721 = 24.217$ kPa (permissible 27.150) — safe

Bearing pressure = $44.316 \times 2.819 = 124.927$ mm (permissible 198.984) — safe

Deflection = $12.823 \times 4.787 = 61.384$ mm (permissible 111.158) — safe

Moment = $4.149 \times 8.012 = 33.247$ kN (permissible 45.966) — safe

Shear = $86.218 \times 3.398 = 292.987$ kPa (permissible 324.755) — revise section

Moment = $33.046 \times 4.350 = 143.756$ kPa (permissible 193.748) — revise section

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

3.16.2 Quality Assurance Records — Sheet 1

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. HY-641, revision B.

3.16.2 Quality Assurance Records — Sheet 2

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
314/5	Dhubri	0.662	Homestead	Under award
437/2	Agomani	2.158	Agricultural	Notified
326/4	Dhubri	2.213	Agricultural	Notified
243/7	Dhubri	1.937	Homestead	Awarded
60/7	Golakganj	1.085	Homestead	Notified
478/9	Gauripur	1.911	Homestead	Awarded
406/9	Dhubri	1.933	Homestead	Awarded
80/3	Golakganj	0.296	Waste	Under award
13/8	Golakganj	1.446	Waste	Under award
432/9	Golakganj	0.917	Waste	Awarded
465/1	Gauripur	0.265	Homestead	Under award
241/2	Agomani	2.248	Homestead	Under award
240/7	Agomani	1.778	Homestead	Under award
164/5	Dhubri	0.232	Homestead	Notified
395/7	Dhubri	1.138	Homestead	Notified
241/1	Gauripur	1.076	Homestead	Awarded
53/3	Gauripur	1.061	Homestead	Awarded
113/8	Dhubri	1.058	Waste	Awarded

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

3.16.2 Quality Assurance Records — Sheet 3

Design Computation

Moment = $56.249 \times 3.761 = 211.573$ mm (permissible 394.520) — safe

Deflection = $46.314 \times 1.114 = 51.589$ kN (permissible 56.982) — safe

Deflection = $42.472 \times 6.234 = 264.761$ mm (permissible 404.504) — safe

Settlement = $66.674 \times 3.950 = 263.362$ mm (permissible 421.140) — safe

Shear = $74.515 \times 11.254 = 838.574$ kN (permissible 908.457) — safe

Moment = $11.302 \times 7.255 = 81.999$ mm (permissible 136.102) — safe

Shear = $62.069 \times 4.223 = 262.146$ kPa (permissible 466.881) — safe

Settlement = $65.097 \times 7.777 = 506.246$ kPa (permissible 709.346) — revise section

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

3.16.2 Quality Assurance Records — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. LA-707, revision C.

3.16.2 Quality Assurance Records — Sheet 5

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
09:00	154	80	5	28	50	145
05:00	104	130	35	13	90	1301
16:00	66	233	2	75	55	175
22:00	369	45	39	41	31	1189
23:00	299	223	2	35	24	383
07:00	168	41	37	56	16	227
22:00	250	89	35	51	74	540
11:00	398	211	6	86	30	609
04:00	197	127	35	28	71	996
09:00	50	159	28	15	24	1338
02:00	272	42	24	66	35	1081
11:00	251	22	33	54	3	1260
07:00	358	89	35	31	13	898
13:00	344	44	34	29	19	1017
05:00	386	165	30	11	35	902
14:00	322	26	36	86	0	575

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

3.16.2 Quality Assurance Records — Sheet 6

Design Computation

Settlement = $23.105 \times 2.300 = 53.134$ kPa (permissible 67.660) — safe

Shear = $72.593 \times 2.019 = 146.533$ kPa (permissible 254.509) — safe

Settlement = $70.809 \times 6.081 = 430.605$ mm (permissible 543.283) — safe

Moment = $7.281 \times 11.197 = 81.531$ kNm (permissible 90.630) — safe

Moment = $85.349 \times 6.596 = 562.963$ kNm (permissible 635.937) — safe

Deflection = $44.767 \times 4.721 = 211.329$ mm (permissible 391.660) — safe

Shear = $55.201 \times 0.906 = 49.991$ kNm (permissible 79.677) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

ANNEXURE I — KEY MAP OF THE PROJECT LOCATION

Key map showing the project location, the existing ferry crossing and the nearest fixed crossing 62 km upstream. Drawing no. KM-001 rev. B.

Annexure I — Location Plans — Sheet 1

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Reference: drawing no. ST-127, revision B.

Annexure I — Location Plans — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Coarse aggregate	Quarry B	30.1	7,542
Steel	Quarry B	19.6	5,677
Cement	Depot	10.2	8,664
Murram	Depot	37.6	3,560
Bitumen	Quarry B	5.1	1,728
Coarse aggregate	Quarry B	24.3	4,824
Cement	River bed	16.4	2,108
Coarse aggregate	River bed	64.1	3,779
Bitumen	Quarry B	6.9	5,107
Cement	Quarry B	63.1	5,682
Bitumen	Depot	81.8	4,665
Bitumen	Depot	15.8	6,828
Boulder	Quarry A	62.7	7,201
Cement	Quarry B	12.2	1,029
Bitumen	Quarry B	40.4	3,003
Murram	Quarry A	12.1	4,988
Coarse aggregate	River bed	61.2	6,767

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Annexure I — Location Plans — Sheet 3

Design Computation

Bearing pressure = $75.388 \times 2.304 = 173.657 \text{ mm}$ (permissible 184.211) — safe

Moment = $10.666 \times 4.260 = 45.438 \text{ mm}$ (permissible 73.341) — safe

Moment = $18.302 \times 6.352 = 116.258 \text{ kN}$ (permissible 124.421) — safe

Deflection = $47.930 \times 1.968 = 94.302 \text{ kN}$ (permissible 132.601) — safe

Shear = $9.779 \times 3.182 = 31.113 \text{ kNm}$ (permissible 40.640) — safe

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Annexure I — Location Plans — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. ST-293, revision C.

Annexure I — Location Plans — Sheet 5

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
03:00	171	62	23	79	38	374
21:00	391	35	0	35	7	873
11:00	396	258	18	50	107	1027
03:00	54	157	30	27	117	100
22:00	394	145	4	70	79	777
22:00	27	55	17	37	77	284
00:00	41	64	13	79	51	1311
02:00	318	226	23	36	56	182
20:00	90	235	39	48	115	1231
12:00	221	75	0	58	11	1271
15:00	183	68	12	37	57	769
12:00	125	169	3	44	17	266
23:00	208	15	7	61	30	1189
04:00	171	208	0	17	87	824
11:00	291	47	1	29	55	1158
14:00	146	246	25	49	88	1177
22:00	168	139	38	83	91	687
19:00	62	177	31	34	45	337
14:00	357	198	15	62	16	804
10:00	341	241	35	18	81	1183
03:00	157	175	2	67	24	838
12:00	129	149	32	90	39	1044

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Annexure I — Location Plans — Sheet 6

Design Computation

Settlement = $12.466 \times 10.601 = 132.155$ kNm (permissible 237.806) — safe

Deflection = $25.441 \times 11.961 = 304.305$ kNm (permissible 357.284) — safe

Bearing pressure = $12.781 \times 11.989 = 153.226$ kNm (permissible 226.532) — safe

Moment = $35.835 \times 10.496 = 376.133$ kN (permissible 657.454) — safe

Shear = $49.941 \times 11.256 = 562.145$ kPa (permissible 801.943) — safe

Shear = $47.144 \times 10.070 = 474.752$ kN (permissible 579.699) — safe

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Annexure I — Location Plans — Sheet 7

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. GA-917, revision A.

ANNEXURE II — APPROVED ALIGNMENT DRAWING

Alignment drawing showing the approved centre line, chainages and land boundaries. Drawing no. AL-002 to AL-009.

Annexure II — Alignment Sheets — Sheet 1

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. LA-963, revision A.

Annexure II — Alignment Sheets — Sheet 2

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Fine sand	Plant	48.4	232
Coarse aggregate	River bed	84.1	7,195
Coarse aggregate	Quarry B	71.9	3,985
Fine sand	Quarry A	40.4	6,844
Bitumen	Depot	5.0	5,569
Bitumen	Depot	63.1	6,216
Coarse aggregate	Quarry B	67.2	1,629
Coarse aggregate	Quarry A	41.9	9,224
Boulder	River bed	37.3	2,655
Fine sand	Quarry A	43.4	4,028
Coarse aggregate	Quarry B	20.4	1,342
Coarse aggregate	Plant	43.9	6,537
Fine sand	River bed	83.2	8,958
Steel	River bed	34.1	7,574
Bitumen	Depot	30.0	8,117
Steel	River bed	4.4	1,708
Fine sand	Depot	2.2	1,328
Bitumen	Plant	67.8	3,817
Bitumen	Plant	30.0	2,026
Fine sand	Quarry B	57.1	5,625
Cement	Plant	83.8	4,206
Coarse aggregate	Quarry B	7.8	3,031
Murram	Depot	35.0	781
Boulder	Plant	81.1	4,137

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Annexure II — Alignment Sheets — Sheet 3

Design Computation

Deflection = $86.567 \times 7.250 = 627.614$ kPa (permissible 929.003) — safe

Settlement = $84.330 \times 3.252 = 274.250$ kNm (permissible 390.353) — safe

Deflection = $66.986 \times 1.066 = 71.420$ kNm (permissible 126.793) — safe

Bearing pressure = $4.094 \times 3.284 = 13.443$ kNm (permissible 24.060) — safe

Bearing pressure = $14.004 \times 2.154 = 30.164$ kNm (permissible 38.139) — safe

Moment = $47.781 \times 8.852 = 422.950$ kN (permissible 767.452) — safe

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Annexure II — Alignment Sheets — Sheet 4

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. HY-242, revision C.

Annexure II — Alignment Sheets — Sheet 5

Land Schedule

Survey No.	Village	Area (ha)	Class	Status
81/1	Golakganj	1.660	Waste	Awarded
145/6	Gauripur	2.051	Waste	Under award
141/7	Gauripur	0.364	Agricultural	Under award
58/3	Dhubri	1.558	Agricultural	Awarded
454/2	Agomani	0.347	Waste	Under award
212/8	Gauripur	1.510	Agricultural	Awarded
449/7	Gauripur	1.718	Waste	Awarded
319/9	Dhubri	1.908	Agricultural	Under award
247/2	Golakganj	1.845	Agricultural	Notified
129/5	Agomani	1.803	Agricultural	Under award
89/5	Agomani	1.971	Waste	Under award
303/2	Dhubri	1.244	Waste	Under award
274/5	Agomani	1.694	Waste	Notified
107/2	Agomani	2.346	Homestead	Awarded
172/7	Dhubri	2.090	Waste	Awarded
330/8	Dhubri	1.648	Waste	Under award
251/9	Golakganj	2.108	Agricultural	Awarded
22/1	Golakganj	0.149	Homestead	Notified
311/7	Golakganj	1.122	Agricultural	Under award
207/6	Golakganj	0.585	Waste	Awarded

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Annexure II — Alignment Sheets — Sheet 6

Design Computation

Settlement = $16.383 \times 4.727 = 77.447$ kN (permissible 132.634) — safe

Settlement = $68.509 \times 8.149 = 558.287$ mm (permissible 702.610) — safe

Settlement = $35.350 \times 7.531 = 266.206$ kNm (permissible 316.144) — safe

Moment = $78.934 \times 8.710 = 687.550$ kN (permissible 768.398) — safe

Bearing pressure = $84.974 \times 9.011 = 765.725$ kNm (permissible 1448.915) — safe

Deflection = $39.561 \times 8.273 = 327.274$ kN (permissible 416.207) — safe

Shear = $29.668 \times 3.349 = 99.363$ kN (permissible 124.465) — safe

Bearing pressure = $64.741 \times 2.223 = 143.925$ kPa (permissible 200.262) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Annexure II — Alignment Sheets — Sheet 7

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Reference: drawing no. ST-320, revision A.

Annexure II — Alignment Sheets — Sheet 8

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
22:00	208	14	4	64	74	160
12:00	331	122	28	30	0	1194
04:00	68	259	19	24	58	838
00:00	209	52	8	73	101	965
16:00	45	89	12	53	51	1210
07:00	135	157	37	59	113	116
21:00	318	28	5	55	115	465
04:00	50	137	33	77	68	1350
15:00	261	183	21	19	34	394
11:00	167	86	40	80	1	225
21:00	54	57	32	12	86	468
11:00	158	100	3	20	4	1278
15:00	160	184	25	20	4	293
07:00	203	147	16	52	25	476
23:00	288	118	5	41	105	528
06:00	277	20	22	77	68	1376
20:00	349	54	1	90	47	1091
15:00	102	231	28	2	116	652
21:00	48	65	39	68	77	308
09:00	219	63	29	89	101	831
00:00	200	54	39	33	77	131
12:00	351	201	10	23	119	1178
16:00	231	62	14	65	71	665
14:00	398	206	23	10	30	910
19:00	111	100	34	85	104	821
18:00	87	16	27	74	96	133

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Annexure II — Alignment Sheets — Sheet 9

Design Computation

Deflection = $19.928 \times 6.577 = 131.068$ mm (permissible 151.308) — safe

Bearing pressure = $38.759 \times 7.834 = 303.642$ kN (permissible 486.787) — safe

Bearing pressure = $39.282 \times 4.785 = 187.972$ kPa (permissible 239.790) — safe

Deflection = $41.832 \times 0.812 = 33.984$ kPa (permissible 58.872) — safe

Bearing pressure = $66.044 \times 11.924 = 787.519$ kN (permissible 1090.375) — revise section

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Annexure II — Alignment Sheets — Sheet 10

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. LA-788, revision A.

Annexure II — Alignment Sheets — Sheet 11

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 235	20.99	2.93	1.29	21	179.728
Item 835	28.34	5.23	2.30	12	262.707
Item 851	31.71	4.82	2.80	7	238.560
Item 441	35.84	0.74	1.06	21	52.824
Item 869	33.58	1.81	2.11	10	258.745
Item 626	9.86	3.14	2.20	7	13.832
Item 443	37.01	3.28	0.23	5	114.207
Item 368	28.46	2.66	0.55	20	262.264
Item 121	39.86	1.63	2.17	16	200.751
Item 661	16.21	3.93	1.98	12	57.909
Item 511	30.69	3.92	1.48	17	234.899
Item 549	32.00	0.41	2.00	15	337.581
Item 893	24.30	3.62	0.44	16	62.603
Item 113	32.37	2.16	1.12	7	385.858
Item 567	9.09	1.16	1.95	5	89.185
Item 214	8.33	3.97	1.96	4	304.587
Item 308	28.62	1.49	0.25	8	88.611
Item 681	5.66	1.46	1.81	17	279.690
Item 109	17.19	5.77	2.46	22	143.649
Item 527	3.30	3.14	2.70	19	193.381
Item 138	37.20	0.75	2.90	16	288.163

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

ANNEXURE III — GENERAL ARRANGEMENT DRAWING

General arrangement of the bridge showing span arrangement, longitudinal section and typical cross section.
Drawing no. GA-010 to GA-021.

Annexure III — General Arrangement — Sheet 1

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Reference: drawing no. ST-818, revision A.

Annexure III — General Arrangement — Sheet 2

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
07:00	157	255	3	40	76	446
17:00	370	197	27	90	113	179
18:00	30	160	16	39	53	469
05:00	77	159	33	72	7	748
19:00	191	76	11	50	39	1135
04:00	163	221	17	85	88	458
13:00	45	97	34	4	53	595
16:00	165	142	26	50	84	458
09:00	308	105	25	74	3	688
18:00	324	153	34	67	55	1161
13:00	65	33	28	45	12	964
12:00	49	247	12	65	30	900
00:00	69	127	12	21	3	1199
12:00	198	55	21	45	80	471
14:00	38	26	34	17	101	492
20:00	375	35	3	24	52	329
04:00	320	127	2	15	63	1153

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Annexure III — General Arrangement — Sheet 3

Design Computation

Bearing pressure = $89.224 \times 9.027 = 805.398$ kPa (permissible 1343.538) — safe

Settlement = $32.486 \times 9.496 = 308.499$ mm (permissible 571.130) — revise section

Deflection = $45.501 \times 7.761 = 353.136$ kNm (permissible 422.234) — safe

Moment = $39.662 \times 3.520 = 139.623$ kNm (permissible 261.383) — safe

Deflection = $11.510 \times 2.910 = 33.490$ kPa (permissible 59.822) — safe

Bearing pressure = $25.597 \times 1.529 = 39.145$ kN (permissible 54.503) — safe

Shear = $81.827 \times 6.487 = 530.841$ kNm (permissible 664.637) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Annexure III — General Arrangement — Sheet 4

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Reference: drawing no. GA-868, revision C.

Annexure III — General Arrangement — Sheet 5

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Cement	Quarry B	80.3	2,024
Cement	Quarry A	83.9	2,765
Fine sand	River bed	7.0	8,351
Fine sand	Plant	26.6	8,154
Boulder	Quarry A	26.4	246
Fine sand	Depot	76.5	4,002
Steel	Plant	39.8	5,999
Cement	River bed	78.0	329
Boulder	Depot	55.5	1,936
Fine sand	Plant	50.5	6,112
Cement	Plant	25.2	4,427
Cement	Plant	16.0	8,533
Steel	Depot	73.9	9,010
Fine sand	Depot	37.7	1,451
Steel	Plant	19.2	2,830
Coarse aggregate	Quarry A	83.0	5,722
Fine sand	Quarry A	47.6	2,684
Fine sand	Plant	33.9	190
Bitumen	River bed	69.6	537
Coarse aggregate	Plant	62.4	1,247
Coarse aggregate	Plant	7.9	3,102
Cement	Plant	28.7	5,301
Steel	Plant	56.8	9,175

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Annexure III — General Arrangement — Sheet 6

Design Computation

Moment = $87.374 \times 11.325 = 989.494$ kN (permissible 1712.520) — safe

Shear = $29.599 \times 3.664 = 108.449$ kN (permissible 119.160) — safe

Deflection = $27.553 \times 0.677 = 18.652$ kNm (permissible 21.220) — safe

Bearing pressure = $30.568 \times 2.436 = 74.450$ mm (permissible 104.426) — safe

Settlement = $73.380 \times 3.959 = 290.488$ kN (permissible 521.790) — safe

Deflection = $34.353 \times 0.870 = 29.876$ kN (permissible 33.728) — safe

Bearing pressure = $67.884 \times 6.852 = 465.137$ kNm (permissible 590.074) — safe

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Annexure III — General Arrangement — Sheet 7

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. LA-170, revision C.

Annexure III — General Arrangement — Sheet 8

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
12:00	87	244	37	58	55	1371
17:00	44	111	29	85	56	819
06:00	166	186	0	10	14	396
14:00	60	65	36	60	101	1382
08:00	342	42	1	20	74	701
17:00	142	160	12	43	94	895
14:00	305	193	11	51	85	1004
17:00	237	77	13	31	28	1397
16:00	158	184	18	73	5	1312
07:00	332	151	15	45	58	843
03:00	317	201	14	44	53	846
04:00	397	45	2	70	77	1112
09:00	353	208	6	16	103	1122
22:00	139	111	8	13	47	1114
20:00	399	76	0	1	21	984
10:00	100	133	10	65	76	408
22:00	71	76	23	15	60	428
05:00	385	31	32	85	32	990
17:00	250	93	36	41	29	448

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Annexure III — General Arrangement — Sheet 9

Design Computation

Settlement = $33.325 \times 0.516 = 17.195$ kPa (permissible 27.208) — safe

Shear = $39.313 \times 8.895 = 349.706$ kPa (permissible 445.597) — safe

Deflection = $55.654 \times 3.572 = 198.786$ kPa (permissible 295.124) — safe

Shear = $62.405 \times 6.641 = 414.447$ kPa (permissible 770.829) — safe

Moment = $41.226 \times 4.864 = 200.543$ mm (permissible 316.116) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Annexure III — General Arrangement — Sheet 10

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. HY-757, revision A.

Annexure III — General Arrangement — Sheet 11

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 307	13.99	5.04	1.47	6	26.679
Item 216	9.33	2.70	2.01	16	252.859
Item 414	25.10	2.34	0.96	10	33.540
Item 580	39.01	1.11	1.38	5	30.806
Item 892	12.67	5.42	2.26	1	324.827
Item 570	21.97	5.34	1.54	1	85.492
Item 851	13.80	5.19	2.52	4	219.320
Item 327	6.24	1.60	1.19	14	69.011
Item 365	26.99	1.68	2.69	16	274.846
Item 217	31.10	2.27	0.92	4	309.539
Item 340	36.72	4.19	1.18	9	246.494
Item 784	13.58	2.54	2.75	7	254.337
Item 236	7.93	1.74	2.89	22	91.803
Item 151	24.17	5.06	1.02	4	167.834
Item 887	36.44	4.53	0.26	21	314.790
Item 598	30.51	2.65	2.10	17	306.283
Item 278	14.20	0.56	0.64	2	302.679
Item 552	10.98	3.47	1.61	8	389.205
Item 380	24.65	4.16	2.13	17	26.577
Item 794	2.93	2.05	2.44	13	155.181
Item 463	17.99	5.05	0.47	4	379.171
Item 722	35.15	2.53	0.89	5	38.629
Item 402	4.12	4.44	2.50	13	130.976

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

ANNEXURE IV — DETAILED ESTIMATE

Item-wise detailed estimate with quantities, rates and amounts, supported by measurement sheets and rate analysis.

Sl.	Item of work	Unit	Rate (Rs.)
1	Earthwork in excavation in ordinary soil	cum	312
2	Plain cement concrete M15	cum	5,480
3	Reinforced cement concrete M35 in substructure	cum	8,940
4	HYSD reinforcement steel Fe500D	MT	72,600
5	Structural steel in built-up sections	MT	94,200
6	Formwork for foundations	sqm	486
7	Formwork for piers and abutments	sqm	612
8	Granular sub-base, close graded	cum	1,842
9	Wet mix macadam	cum	2,110
10	Prime coat with bituminous primer	sqm	48
11	Tack coat	sqm	22
12	Dense bituminous macadam	cum	7,320
13	Brick masonry in cement mortar 1:6	cum	6,180
14	Cement plaster 12 mm thick	sqm	268
15	Providing and launching PSC girders	MT	88,400
16	Pot-PTFE bearings	each	42,600
17	Strip seal expansion joint	rm	18,900
18	Bituminous concrete, grading II	cum	9,240
19	Thermoplastic road marking	sqm	412

ANNEXURE IV-A — YEAR-WISE CASH FLOW STATEMENT

All figures in Rs. crore. Year 0 is the year of commencement.

Year	Capital cost	O&M cost	Gross benefit	Net cash flow
0	148.00	-	-	-148.00
1	165.00	-	-	-165.00
2	99.50	-	-	-99.50
3	-	5.80	47.83	42.03
4	-	5.80	47.64	41.84
5	-	5.80	47.45	41.65
6	-	5.80	47.26	41.46
7	-	5.80	47.06	41.26
8	-	5.80	46.87	41.07
9	-	5.80	46.68	40.88
10	-	5.80	46.49	40.69
11	-	5.80	46.30	40.50
12	-	5.80	46.11	40.31
13	-	5.80	45.92	40.12
14	-	5.80	45.73	39.93
15	-	5.80	45.53	39.73
16	-	5.80	45.34	39.54
17	-	5.80	45.15	39.35
18	-	5.80	44.96	39.16
19	-	5.80	44.77	38.97
20	-	5.80	44.58	38.78
21	-	5.80	44.39	38.59
22	-	5.80	44.19	38.39
23	-	5.80	44.00	38.20
24	-	5.80	43.81	38.01
25	-	5.80	43.62	37.82
26	-	5.80	43.43	37.63
27	-	5.80	43.24	37.44
28	-	5.80	43.05	37.25
29	-	5.80	42.86	37.06
30	-	5.80	42.66	36.86

Annexure IV — Measurement Sheets — Lead Statement, page 1

Material	Source	Lead (km)	Rate (Rs)
Murram	Quarry A	3.5	5,205
Steel	Depot	48.6	2,314
Boulder	Quarry B	25.5	410
Fine sand	Plant	42.0	585
Coarse aggregate	Quarry A	88.4	8,343
Steel	Quarry A	47.1	7,246
Cement	Quarry A	20.1	758
Bitumen	River bed	59.9	3,719
Boulder	River bed	51.8	6,220
Bitumen	River bed	60.1	5,330
Bitumen	Quarry A	73.7	3,090
Boulder	Depot	54.0	7,511
Boulder	Quarry B	18.0	3,105
Bitumen	River bed	61.3	8,700
Fine sand	Plant	12.1	1,795
Steel	Quarry A	52.0	4,777
Murram	Plant	31.1	6,236
Cement	Plant	28.8	7,630
Bitumen	Quarry B	24.7	1,935
Bitumen	Quarry A	64.2	8,257
Fine sand	River bed	50.0	1,824
Steel	Quarry B	41.5	733
Coarse aggregate	Depot	36.6	2,843
Murram	Quarry B	62.5	3,820
Murram	Quarry A	49.4	7,472
Boulder	Quarry A	32.5	6,356

Annexure IV — Measurement Sheets — Land Schedule, page 2

Survey No.	Village	Area (ha)	Class	Status
108/1	Dhubri	0.671	Waste	Awarded
396/8	Agomani	0.557	Agricultural	Notified
280/3	Gauripur	0.645	Homestead	Awarded
471/1	Golakganj	1.841	Waste	Under award
426/3	Golakganj	0.214	Waste	Notified
132/9	Agomani	1.049	Homestead	Notified
322/9	Dhubri	1.992	Agricultural	Awarded
333/3	Agomani	2.318	Waste	Under award
137/3	Dhubri	0.844	Waste	Notified
68/4	Golakganj	1.929	Waste	Under award
12/6	Golakganj	0.106	Agricultural	Notified
454/7	Gauripur	1.915	Agricultural	Awarded
345/8	Golakganj	2.294	Homestead	Notified
151/8	Gauripur	1.286	Homestead	Under award
470/2	Gauripur	2.110	Homestead	Under award
407/1	Golakganj	1.472	Homestead	Notified
173/7	Agomani	1.941	Homestead	Under award
92/7	Dhubri	1.036	Waste	Under award
397/9	Agomani	1.316	Waste	Awarded
206/4	Agomani	2.372	Agricultural	Notified
199/2	Golakganj	2.376	Homestead	Notified
404/8	Dhubri	2.067	Homestead	Awarded
136/8	Dhubri	1.006	Agricultural	Under award
112/6	Gauripur	1.512	Agricultural	Awarded
387/9	Golakganj	1.485	Homestead	Under award
50/3	Dhubri	2.344	Waste	Awarded
19/3	Gauripur	2.241	Homestead	Under award
469/5	Dhubri	1.981	Waste	Notified
62/6	Dhubri	0.365	Agricultural	Awarded

Annexure IV — Measurement Sheets — Detailed Measurement Sheet, page 3

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 311	38.74	5.95	2.32	8	213.805
Item 546	27.21	3.33	0.81	18	370.261
Item 419	19.35	4.06	2.88	21	52.549
Item 489	39.32	2.07	2.78	16	158.631
Item 841	9.99	4.01	0.65	11	163.316
Item 159	3.55	4.49	1.52	6	236.145
Item 524	15.93	5.74	0.30	10	178.738
Item 277	17.15	2.39	1.60	1	215.018
Item 738	25.99	4.94	0.71	15	392.535
Item 776	31.38	5.89	1.48	1	157.177
Item 491	11.67	4.50	1.38	12	389.917
Item 858	27.52	5.89	2.65	24	68.293
Item 716	39.39	4.58	2.47	15	217.457
Item 224	37.26	2.71	2.12	21	319.657
Item 220	25.58	4.58	1.90	18	258.573
Item 455	31.75	2.08	2.17	14	197.820
Item 659	37.63	3.39	1.15	24	9.868
Item 784	23.65	5.44	0.63	23	312.491
Item 243	27.35	0.94	2.78	9	45.211
Item 696	5.01	0.76	2.43	24	191.270
Item 747	3.34	1.81	1.72	14	396.991
Item 762	2.26	5.54	2.78	22	292.979
Item 172	26.97	1.05	2.54	16	25.424
Item 778	36.11	1.78	2.79	20	323.987
Item 877	3.73	4.62	1.38	2	254.909

Annexure IV — Measurement Sheets — Bore Log Summary, page 4

Depth (m)	Strata	N value	Sample
4.8	Silty clay	26	DS
17.7	Clayey silt	39	SPT
0.7	Weathered rock	9	SPT
10.1	Gravelly sand	49	DS
1.5	Medium sand	17	SPT
24.2	Silty clay	13	UDS
4.6	Clayey silt	46	UDS
18.6	Weathered rock	52	DS
6.3	Medium sand	37	DS
16.3	Clayey silt	42	UDS
25.3	Coarse sand	46	DS
3.4	Clayey silt	49	UDS
3.9	Weathered rock	26	UDS
6.6	Clayey silt	28	SPT
23.2	Silty clay	49	UDS
4.9	Gravelly sand	60	SPT
14.1	Medium sand	39	DS
23.9	Coarse sand	6	DS
18.0	Silty clay	53	UDS
22.2	Medium sand	8	DS
11.5	Silty clay	52	UDS
13.7	Silty clay	14	SPT
10.8	Medium sand	33	DS
23.5	Coarse sand	23	UDS

Annexure IV — Measurement Sheets — Traffic Volume Count, page 5

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
15:00	73	113	34	85	48	850
20:00	395	14	11	72	112	218
00:00	38	229	21	88	18	296
23:00	362	207	40	42	50	1377
22:00	93	114	19	16	67	642
16:00	298	253	38	81	103	1352
18:00	338	171	32	51	86	444
19:00	157	113	21	19	104	102
10:00	173	150	5	62	61	798
22:00	47	209	26	85	96	241
08:00	110	39	1	78	78	545
11:00	237	239	15	77	11	1364
16:00	374	76	16	25	27	432
17:00	95	135	20	52	43	838
17:00	133	160	27	6	95	848
10:00	322	212	7	8	7	340
03:00	114	39	27	36	110	337
11:00	106	35	23	42	37	1249
23:00	253	244	39	26	109	1237
12:00	326	201	38	36	64	155
20:00	103	196	40	4	105	1016
17:00	222	173	2	16	61	107
23:00	205	243	40	56	99	550
17:00	92	178	12	90	4	654
18:00	119	38	19	62	75	86
08:00	151	134	33	61	22	1334
13:00	235	212	12	14	90	500
12:00	20	232	26	15	94	635

Annexure IV — Measurement Sheets — Lead Statement, page 6

Material	Source	Lead (km)	Rate (Rs)
Cement	Quarry B	21.2	297
Boulder	River bed	47.3	6,974
Coarse aggregate	Quarry B	38.2	3,424
Coarse aggregate	Plant	81.2	1,346
Boulder	Quarry A	2.3	6,265
Coarse aggregate	Plant	4.3	6,919
Boulder	River bed	52.7	1,489
Murram	Quarry B	25.8	2,357
Coarse aggregate	Quarry A	7.3	1,956
Coarse aggregate	Quarry B	85.1	1,597
Boulder	Depot	48.5	8,315
Steel	Plant	65.1	932
Murram	Quarry A	50.7	2,982
Coarse aggregate	River bed	46.5	7,970
Murram	Quarry A	52.2	4,142
Cement	Plant	53.1	720
Cement	Plant	50.3	8,441
Murram	Quarry B	44.9	1,615
Fine sand	River bed	76.2	5,335
Boulder	Plant	67.9	4,986
Murram	Quarry B	39.0	743
Coarse aggregate	Quarry A	78.5	6,680
Boulder	Plant	35.1	6,847
Fine sand	Plant	10.2	8,571

Annexure IV — Measurement Sheets — Land Schedule, page 7

Survey No.	Village	Area (ha)	Class	Status
124/4	Golakganj	1.681	Waste	Awarded
239/1	Golakganj	0.677	Homestead	Awarded
278/4	Dhubri	0.852	Agricultural	Notified
383/7	Dhubri	1.709	Waste	Under award
94/4	Agomani	0.414	Agricultural	Notified
90/6	Golakganj	0.422	Waste	Under award
22/1	Gauripur	2.146	Homestead	Awarded
302/4	Agomani	0.396	Agricultural	Notified
453/9	Golakganj	0.493	Waste	Under award
400/9	Golakganj	0.226	Homestead	Under award
339/9	Golakganj	0.847	Agricultural	Under award
92/5	Agomani	2.105	Agricultural	Awarded
191/8	Agomani	2.124	Waste	Awarded
112/2	Gauripur	1.460	Agricultural	Under award
50/4	Dhubri	1.360	Agricultural	Under award
279/2	Agomani	0.593	Waste	Notified
69/4	Agomani	0.829	Waste	Awarded
246/3	Agomani	1.819	Agricultural	Under award
373/8	Gauripur	0.476	Waste	Notified
231/9	Agomani	1.231	Waste	Awarded
355/7	Golakganj	1.450	Homestead	Under award
453/3	Golakganj	0.082	Agricultural	Notified
363/2	Dhubri	0.056	Waste	Awarded
20/7	Gauripur	2.398	Waste	Under award

Annexure IV — Measurement Sheets — Detailed Measurement Sheet, page 8

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 419	36.52	3.54	2.47	12	308.556
Item 362	28.62	4.64	2.14	5	258.339
Item 425	16.63	1.05	2.17	14	57.467
Item 462	36.88	2.49	1.35	17	32.165
Item 826	39.81	4.58	2.46	23	398.984
Item 567	12.38	1.65	0.26	21	237.519
Item 409	15.76	4.57	0.86	9	341.999
Item 608	10.94	5.15	1.57	24	2.563
Item 295	27.32	2.36	2.91	3	91.769
Item 806	35.63	4.19	1.19	2	303.389
Item 840	6.33	5.72	2.18	1	336.168
Item 625	6.44	3.84	2.79	19	142.022
Item 580	25.46	2.16	2.99	11	50.977
Item 326	33.77	2.43	1.85	8	124.355
Item 109	18.69	4.36	1.62	6	7.302
Item 222	21.61	3.29	1.45	18	262.057
Item 777	6.93	5.01	2.52	10	271.858
Item 548	33.53	2.24	1.84	21	20.597
Item 467	6.63	3.20	0.78	5	93.986
Item 557	5.93	3.17	0.94	2	41.190
Item 558	11.98	3.09	1.83	12	389.130
Item 795	10.40	2.30	2.89	22	78.765
Item 662	16.86	5.43	0.40	20	174.237
Item 146	33.52	3.35	1.49	6	227.119
Item 750	7.56	2.01	0.45	1	127.794
Item 145	20.70	2.33	2.95	24	186.103
Item 393	24.33	5.41	2.39	3	22.728
Item 164	38.92	1.65	2.05	19	390.233

Annexure IV — Measurement Sheets — Bore Log Summary, page 9

Depth (m)	Strata	N value	Sample
25.7	Weathered rock	21	DS
13.9	Silty clay	7	UDS
28.2	Silty clay	53	UDS
10.3	Coarse sand	25	UDS
25.6	Coarse sand	24	SPT
1.4	Medium sand	60	SPT
18.1	Gravelly sand	54	DS
26.0	Medium sand	52	SPT
8.0	Silty clay	54	UDS
12.5	Coarse sand	19	SPT
15.5	Coarse sand	45	DS
22.9	Coarse sand	24	DS
12.7	Medium sand	22	UDS
6.9	Silty clay	60	DS
26.2	Weathered rock	52	UDS
1.6	Clayey silt	42	DS
23.6	Silty clay	13	DS
26.7	Silty clay	16	SPT
24.6	Weathered rock	59	SPT
11.4	Gravelly sand	40	SPT
12.7	Weathered rock	35	SPT
9.6	Clayey silt	29	UDS
29.2	Coarse sand	60	UDS
11.0	Coarse sand	44	UDS
21.8	Medium sand	38	SPT
13.9	Gravelly sand	21	SPT
8.6	Coarse sand	42	DS
16.7	Gravelly sand	9	SPT
5.8	Coarse sand	34	UDS
5.5	Clayey silt	26	DS

Annexure IV — Measurement Sheets — Traffic Volume Count, page 10

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
16:00	345	119	25	27	68	812
17:00	276	56	2	16	104	276
01:00	163	256	28	27	20	1000
14:00	245	10	33	51	83	445
21:00	297	22	10	30	110	318
10:00	75	126	1	1	75	1135
04:00	327	199	17	73	34	193
09:00	271	121	40	86	10	1262
02:00	85	39	17	26	98	1275
05:00	280	91	34	44	75	680
02:00	204	12	10	37	78	352
08:00	347	131	1	61	31	306
10:00	79	96	30	48	83	846
19:00	360	202	18	40	109	757
11:00	151	193	14	17	65	512
17:00	117	98	5	55	3	497
06:00	70	49	28	79	75	554
17:00	349	158	26	45	9	304
23:00	394	206	9	40	39	256
22:00	183	51	0	49	46	869
20:00	299	13	9	81	48	100
17:00	110	179	11	54	100	1357
05:00	263	72	29	7	9	1392
13:00	155	50	28	55	63	584
12:00	200	87	18	11	79	316
03:00	22	192	25	21	75	1306
05:00	386	139	38	60	87	735
08:00	192	167	20	30	4	575
22:00	384	196	33	10	9	1166
00:00	84	158	18	87	69	1306

Annexure IV — Measurement Sheets — Lead Statement, page 11

Material	Source	Lead (km)	Rate (Rs)
Bitumen	Quarry B	15.3	3,113
Fine sand	Quarry A	42.9	1,274
Murram	Quarry A	3.2	3,392
Fine sand	Plant	46.4	871
Fine sand	Quarry A	31.2	2,654
Murram	Depot	81.5	3,054
Murram	Plant	58.8	5,135
Cement	Quarry B	4.8	2,985
Murram	River bed	47.6	8,266
Bitumen	Depot	39.6	5,789
Fine sand	Plant	77.3	8,278
Murram	Depot	47.3	8,215
Murram	River bed	15.0	6,515
Bitumen	River bed	49.0	5,541
Bitumen	Plant	27.8	210
Murram	Depot	81.2	6,831
Steel	Depot	23.1	4,402
Coarse aggregate	Depot	2.2	1,805
Murram	Quarry A	63.5	610
Coarse aggregate	Plant	67.3	7,864
Bitumen	River bed	77.6	7,785
Cement	Plant	12.5	4,462
Steel	Quarry A	2.5	8,692
Bitumen	Plant	19.9	4,172
Coarse aggregate	Quarry A	22.7	620
Murram	River bed	86.6	438
Bitumen	Plant	43.8	1,149
Bitumen	River bed	16.8	4,287
Coarse aggregate	Plant	47.8	1,999

Annexure IV — Measurement Sheets — Land Schedule, page 12

Survey No.	Village	Area (ha)	Class	Status
63/9	Gauripur	2.077	Waste	Under award
408/5	Golakganj	1.403	Waste	Notified
229/8	Agomani	2.081	Agricultural	Under award
95/4	Golakganj	1.756	Homestead	Notified
348/1	Gauripur	0.346	Agricultural	Awarded
121/3	Golakganj	0.392	Waste	Under award
365/5	Golakganj	1.795	Homestead	Notified
129/5	Golakganj	1.676	Homestead	Under award
207/4	Gauripur	0.352	Homestead	Under award
18/8	Golakganj	0.832	Waste	Notified
221/9	Dhubri	1.420	Waste	Under award
396/7	Golakganj	1.099	Waste	Notified
222/8	Agomani	1.482	Homestead	Notified
239/7	Golakganj	0.805	Agricultural	Under award
44/8	Gauripur	1.542	Agricultural	Under award
372/3	Golakganj	2.134	Agricultural	Notified
256/3	Dhubri	1.443	Waste	Notified
184/5	Gauripur	2.369	Waste	Notified
178/8	Dhubri	1.688	Agricultural	Under award
421/2	Dhubri	0.095	Agricultural	Under award
464/9	Agomani	1.119	Homestead	Notified
254/9	Gauripur	0.561	Homestead	Under award

Annexure IV — Measurement Sheets — Detailed Measurement Sheet, page 13

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 665	2.71	3.56	0.38	18	107.925
Item 438	9.03	2.91	2.33	8	306.542
Item 226	18.55	3.25	1.73	10	339.423
Item 113	31.61	3.03	2.52	13	344.592
Item 341	4.06	4.94	2.49	11	332.245
Item 431	3.87	1.74	1.40	20	125.640
Item 407	38.34	0.35	2.78	19	267.649
Item 652	17.41	5.43	1.74	19	202.373
Item 367	13.82	4.17	1.77	24	288.705
Item 127	23.39	2.02	1.02	10	2.652
Item 253	14.86	4.26	0.71	8	198.801
Item 510	39.33	5.44	1.28	1	374.005
Item 245	27.04	3.79	1.21	14	122.544
Item 329	8.41	4.51	1.64	4	57.252
Item 373	19.36	4.05	2.98	4	268.065
Item 627	11.47	3.23	1.07	13	381.892
Item 218	27.18	1.68	2.66	12	116.883
Item 468	7.67	5.70	2.19	14	69.864
Item 178	4.15	1.08	2.69	4	162.861
Item 174	25.25	0.38	1.38	2	167.086
Item 174	6.39	5.80	2.70	22	393.315
Item 615	28.26	5.43	0.50	14	308.489
Item 135	9.91	3.62	1.42	22	136.483
Item 181	33.98	1.73	0.70	10	92.581
Item 499	21.15	4.65	2.36	11	232.706
Item 440	31.91	3.21	1.67	16	279.765
Item 883	10.00	3.57	2.60	9	367.634
Item 266	16.18	1.99	2.79	24	323.103
Item 156	21.63	2.29	1.22	4	151.913

Annexure IV — Measurement Sheets — Bore Log Summary, page 14

Depth (m)	Strata	N value	Sample
20.3	Weathered rock	11	DS
15.8	Weathered rock	12	DS
21.7	Medium sand	54	DS
16.0	Coarse sand	9	UDS
20.5	Silty clay	21	DS
11.4	Coarse sand	25	DS
29.0	Medium sand	8	UDS
19.4	Weathered rock	27	UDS
10.9	Coarse sand	47	DS
5.5	Clayey silt	33	UDS
5.7	Medium sand	36	DS
2.0	Clayey silt	27	SPT
3.9	Clayey silt	26	DS
4.2	Coarse sand	15	UDS
18.4	Medium sand	51	DS
9.5	Medium sand	4	SPT
25.7	Coarse sand	23	SPT
13.8	Weathered rock	26	UDS
9.9	Coarse sand	13	SPT
12.4	Weathered rock	21	SPT
20.0	Clayey silt	12	DS
16.8	Silty clay	20	SPT
15.8	Clayey silt	60	DS
0.8	Medium sand	34	UDS
13.2	Clayey silt	28	UDS
25.1	Medium sand	4	DS
19.8	Weathered rock	20	DS

Annexure IV — Measurement Sheets — Traffic Volume Count, page 15

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
01:00	125	176	26	74	46	1347
11:00	117	97	27	23	113	425
19:00	43	53	28	19	51	651
07:00	325	180	26	56	47	196
19:00	71	71	36	87	112	353
13:00	231	156	37	18	23	979
03:00	79	107	38	26	54	1047
19:00	115	172	23	23	110	502
17:00	306	193	40	21	0	246
07:00	202	237	3	9	57	1214
19:00	294	180	4	77	57	1127
15:00	143	228	19	30	2	685
13:00	369	126	6	62	108	873
13:00	335	196	2	61	2	800
19:00	267	252	26	3	84	99
20:00	20	138	33	45	53	982
11:00	195	57	28	22	88	610
18:00	187	196	20	61	14	398
02:00	128	209	19	27	105	1010
00:00	34	72	5	74	90	1022
22:00	307	68	11	45	75	649
23:00	380	174	34	52	8	929
00:00	258	95	25	69	23	537
03:00	265	173	5	36	34	1252
22:00	111	232	39	44	97	240
19:00	238	53	3	73	53	958
20:00	378	204	4	4	55	481
23:00	162	79	37	72	8	754
04:00	331	259	2	21	85	872

Annexure IV — Measurement Sheets — Lead Statement, page 16

Material	Source	Lead (km)	Rate (Rs)
Coarse aggregate	Depot	2.3	8,289
Coarse aggregate	Depot	52.2	5,446
Coarse aggregate	Depot	79.2	5,433
Bitumen	Quarry B	39.8	1,285
Fine sand	River bed	50.3	2,580
Fine sand	Plant	25.5	3,234
Cement	River bed	48.7	6,299
Bitumen	Depot	27.0	6,260
Coarse aggregate	Depot	17.4	3,483
Murram	Plant	65.4	5,554
Steel	River bed	33.7	5,016
Murram	River bed	36.9	7,397
Steel	Quarry B	87.6	2,361
Coarse aggregate	Depot	7.3	532
Bitumen	Plant	51.4	809
Fine sand	Quarry B	62.7	2,315
Fine sand	River bed	29.6	6,065
Cement	Plant	30.0	5,502
Cement	River bed	24.4	9,338
Murram	Depot	9.0	5,958
Fine sand	River bed	78.5	6,677
Bitumen	Quarry B	18.2	1,425
Coarse aggregate	River bed	47.4	2,086
Boulder	River bed	32.1	8,605
Coarse aggregate	Quarry B	13.5	5,975
Fine sand	Quarry B	85.6	7,091
Coarse aggregate	Plant	41.8	1,029

Annexure IV — Measurement Sheets — Land Schedule, page 17

Survey No.	Village	Area (ha)	Class	Status
252/1	Agomani	0.420	Homestead	Awarded
452/3	Golakganj	0.296	Homestead	Under award
86/1	Agomani	0.712	Waste	Notified
167/9	Gauripur	0.414	Waste	Awarded
470/5	Agomani	0.197	Agricultural	Awarded
86/5	Golakganj	1.898	Waste	Notified
237/9	Agomani	1.224	Waste	Under award
394/6	Golakganj	0.839	Agricultural	Notified
32/2	Golakganj	2.231	Homestead	Awarded
403/2	Dhubri	1.065	Waste	Under award
30/7	Agomani	1.054	Agricultural	Under award
274/9	Dhubri	2.108	Waste	Under award
35/8	Agomani	0.052	Homestead	Notified
348/9	Agomani	0.729	Agricultural	Awarded
211/1	Golakganj	0.099	Agricultural	Notified
171/5	Agomani	1.577	Waste	Notified
84/7	Dhubri	1.217	Agricultural	Awarded
317/7	Dhubri	2.261	Waste	Notified
216/9	Golakganj	1.711	Agricultural	Awarded
335/4	Agomani	2.378	Homestead	Under award
303/8	Agomani	1.935	Homestead	Under award
474/2	Dhubri	0.189	Agricultural	Awarded
256/9	Gauripur	0.459	Homestead	Awarded
408/9	Dhubri	2.393	Homestead	Awarded

Annexure IV — Measurement Sheets — Detailed Measurement Sheet, page 18

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 381	35.69	3.32	1.59	18	251.982
Item 481	14.29	0.76	0.98	24	399.584
Item 195	19.29	4.03	1.34	5	162.267
Item 595	22.99	2.60	2.02	22	186.186
Item 403	18.74	2.79	2.74	7	105.251
Item 548	29.44	3.84	0.44	1	147.759
Item 818	16.31	0.89	0.66	7	49.109
Item 321	20.22	1.58	1.07	4	83.139
Item 565	29.33	2.43	2.74	8	67.135
Item 442	37.00	2.94	0.47	10	175.074
Item 727	14.13	3.40	0.66	23	283.423
Item 237	26.84	2.48	2.24	16	263.634
Item 243	31.89	3.81	0.60	11	377.010
Item 308	5.66	5.21	2.43	12	43.040
Item 516	5.57	3.87	0.87	9	48.632
Item 130	39.16	0.46	1.97	21	36.080
Item 551	39.07	3.78	0.34	24	237.024
Item 227	24.60	0.48	0.58	20	226.348
Item 878	7.83	2.77	2.90	17	127.796
Item 744	7.57	3.18	0.84	1	315.637
Item 849	22.81	1.44	0.35	14	226.620
Item 592	5.73	4.80	2.07	9	323.065
Item 399	11.52	1.01	0.63	17	166.973
Item 183	8.79	4.28	0.70	2	9.120
Item 414	32.80	3.14	2.46	24	226.439
Item 794	4.23	3.70	0.37	11	161.502

Annexure IV — Measurement Sheets — Bore Log Summary, page 19

Depth (m)	Strata	N value	Sample
15.8	Clayey silt	45	UDS
13.7	Medium sand	13	DS
18.4	Silty clay	20	DS
27.4	Gravelly sand	16	SPT
6.2	Medium sand	15	DS
26.2	Clayey silt	26	UDS
23.9	Weathered rock	22	DS
17.3	Gravelly sand	11	SPT
2.8	Silty clay	44	DS
28.4	Coarse sand	60	DS
12.2	Weathered rock	35	SPT
3.3	Silty clay	9	UDS
4.6	Weathered rock	56	DS
0.7	Gravelly sand	17	UDS
9.3	Coarse sand	16	SPT
6.0	Medium sand	22	UDS
17.9	Medium sand	47	DS
29.1	Silty clay	49	SPT
12.9	Medium sand	12	DS
29.9	Clayey silt	58	UDS
2.6	Gravelly sand	6	SPT
15.4	Clayey silt	45	UDS
5.5	Silty clay	12	SPT
2.0	Weathered rock	48	SPT
18.3	Clayey silt	14	UDS

Annexure IV — Measurement Sheets — Traffic Volume Count, page 20

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
01:00	31	92	37	71	114	446
18:00	229	129	7	34	59	905
10:00	273	22	31	68	100	636
21:00	226	19	9	55	16	184
03:00	38	230	6	52	105	539
08:00	152	63	2	17	11	89
07:00	50	215	30	16	56	1217
18:00	280	165	8	80	26	601
01:00	268	66	2	74	47	1048
01:00	368	220	13	45	89	714
03:00	277	133	17	67	45	962
07:00	111	252	37	85	24	520
14:00	265	108	26	83	118	848
14:00	132	39	30	86	103	259
09:00	229	190	39	23	111	706
02:00	31	194	32	70	27	755
08:00	182	107	7	13	119	581
14:00	312	87	34	8	65	169
11:00	63	17	13	19	93	1090
06:00	381	88	10	60	50	435
01:00	141	204	31	43	104	966
18:00	204	62	40	40	82	319
17:00	175	62	24	61	44	1182
02:00	129	175	13	57	116	1073
16:00	344	131	26	48	41	645
11:00	102	47	29	21	97	139

Annexure IV — Measurement Sheets — Lead Statement, page 21

Material	Source	Lead (km)	Rate (Rs)
Boulder	Depot	76.1	667
Murram	River bed	22.8	8,759
Coarse aggregate	Plant	79.4	6,246
Cement	River bed	66.6	4,971
Boulder	River bed	56.6	3,539
Coarse aggregate	Quarry B	56.4	5,381
Steel	Quarry B	55.7	5,696
Steel	Depot	8.0	8,282
Murram	Plant	74.2	4,287
Steel	Plant	11.8	3,138
Fine sand	Plant	23.3	9,041
Coarse aggregate	Quarry B	27.5	4,538
Cement	Depot	29.3	5,329
Cement	Quarry B	82.3	7,113
Steel	Depot	19.4	2,650
Bitumen	Quarry A	43.4	1,036
Fine sand	Plant	45.3	3,379
Steel	Depot	58.6	7,901
Coarse aggregate	Quarry A	47.7	8,082
Murram	Plant	51.5	4,959
Steel	Quarry B	59.6	8,548
Cement	Plant	89.0	1,569
Steel	Quarry B	86.3	6,055
Steel	Quarry A	19.2	5,041
Cement	Quarry B	38.4	3,970
Boulder	Depot	80.1	4,273
Fine sand	River bed	51.0	1,573
Cement	Quarry A	80.8	6,891

Annexure IV — Measurement Sheets — Land Schedule, page 22

Survey No.	Village	Area (ha)	Class	Status
156/9	Dhubri	1.783	Agricultural	Awarded
387/7	Golakganj	1.541	Waste	Notified
325/3	Agomani	2.019	Waste	Notified
171/1	Gauripur	2.229	Agricultural	Notified
263/3	Golakganj	0.748	Waste	Awarded
265/6	Agomani	2.081	Homestead	Under award
242/5	Golakganj	0.167	Homestead	Awarded
355/2	Gauripur	2.272	Homestead	Awarded
51/2	Gauripur	0.162	Agricultural	Awarded
378/1	Dhubri	0.615	Homestead	Awarded
303/4	Agomani	2.120	Homestead	Awarded
333/7	Dhubri	1.339	Waste	Under award
162/7	Gauripur	2.282	Agricultural	Under award
253/1	Golakganj	1.300	Agricultural	Under award
46/8	Dhubri	1.023	Agricultural	Awarded
450/1	Dhubri	0.613	Agricultural	Awarded
60/9	Golakganj	2.399	Waste	Under award
274/4	Gauripur	0.253	Waste	Notified
243/3	Gauripur	2.022	Homestead	Notified
419/4	Agomani	1.441	Waste	Under award
31/2	Golakganj	1.405	Homestead	Awarded
220/2	Agomani	0.069	Waste	Under award
46/2	Dhubri	2.320	Agricultural	Notified
182/1	Agomani	2.074	Homestead	Notified
46/2	Agomani	0.797	Waste	Notified
273/5	Dhubri	2.344	Homestead	Awarded

Annexure IV — Measurement Sheets — Detailed Measurement Sheet, page 23

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 563	39.56	1.02	1.44	22	319.824
Item 811	12.36	3.95	2.09	3	167.221
Item 312	20.29	5.31	1.53	23	229.240
Item 334	21.36	1.46	1.14	22	338.013
Item 750	18.36	0.72	1.24	6	173.244
Item 516	12.23	4.14	2.23	15	133.423
Item 427	16.12	2.96	2.84	7	37.126
Item 142	3.22	1.68	1.57	5	348.324
Item 780	7.19	3.60	0.99	3	391.529
Item 698	18.65	3.61	1.17	11	273.931
Item 665	30.28	4.42	1.02	10	64.362
Item 185	10.25	3.21	0.40	18	213.168
Item 482	2.08	1.56	1.08	10	185.628
Item 467	26.83	4.08	0.72	14	245.274
Item 233	35.90	1.20	0.85	15	245.215
Item 155	39.37	4.31	1.25	20	205.428
Item 191	37.33	4.86	1.59	4	152.654
Item 766	32.09	0.32	2.49	15	145.551
Item 672	5.29	0.77	0.38	9	139.422
Item 694	16.11	3.09	0.89	19	397.715
Item 163	14.12	1.12	2.95	14	135.534
Item 416	6.24	4.44	1.42	24	164.000
Item 251	30.73	5.80	3.00	3	250.821
Item 306	32.22	2.69	1.00	1	314.193
Item 509	7.04	1.84	0.85	15	103.302
Item 542	9.05	3.47	1.66	13	30.967
Item 664	38.20	5.06	0.38	12	134.475
Item 542	36.68	4.23	1.24	23	164.110
Item 187	2.02	3.26	2.39	2	20.342
Item 334	13.77	5.25	1.92	2	258.582

Annexure IV — Measurement Sheets — Bore Log Summary, page 24

Depth (m)	Strata	N value	Sample
5.3	Clayey silt	60	UDS
16.6	Silty clay	59	UDS
21.5	Silty clay	16	UDS
22.6	Medium sand	18	SPT
30.0	Clayey silt	57	DS
10.5	Silty clay	18	DS
19.1	Gravelly sand	35	SPT
22.8	Coarse sand	28	SPT
6.7	Coarse sand	42	DS
18.3	Weathered rock	7	UDS
18.2	Gravelly sand	33	SPT
12.7	Clayey silt	35	UDS
13.3	Silty clay	43	DS
2.2	Weathered rock	55	UDS
27.9	Clayey silt	25	UDS
0.9	Weathered rock	46	SPT
25.7	Silty clay	34	DS
6.5	Medium sand	57	DS
10.0	Silty clay	8	DS
27.9	Silty clay	47	DS
14.5	Medium sand	49	SPT
15.3	Clayey silt	46	UDS
0.8	Medium sand	12	SPT
4.6	Clayey silt	6	SPT
28.4	Clayey silt	13	SPT
21.5	Coarse sand	53	SPT

Annexure IV — Measurement Sheets — Traffic Volume Count, page 25

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
22:00	203	47	21	63	81	862
02:00	104	138	9	79	40	277
04:00	229	152	29	2	109	628
11:00	370	127	5	25	54	710
17:00	244	49	2	38	21	1369
17:00	323	146	2	62	61	1337
12:00	33	200	2	65	68	878
13:00	76	78	22	12	31	715
05:00	189	115	20	79	37	319
21:00	189	104	24	7	46	403
18:00	203	110	0	57	32	300
08:00	363	115	0	71	102	976
09:00	59	225	23	58	42	1188
09:00	379	117	34	11	64	958
06:00	297	88	6	58	17	577
15:00	309	81	12	23	18	1293
02:00	366	258	13	79	35	616
08:00	65	228	10	90	64	422
08:00	117	162	20	77	87	840
06:00	186	31	7	37	37	1121
01:00	118	85	40	19	85	383
22:00	298	256	8	20	50	1218

Annexure IV — Measurement Sheets — Lead Statement, page 26

Material	Source	Lead (km)	Rate (Rs)
Boulder	River bed	72.9	3,771
Coarse aggregate	River bed	68.1	2,968
Steel	Quarry B	35.0	9,155
Steel	Depot	81.3	929
Murram	Plant	64.7	7,401
Murram	Quarry A	43.5	8,160
Fine sand	River bed	54.0	6,394
Fine sand	Quarry B	22.3	7,262
Murram	River bed	24.9	8,871
Bitumen	River bed	40.8	1,736
Bitumen	River bed	39.4	6,360
Fine sand	Plant	30.9	791
Boulder	Quarry A	44.0	8,446
Steel	Plant	35.1	3,187
Fine sand	River bed	25.1	8,937
Coarse aggregate	Depot	76.0	9,123
Steel	Quarry A	77.5	2,431
Murram	River bed	56.9	5,127
Murram	Plant	30.0	2,157
Coarse aggregate	Quarry A	67.9	2,004
Bitumen	River bed	8.4	9,111
Bitumen	Quarry B	83.2	3,925
Boulder	Depot	60.1	4,764
Coarse aggregate	Plant	50.1	8,601
Coarse aggregate	Plant	29.7	2,606

Annexure IV — Measurement Sheets — Land Schedule, page 27

Survey No.	Village	Area (ha)	Class	Status
136/5	Golakganj	1.867	Agricultural	Notified
150/8	Dhubri	1.006	Waste	Awarded
178/4	Gauripur	0.558	Homestead	Under award
449/7	Gauripur	0.960	Agricultural	Awarded
378/4	Dhubri	1.351	Homestead	Notified
476/4	Agomani	0.323	Agricultural	Awarded
470/2	Gauripur	1.467	Homestead	Under award
375/3	Gauripur	2.302	Agricultural	Under award
424/7	Agomani	0.545	Waste	Under award
75/4	Dhubri	1.001	Waste	Awarded
415/7	Golakganj	0.155	Agricultural	Awarded
198/3	Dhubri	1.576	Homestead	Notified
156/1	Dhubri	0.584	Agricultural	Awarded
29/1	Dhubri	0.392	Waste	Notified
59/6	Agomani	0.161	Homestead	Awarded
475/4	Dhubri	1.654	Agricultural	Awarded
273/8	Golakganj	2.346	Waste	Awarded
268/6	Gauripur	1.503	Agricultural	Notified
61/1	Golakganj	0.123	Homestead	Under award
34/9	Golakganj	2.283	Agricultural	Notified
379/9	Agomani	1.707	Waste	Under award
377/3	Gauripur	2.138	Agricultural	Notified
234/3	Dhubri	0.847	Agricultural	Notified
93/1	Agomani	0.667	Waste	Under award
14/9	Dhubri	1.321	Homestead	Awarded
261/5	Dhubri	1.212	Agricultural	Notified
330/2	Golakganj	0.651	Homestead	Under award
102/2	Agomani	1.924	Agricultural	Notified
459/8	Gauripur	2.047	Waste	Awarded

Annexure IV — Measurement Sheets — Detailed Measurement Sheet, page 28

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 832	14.33	0.83	1.29	4	158.738
Item 853	4.80	6.00	0.71	16	95.426
Item 113	22.98	3.39	2.77	2	204.442
Item 409	22.37	5.13	0.70	23	241.055
Item 899	29.02	0.96	2.81	3	271.856
Item 673	28.74	3.75	1.19	9	92.448
Item 424	9.55	4.70	1.29	9	21.734
Item 268	27.13	3.87	1.37	4	217.042
Item 517	11.43	1.94	0.57	7	75.501
Item 733	24.38	0.57	2.61	14	170.038
Item 220	5.00	0.47	2.53	14	152.559
Item 744	32.06	3.89	2.60	22	364.896
Item 430	30.08	5.02	1.00	12	47.732
Item 527	6.65	0.75	0.59	12	19.884
Item 847	31.31	3.65	2.18	14	117.458
Item 891	35.90	3.59	0.27	5	372.032
Item 499	32.41	2.52	1.46	12	172.233
Item 463	32.99	5.62	0.93	12	359.743
Item 523	3.37	3.29	2.08	12	332.652
Item 196	13.54	5.38	1.62	11	25.947
Item 122	37.53	3.93	0.23	21	77.037
Item 693	6.34	4.23	2.05	23	194.507
Item 495	14.44	3.81	1.03	3	254.300
Item 623	15.46	4.89	2.52	15	2.496
Item 209	23.87	1.29	1.46	4	280.593
Item 867	19.17	4.37	1.59	9	384.913
Item 854	10.78	3.51	1.33	23	302.309
Item 673	37.56	0.44	1.18	14	127.357
Item 332	25.95	5.43	2.82	23	394.009

Annexure IV — Measurement Sheets — Bore Log Summary, page 29

Depth (m)	Strata	N value	Sample
13.5	Medium sand	9	DS
5.7	Weathered rock	27	UDS
27.2	Gravelly sand	50	DS
14.9	Clayey silt	28	DS
22.2	Weathered rock	34	DS
8.0	Gravelly sand	55	DS
15.0	Weathered rock	57	SPT
10.5	Gravelly sand	12	UDS
26.3	Silty clay	5	SPT
27.7	Silty clay	45	DS
2.8	Silty clay	58	DS
20.4	Coarse sand	38	UDS
14.6	Medium sand	52	UDS
11.0	Clayey silt	47	DS
12.1	Coarse sand	34	SPT
14.7	Weathered rock	43	DS
15.5	Coarse sand	45	DS
17.9	Coarse sand	13	DS
14.4	Silty clay	30	UDS
24.9	Silty clay	5	UDS
23.9	Clayey silt	11	UDS
14.9	Medium sand	35	UDS
11.9	Weathered rock	43	DS
10.3	Silty clay	12	UDS

Annexure IV — Measurement Sheets — Traffic Volume Count, page 30

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
07:00	255	179	24	69	116	1264
05:00	95	58	33	73	114	1294
03:00	56	75	24	29	95	726
07:00	332	239	27	79	79	1367
04:00	82	108	16	67	111	590
00:00	85	42	33	19	98	191
19:00	380	106	9	25	47	1206
13:00	355	152	17	80	52	1137
15:00	121	195	22	6	10	836
10:00	201	59	11	42	120	535
18:00	178	115	24	83	113	445
00:00	319	35	16	17	112	259
17:00	313	230	29	67	24	668
13:00	339	66	10	3	38	1265
01:00	277	76	6	88	102	1067
16:00	24	25	13	8	50	365
03:00	165	93	17	83	39	1281
17:00	72	149	8	66	39	1023
16:00	109	190	22	49	47	129
05:00	147	196	40	62	44	203
05:00	248	122	29	83	111	891
02:00	330	95	28	20	113	189
08:00	241	96	7	70	57	283
12:00	339	28	9	50	32	1367
19:00	322	208	34	19	43	1176
06:00	113	117	3	9	103	1066

ANNEXURE V — GEO-TECHNICAL INVESTIGATION REPORT

Bore logs, laboratory test results and foundation recommendations for eighteen boreholes.

Annexure V — Bore Logs — Land Schedule, page 1

Survey No.	Village	Area (ha)	Class	Status
439/3	Gauripur	2.396	Agricultural	Awarded
353/2	Gauripur	1.945	Homestead	Notified
402/1	Golakganj	1.149	Waste	Notified
69/1	Gauripur	2.302	Waste	Under award
359/5	Golakganj	2.290	Agricultural	Notified
399/5	Gauripur	1.651	Homestead	Awarded
287/6	Dhubri	2.223	Homestead	Under award
468/2	Agomani	0.278	Waste	Notified
419/3	Golakganj	2.070	Homestead	Notified
390/5	Gauripur	0.270	Homestead	Awarded
344/3	Agomani	1.177	Waste	Notified
411/3	Golakganj	1.238	Waste	Awarded
282/4	Dhubri	1.533	Homestead	Under award
339/9	Gauripur	0.969	Homestead	Under award
44/8	Agomani	2.004	Homestead	Awarded
390/7	Agomani	2.104	Agricultural	Under award
27/1	Golakganj	1.702	Homestead	Under award
412/2	Dhubri	0.242	Homestead	Notified
219/7	Golakganj	1.404	Homestead	Under award
439/4	Agomani	0.762	Homestead	Awarded
182/2	Dhubri	2.190	Waste	Awarded
364/7	Golakganj	1.537	Homestead	Under award
240/7	Agomani	0.799	Waste	Awarded
213/1	Golakganj	1.137	Agricultural	Awarded
411/4	Gauripur	1.294	Waste	Awarded
238/3	Gauripur	1.989	Agricultural	Awarded
36/4	Dhubri	1.358	Waste	Under award
35/3	Agomani	1.915	Agricultural	Awarded

Annexure V — Bore Logs — Detailed Measurement Sheet, page 2

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 244	31.61	4.66	1.87	17	199.078
Item 820	13.63	4.21	2.77	15	259.695
Item 520	36.87	4.06	2.27	5	137.159
Item 295	22.18	1.34	1.43	11	113.573
Item 337	26.37	1.07	0.48	24	255.079
Item 719	37.94	1.65	0.96	3	75.953
Item 485	39.26	3.51	1.53	11	396.830
Item 566	28.19	1.19	0.33	6	68.344
Item 294	29.85	1.59	0.86	10	190.817
Item 736	30.44	1.66	1.12	8	117.846
Item 585	22.97	5.66	0.88	21	239.067
Item 400	37.42	2.38	0.84	17	157.693
Item 176	5.24	2.83	2.79	12	384.396
Item 582	23.45	1.66	0.50	17	13.179
Item 890	35.13	2.02	2.40	3	233.713
Item 468	39.23	5.49	2.33	20	86.266
Item 244	38.03	3.98	1.66	7	207.221
Item 725	25.94	2.41	0.98	15	115.728
Item 267	12.82	3.60	1.27	1	356.734
Item 177	11.18	5.51	0.89	16	381.437
Item 884	20.47	5.48	2.24	20	140.726
Item 198	25.00	4.01	1.16	24	72.527
Item 718	39.38	1.60	1.82	11	70.743

Annexure V — Bore Logs — Bore Log Summary, page 3

Depth (m)	Strata	N value	Sample
7.3	Coarse sand	34	SPT
17.0	Weathered rock	38	SPT
9.6	Weathered rock	57	DS
23.8	Gravelly sand	19	DS
5.3	Silty clay	21	DS
4.6	Silty clay	36	DS
16.1	Clayey silt	19	DS
27.7	Weathered rock	53	DS
27.2	Clayey silt	16	UDS
0.5	Clayey silt	59	SPT
18.8	Medium sand	46	UDS
10.6	Coarse sand	31	UDS
19.5	Silty clay	18	SPT
27.8	Clayey silt	22	DS
6.4	Medium sand	17	SPT
13.3	Clayey silt	40	DS
23.6	Medium sand	50	DS
3.9	Silty clay	50	UDS
4.2	Gravelly sand	40	SPT
4.7	Silty clay	26	DS
6.0	Clayey silt	22	DS
4.5	Coarse sand	8	UDS
25.5	Weathered rock	54	SPT
4.8	Coarse sand	50	UDS
26.4	Medium sand	29	UDS
25.4	Clayey silt	54	SPT
26.6	Gravelly sand	6	UDS
18.8	Coarse sand	30	SPT

Annexure V — Bore Logs — Traffic Volume Count, page 4

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
15:00	70	82	23	76	77	968
12:00	219	237	10	18	38	274
15:00	24	219	9	2	73	986
20:00	36	75	4	85	23	1300
09:00	224	53	23	15	73	1294
08:00	81	175	22	65	41	599
21:00	348	155	34	45	84	766
04:00	363	260	35	30	16	375
23:00	137	260	26	49	52	1261
03:00	238	75	25	6	39	895
06:00	213	39	17	16	24	1025
20:00	186	50	40	72	3	527
23:00	107	160	20	82	10	424
03:00	305	235	35	75	29	546
06:00	389	99	32	72	56	577
20:00	246	205	15	61	110	681
21:00	318	61	34	51	116	81
01:00	361	93	8	16	24	899
05:00	127	82	7	38	44	1046
08:00	183	30	8	42	7	920
14:00	283	120	29	90	96	723
04:00	305	238	9	37	5	872
02:00	221	219	24	54	13	407
07:00	327	16	16	35	115	444
03:00	368	49	4	16	37	126
06:00	60	134	38	10	114	1375
07:00	372	195	26	20	80	630
13:00	161	104	7	14	6	384
20:00	211	82	10	59	61	377
20:00	79	253	23	33	4	753

Annexure V — Bore Logs — Lead Statement, page 5

Material	Source	Lead (km)	Rate (Rs)
Cement	River bed	69.7	7,813
Fine sand	Plant	43.3	1,743
Cement	Quarry A	4.1	4,443
Coarse aggregate	Depot	63.0	3,221
Boulder	Quarry B	59.5	1,706
Fine sand	Plant	72.1	608
Fine sand	Plant	69.6	7,518
Murram	Quarry A	53.9	6,267
Boulder	Plant	5.0	2,686
Coarse aggregate	Quarry A	65.5	1,238
Bitumen	Quarry A	4.0	6,746
Cement	River bed	87.0	1,999
Cement	Quarry B	76.0	573
Coarse aggregate	Quarry B	39.2	1,060
Cement	River bed	54.2	5,500
Boulder	Plant	14.9	3,099
Steel	Quarry B	70.3	3,370
Fine sand	River bed	36.0	4,979
Coarse aggregate	Depot	60.6	6,749
Bitumen	Depot	33.1	6,415
Bitumen	Depot	83.1	2,325
Cement	River bed	43.0	3,018
Cement	Quarry B	15.8	2,135
Bitumen	Plant	58.9	5,009

Annexure V — Bore Logs — Land Schedule, page 6

Survey No.	Village	Area (ha)	Class	Status
44/9	Golakganj	0.839	Agricultural	Under award
69/7	Dhubri	1.583	Agricultural	Under award
416/7	Golakganj	0.390	Agricultural	Under award
169/9	Agomani	2.152	Homestead	Awarded
161/9	Gauripur	0.141	Homestead	Under award
30/2	Gauripur	2.249	Homestead	Under award
299/4	Gauripur	2.146	Homestead	Awarded
340/6	Agomani	1.746	Homestead	Notified
90/1	Agomani	1.972	Waste	Awarded
270/5	Agomani	1.650	Homestead	Notified
215/5	Dhubri	1.561	Homestead	Awarded
131/5	Agomani	1.408	Waste	Under award
294/2	Golakganj	1.824	Waste	Under award
433/7	Gauripur	0.266	Agricultural	Notified
72/7	Agomani	0.596	Agricultural	Awarded
370/4	Agomani	2.356	Waste	Awarded
15/3	Golakganj	1.029	Waste	Notified
228/2	Agomani	1.107	Waste	Notified
239/5	Dhubri	1.493	Agricultural	Under award
360/2	Agomani	0.567	Homestead	Notified
156/9	Agomani	0.756	Waste	Awarded
162/8	Dhubri	0.348	Agricultural	Under award
331/1	Dhubri	1.594	Agricultural	Under award
236/3	Gauripur	2.254	Homestead	Notified
24/1	Golakganj	1.210	Waste	Notified
405/8	Agomani	0.375	Homestead	Awarded
69/6	Golakganj	1.849	Homestead	Awarded

Annexure V — Bore Logs — Detailed Measurement Sheet, page 7

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 607	33.40	4.17	2.17	4	343.704
Item 810	29.96	2.25	2.65	23	104.370
Item 227	34.09	5.99	1.02	1	127.423
Item 820	26.93	2.12	2.77	22	215.128
Item 251	20.85	5.09	2.18	10	379.677
Item 272	3.07	5.46	1.98	22	200.800
Item 402	25.81	3.74	1.00	16	224.646
Item 405	35.42	4.73	0.71	13	165.618
Item 600	9.53	0.52	0.38	24	237.823
Item 614	27.60	3.10	0.92	22	145.423
Item 324	9.34	1.85	1.60	8	290.747
Item 547	15.97	5.67	0.63	16	67.455
Item 143	28.21	3.47	2.99	9	359.198
Item 399	15.31	3.22	2.77	1	379.692
Item 445	22.89	3.48	0.51	1	137.760
Item 150	32.74	0.56	1.56	12	320.675
Item 815	19.19	1.65	1.50	8	87.594
Item 855	31.59	4.42	1.47	3	92.266
Item 576	8.81	0.71	0.97	12	34.952
Item 289	29.51	4.92	1.26	16	144.400
Item 655	39.53	4.64	0.74	18	285.274
Item 350	36.34	5.27	1.50	16	314.271
Item 636	15.57	5.01	0.66	22	152.864
Item 510	19.67	1.87	0.95	12	57.953
Item 606	19.78	2.19	2.91	19	22.811
Item 616	31.29	1.97	1.85	17	227.736
Item 856	11.99	0.99	1.35	24	173.414
Item 216	33.30	1.81	1.01	3	55.421
Item 128	21.92	3.87	2.58	8	103.689
Item 567	39.33	5.75	2.46	5	166.369

Annexure V — Bore Logs — Bore Log Summary, page 8

Depth (m)	Strata	N value	Sample
14.7	Coarse sand	15	UDS
19.9	Gravelly sand	45	SPT
0.9	Coarse sand	30	DS
8.6	Weathered rock	5	UDS
17.4	Gravelly sand	48	UDS
23.6	Clayey silt	43	UDS
10.7	Coarse sand	5	DS
25.2	Clayey silt	46	SPT
15.3	Coarse sand	24	SPT
20.8	Silty clay	20	DS
17.1	Clayey silt	35	UDS
26.4	Weathered rock	34	UDS
19.5	Weathered rock	20	SPT
19.9	Gravelly sand	20	SPT
6.1	Silty clay	19	SPT
28.3	Gravelly sand	10	UDS
26.3	Medium sand	28	DS
13.6	Silty clay	7	SPT
24.5	Medium sand	52	SPT
17.6	Gravelly sand	16	SPT
1.3	Medium sand	24	SPT
5.8	Weathered rock	58	SPT
12.4	Coarse sand	50	UDS
20.1	Coarse sand	39	UDS
7.7	Medium sand	60	SPT
25.5	Clayey silt	50	SPT
9.0	Clayey silt	50	UDS
4.0	Gravelly sand	15	SPT
7.2	Silty clay	18	SPT

Annexure V — Bore Logs — Traffic Volume Count, page 9

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
17:00	107	122	5	52	39	245
23:00	169	141	23	38	106	1129
13:00	206	200	8	50	29	1314
22:00	370	80	32	7	46	414
02:00	151	141	30	65	75	569
12:00	239	31	38	3	13	459
05:00	63	257	36	65	7	976
14:00	202	32	22	73	57	964
07:00	377	152	34	53	88	266
07:00	250	85	39	90	77	1220
06:00	83	86	40	55	84	262
11:00	164	259	32	75	73	1141
02:00	321	31	10	17	62	143
15:00	213	117	0	56	3	832
21:00	388	78	25	16	82	274
08:00	330	68	19	24	107	378
17:00	94	104	40	71	80	1023
12:00	91	187	24	23	29	1257
09:00	321	21	19	10	66	335
19:00	24	243	30	20	104	791
16:00	275	44	19	46	75	655
08:00	148	214	38	77	64	1252
09:00	78	117	8	16	95	673
13:00	40	112	4	34	2	497
06:00	93	162	37	20	65	379
04:00	229	170	26	57	10	341
13:00	80	43	16	86	115	1008
11:00	109	201	9	56	22	1373
20:00	44	45	21	1	10	989

Annexure V — Bore Logs — Lead Statement, page 10

Material	Source	Lead (km)	Rate (Rs)
Murram	Plant	50.6	8,399
Cement	Quarry A	66.4	6,722
Boulder	Depot	74.0	2,774
Boulder	River bed	52.8	6,747
Bitumen	Plant	69.8	6,385
Fine sand	River bed	79.6	4,226
Murram	Depot	73.4	4,730
Steel	River bed	7.3	2,226
Steel	Quarry B	46.5	8,802
Cement	Depot	75.7	5,338
Boulder	Depot	76.3	6,385
Cement	Quarry B	69.7	1,133
Coarse aggregate	Plant	30.6	6,381
Coarse aggregate	Depot	47.8	1,164
Fine sand	River bed	48.1	7,855
Steel	Plant	9.6	6,968
Coarse aggregate	Quarry A	24.7	4,202
Coarse aggregate	Depot	14.2	7,847
Fine sand	Quarry B	32.2	256
Bitumen	Quarry B	38.3	4,491
Cement	Quarry A	57.6	3,122
Cement	Quarry B	20.6	356
Steel	Quarry A	52.1	3,042
Boulder	Quarry A	41.6	5,916
Murram	Quarry A	35.2	1,172

Annexure V — Bore Logs — Land Schedule, page 11

Survey No.	Village	Area (ha)	Class	Status
114/3	Dhubri	1.582	Waste	Awarded
302/6	Gauripur	1.381	Homestead	Notified
313/9	Golakganj	1.736	Homestead	Awarded
202/1	Golakganj	0.247	Agricultural	Under award
321/1	Dhubri	0.304	Waste	Notified
297/8	Gauripur	1.266	Waste	Notified
460/6	Agomani	1.924	Homestead	Under award
236/2	Golakganj	0.571	Agricultural	Awarded
476/1	Agomani	2.358	Waste	Notified
270/1	Gauripur	1.767	Homestead	Awarded
262/4	Agomani	2.130	Waste	Notified
463/6	Golakganj	1.599	Agricultural	Awarded
265/9	Golakganj	1.785	Waste	Notified
416/1	Golakganj	0.802	Waste	Under award
154/6	Gauripur	0.176	Homestead	Awarded
282/6	Dhubri	1.989	Homestead	Notified
270/6	Gauripur	0.514	Waste	Notified
423/3	Dhubri	2.023	Agricultural	Notified
203/5	Dhubri	0.339	Waste	Notified
10/7	Dhubri	2.181	Homestead	Under award
395/8	Golakganj	1.342	Homestead	Awarded
41/6	Agomani	0.882	Homestead	Under award
138/1	Gauripur	2.364	Waste	Awarded
128/3	Agomani	1.961	Agricultural	Awarded
373/8	Golakganj	0.319	Waste	Awarded
262/5	Dhubri	0.815	Homestead	Notified
115/6	Golakganj	0.546	Homestead	Under award
248/6	Agomani	2.082	Agricultural	Notified
349/3	Golakganj	2.314	Waste	Under award
368/5	Golakganj	0.218	Agricultural	Awarded

Annexure V — Bore Logs — Detailed Measurement Sheet, page 12

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 482	37.09	5.13	0.46	3	74.740
Item 117	14.91	1.21	0.64	2	154.184
Item 562	26.16	3.21	1.19	15	360.322
Item 303	22.22	4.52	1.84	5	158.321
Item 626	28.85	5.49	2.61	10	44.120
Item 605	3.37	1.91	2.36	14	265.500
Item 228	25.14	2.00	1.79	4	152.519
Item 872	38.98	5.79	2.76	10	6.655
Item 417	24.92	0.65	0.35	15	211.625
Item 135	30.91	3.25	1.39	23	307.011
Item 396	9.90	4.90	1.32	7	247.667
Item 767	18.90	5.80	2.37	18	184.770
Item 537	36.61	5.70	0.92	24	226.594
Item 810	39.18	3.81	0.85	12	13.179
Item 542	24.17	4.45	1.28	6	250.395
Item 354	33.20	5.71	1.30	10	183.989
Item 847	34.48	2.10	1.02	6	42.387
Item 814	35.80	3.25	1.01	2	287.270
Item 198	30.33	0.94	1.96	6	5.795
Item 724	12.65	2.62	0.96	21	211.085
Item 472	34.05	3.98	0.26	20	382.656
Item 754	4.56	2.15	1.01	19	172.490
Item 896	34.62	3.51	0.85	15	176.615
Item 387	4.88	2.19	2.20	15	162.579
Item 890	35.01	1.94	2.89	2	263.676
Item 524	38.96	0.91	1.32	13	204.631
Item 467	17.50	2.99	1.16	23	39.796

Annexure V — Bore Logs — Bore Log Summary, page 13

Depth (m)	Strata	N value	Sample
28.0	Coarse sand	18	DS
21.5	Medium sand	38	DS
10.8	Gravelly sand	5	UDS
6.9	Weathered rock	48	UDS
13.0	Silty clay	60	DS
8.7	Clayey silt	55	DS
18.7	Coarse sand	44	UDS
22.4	Clayey silt	33	SPT
11.7	Medium sand	57	DS
7.8	Weathered rock	30	SPT
26.4	Coarse sand	60	UDS
5.5	Silty clay	5	UDS
14.9	Medium sand	50	DS
28.6	Coarse sand	41	UDS
17.7	Medium sand	11	DS
11.1	Gravelly sand	12	DS
26.2	Silty clay	16	UDS
10.9	Gravelly sand	25	DS
3.2	Gravelly sand	57	SPT
12.7	Silty clay	39	SPT
25.7	Silty clay	35	SPT
16.3	Clayey silt	29	UDS
17.7	Medium sand	14	UDS
11.6	Silty clay	39	DS
14.8	Clayey silt	29	DS
16.3	Clayey silt	34	DS
29.2	Weathered rock	26	UDS
24.5	Coarse sand	31	SPT

Annexure V — Bore Logs — Traffic Volume Count, page 14

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
00:00	223	112	31	69	32	1243
15:00	295	252	39	3	14	399
11:00	52	156	4	62	111	1396
09:00	271	58	11	7	70	907
15:00	55	225	30	70	89	556
03:00	260	33	6	90	64	914
04:00	152	139	5	68	73	989
01:00	228	72	21	32	17	538
18:00	234	142	38	46	110	298
03:00	395	199	6	53	13	1153
18:00	60	122	21	81	99	600
04:00	372	198	29	30	82	180
18:00	191	127	21	55	82	899
05:00	196	54	22	43	114	582
07:00	304	239	35	52	47	284
17:00	388	137	2	21	99	1263
09:00	132	231	16	43	68	205
09:00	136	215	20	87	29	1189
17:00	359	152	5	62	80	974
07:00	120	156	33	17	74	819
16:00	29	177	13	82	68	945
12:00	38	143	34	63	104	919
02:00	321	234	18	54	39	1200
19:00	274	85	29	20	110	784
04:00	45	131	2	16	86	650
03:00	327	48	26	60	81	115
17:00	308	244	28	37	22	1175
19:00	240	215	20	31	104	1070
11:00	121	215	20	85	93	1346

Annexure V — Bore Logs — Lead Statement, page 15

Material	Source	Lead (km)	Rate (Rs)
Boulder	Depot	54.1	4,422
Steel	Quarry A	5.9	1,585
Coarse aggregate	Depot	8.9	4,819
Murram	Plant	62.1	2,478
Cement	Quarry B	60.6	7,035
Boulder	River bed	83.3	2,940
Bitumen	River bed	28.4	7,857
Cement	Quarry B	69.2	4,833
Murram	Plant	44.0	8,118
Murram	Depot	6.0	4,868
Bitumen	Depot	81.3	8,941
Bitumen	Depot	24.7	2,428
Steel	Plant	75.0	2,987
Murram	Quarry B	9.2	7,818
Murram	Plant	21.4	8,275
Fine sand	Depot	7.8	8,265
Steel	Quarry A	79.2	629
Bitumen	River bed	31.6	4,360
Fine sand	Plant	42.2	665
Coarse aggregate	Plant	52.4	2,413
Coarse aggregate	Quarry B	79.7	4,112
Bitumen	Plant	68.7	8,449
Murram	Depot	24.6	4,900
Fine sand	Depot	84.3	5,005
Murram	Depot	53.9	3,625
Cement	Plant	58.5	6,889
Steel	River bed	67.7	1,581
Coarse aggregate	Quarry B	13.2	6,313
Cement	Plant	30.5	4,725
Boulder	River bed	59.7	6,863

Annexure V — Bore Logs — Land Schedule, page 16

Survey No.	Village	Area (ha)	Class	Status
62/2	Golakganj	2.293	Agricultural	Under award
11/1	Gauripur	2.118	Waste	Notified
14/4	Gauripur	2.172	Homestead	Notified
148/1	Agomani	0.689	Agricultural	Notified
41/1	Dhubri	0.808	Homestead	Under award
74/7	Gauripur	2.221	Agricultural	Notified
120/8	Golakganj	2.047	Agricultural	Awarded
198/7	Golakganj	1.530	Agricultural	Notified
468/2	Agomani	1.106	Agricultural	Notified
271/9	Gauripur	0.749	Homestead	Notified
382/6	Golakganj	0.626	Waste	Awarded
164/8	Agomani	1.804	Waste	Notified
326/3	Agomani	1.511	Homestead	Under award
47/8	Golakganj	2.008	Homestead	Under award
158/6	Dhubri	1.060	Waste	Notified
12/2	Golakganj	1.044	Agricultural	Awarded
406/4	Golakganj	0.145	Homestead	Awarded
216/2	Golakganj	0.774	Waste	Under award
44/2	Agomani	1.552	Homestead	Notified
60/4	Dhubri	1.349	Waste	Awarded
267/3	Dhubri	0.613	Agricultural	Awarded
277/8	Gauripur	0.888	Agricultural	Under award
178/2	Agomani	2.121	Waste	Awarded
324/9	Gauripur	1.309	Homestead	Awarded

Annexure V — Bore Logs — Detailed Measurement Sheet, page 17

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 283	13.69	4.62	0.85	3	161.380
Item 158	18.27	1.62	0.35	13	221.189
Item 382	29.63	5.88	1.14	7	196.107
Item 215	22.52	1.09	1.80	19	226.818
Item 207	6.56	5.57	2.38	9	73.280
Item 284	4.59	2.75	1.76	9	1.647
Item 708	21.24	4.08	0.36	16	237.320
Item 668	10.59	2.89	2.32	1	197.923
Item 491	17.84	3.69	1.59	4	162.784
Item 461	7.91	4.53	0.92	8	152.021
Item 853	19.58	4.90	2.06	23	49.740
Item 367	11.46	0.47	0.62	5	337.022
Item 557	3.12	3.24	2.85	3	333.389
Item 613	2.99	5.51	2.22	23	103.887
Item 766	25.10	0.61	1.19	16	71.293
Item 306	29.95	5.22	2.95	20	313.190
Item 652	4.36	5.26	2.36	16	311.053
Item 455	21.61	2.20	0.96	23	43.462
Item 701	15.46	5.50	1.39	16	194.160
Item 840	39.02	3.67	2.85	7	36.556
Item 576	36.83	3.41	2.18	6	383.271
Item 301	20.13	1.88	2.04	16	232.369
Item 691	21.46	0.74	1.10	4	246.878

Annexure V — Bore Logs — Bore Log Summary, page 18

Depth (m)	Strata	N value	Sample
20.0	Coarse sand	21	SPT
14.1	Coarse sand	59	SPT
19.9	Gravelly sand	23	DS
26.7	Clayey silt	57	DS
16.0	Medium sand	5	DS
22.6	Silty clay	36	UDS
6.3	Medium sand	46	SPT
28.4	Coarse sand	56	SPT
28.7	Clayey silt	50	SPT
9.3	Coarse sand	17	DS
2.8	Coarse sand	31	SPT
1.3	Medium sand	34	DS
5.3	Gravelly sand	19	DS
3.5	Coarse sand	32	SPT
28.8	Clayey silt	9	UDS
20.0	Silty clay	52	SPT
22.8	Silty clay	12	DS
20.8	Medium sand	45	SPT
16.8	Silty clay	60	DS
7.7	Silty clay	56	UDS
10.2	Gravelly sand	49	DS
11.0	Clayey silt	35	UDS
9.0	Medium sand	4	DS
4.8	Coarse sand	38	SPT
23.0	Medium sand	20	UDS
12.2	Clayey silt	12	SPT
6.0	Clayey silt	29	SPT
13.8	Medium sand	59	DS

Annexure V — Bore Logs — Traffic Volume Count, page 19

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
02:00	261	123	13	4	108	1375
21:00	238	167	30	49	41	934
04:00	146	84	5	73	87	1143
19:00	61	257	20	10	38	549
06:00	396	15	29	58	41	1250
23:00	70	48	15	70	92	1091
22:00	295	31	3	60	72	641
19:00	342	120	33	62	45	852
06:00	148	230	11	2	39	1326
18:00	63	32	28	18	115	281
21:00	341	241	19	74	108	358
11:00	251	61	36	12	28	508
05:00	250	72	22	58	59	1284
01:00	222	103	21	12	68	465
11:00	62	29	2	75	112	833
23:00	159	248	39	3	43	1287
12:00	319	128	35	2	86	662
02:00	232	161	31	80	32	833
05:00	110	179	29	84	69	559
16:00	104	227	40	52	38	501
06:00	125	14	11	47	22	382
13:00	171	218	22	56	114	417
07:00	221	152	18	55	44	499

Annexure V — Bore Logs — Lead Statement, page 20

Material	Source	Lead (km)	Rate (Rs)
Bitumen	River bed	69.5	6,891
Steel	Quarry A	65.4	9,019
Cement	Quarry A	78.0	4,348
Cement	Plant	46.8	1,192
Fine sand	Plant	17.8	6,919
Boulder	River bed	77.3	5,888
Boulder	Depot	77.4	8,193
Boulder	Quarry A	68.4	3,222
Bitumen	River bed	59.1	9,397
Steel	Quarry A	79.1	2,608
Steel	Plant	8.3	2,339
Murram	Depot	74.3	7,440
Cement	Plant	39.8	327
Steel	Depot	26.1	8,435
Murram	River bed	13.4	6,187
Steel	Quarry B	61.0	1,969
Fine sand	Quarry A	3.7	6,520
Steel	Plant	26.8	359
Cement	Quarry B	8.4	3,661
Fine sand	Quarry B	58.2	4,025
Cement	River bed	86.9	7,364
Murram	Quarry A	28.8	4,810
Bitumen	Plant	28.1	3,095
Bitumen	Plant	11.7	4,047
Fine sand	Plant	29.9	4,317
Boulder	Quarry B	69.7	1,463

Annexure V — Bore Logs — Land Schedule, page 21

Survey No.	Village	Area (ha)	Class	Status
169/4	Agomani	1.310	Agricultural	Under award
450/7	Agomani	1.392	Agricultural	Notified
195/5	Golakganj	0.273	Agricultural	Notified
340/6	Golakganj	2.128	Agricultural	Under award
351/6	Dhubri	1.587	Homestead	Awarded
115/9	Gauripur	2.261	Waste	Awarded
73/7	Golakganj	1.548	Waste	Awarded
28/5	Dhubri	0.909	Waste	Awarded
29/9	Agomani	1.045	Waste	Awarded
342/7	Agomani	2.233	Waste	Notified
148/8	Golakganj	0.671	Agricultural	Notified
161/2	Dhubri	1.483	Homestead	Awarded
252/2	Golakganj	0.763	Waste	Under award
339/7	Golakganj	1.168	Agricultural	Under award
25/4	Gauripur	0.881	Waste	Notified
132/2	Gauripur	1.976	Homestead	Awarded
94/2	Dhubri	0.793	Waste	Notified
428/2	Agomani	1.470	Homestead	Awarded
230/3	Agomani	0.636	Homestead	Under award
371/9	Agomani	2.194	Waste	Awarded
95/8	Golakganj	1.018	Agricultural	Notified
358/1	Agomani	0.674	Waste	Awarded
145/6	Golakganj	1.501	Waste	Under award
374/8	Agomani	1.909	Homestead	Notified
65/9	Golakganj	2.371	Homestead	Notified
36/9	Golakganj	2.011	Homestead	Awarded
76/8	Dhubri	0.269	Homestead	Notified
406/8	Dhubri	1.593	Homestead	Awarded

Annexure V — Bore Logs — Detailed Measurement Sheet, page 22

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 623	37.46	0.73	0.28	9	281.157
Item 625	24.99	3.61	1.37	24	118.679
Item 289	26.62	2.45	1.72	6	169.143
Item 391	37.05	3.00	0.75	10	256.437
Item 321	35.68	4.29	1.03	5	324.426
Item 679	3.77	1.34	1.63	11	264.414
Item 518	31.19	4.23	2.96	6	117.132
Item 165	20.31	4.64	0.21	1	195.241
Item 650	8.79	4.27	1.76	7	181.571
Item 386	32.13	5.63	1.09	7	386.817
Item 417	14.34	0.41	0.46	23	65.458
Item 455	37.64	0.92	2.76	5	154.242
Item 195	23.60	1.51	1.08	16	284.763
Item 843	10.77	1.79	1.75	8	10.274
Item 213	15.31	1.30	2.64	3	396.497
Item 366	34.70	2.82	0.43	16	53.270
Item 502	37.02	1.27	1.75	16	113.879
Item 442	34.84	0.30	2.97	6	206.183
Item 326	28.11	1.98	1.92	7	71.265
Item 500	12.32	4.38	2.29	15	224.411
Item 357	2.75	3.72	2.31	23	99.967
Item 252	32.64	5.92	0.21	21	38.927
Item 129	31.15	5.76	1.23	10	297.372
Item 608	34.61	1.60	1.61	5	88.341
Item 627	17.97	2.55	0.31	10	126.979
Item 685	11.17	5.70	0.84	9	134.697
Item 539	11.13	3.77	0.22	24	23.866

Annexure V — Bore Logs — Bore Log Summary, page 23

Depth (m)	Strata	N value	Sample
24.6	Coarse sand	34	DS
15.4	Coarse sand	46	DS
19.9	Gravelly sand	36	UDS
21.3	Coarse sand	46	DS
18.1	Gravelly sand	30	DS
12.5	Silty clay	22	DS
9.5	Coarse sand	24	DS
28.0	Weathered rock	28	UDS
22.3	Clayey silt	54	UDS
15.9	Coarse sand	25	DS
27.0	Coarse sand	22	UDS
5.6	Coarse sand	24	DS
25.4	Clayey silt	11	SPT
1.7	Weathered rock	21	SPT
15.3	Medium sand	30	UDS
12.4	Gravelly sand	35	DS
18.4	Gravelly sand	12	UDS
11.3	Medium sand	57	UDS
8.4	Clayey silt	4	UDS
26.7	Silty clay	35	UDS
24.4	Weathered rock	16	SPT
25.2	Medium sand	7	UDS
26.9	Coarse sand	11	DS
8.4	Clayey silt	18	SPT
10.4	Medium sand	34	DS
27.2	Clayey silt	23	SPT
8.1	Silty clay	18	UDS
9.2	Gravelly sand	55	DS
8.5	Weathered rock	22	SPT
13.4	Coarse sand	46	UDS

Annexure V — Bore Logs — Traffic Volume Count, page 24

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
11:00	378	107	1	23	113	1322
00:00	229	75	15	64	44	1175
21:00	340	45	2	7	45	940
22:00	239	30	23	1	102	770
06:00	345	82	2	61	90	340
23:00	23	196	30	4	6	1082
06:00	240	105	28	24	74	697
14:00	162	182	6	61	106	1398
05:00	318	61	17	52	77	931
11:00	73	82	2	31	66	1005
03:00	309	114	36	25	87	654
04:00	135	86	10	36	112	306
21:00	222	181	29	54	86	953
10:00	379	221	33	1	109	611
20:00	173	175	36	69	77	85
08:00	390	256	21	67	10	455
04:00	255	42	15	37	10	1241
07:00	317	126	11	81	34	427
14:00	366	165	25	79	55	868
00:00	311	133	3	57	8	137
06:00	189	161	11	27	115	1337
21:00	165	162	3	52	108	357
23:00	242	144	36	32	90	1244
20:00	315	125	10	38	45	173

Annexure V — Bore Logs — Lead Statement, page 25

Material	Source	Lead (km)	Rate (Rs)
Steel	Depot	18.4	2,460
Steel	Depot	74.9	5,622
Boulder	River bed	52.0	5,640
Bitumen	Quarry A	38.8	5,095
Cement	Quarry A	20.7	8,273
Steel	Plant	53.8	8,497
Fine sand	Quarry B	43.3	2,125
Coarse aggregate	Quarry A	42.7	1,316
Fine sand	River bed	45.7	4,556
Bitumen	Depot	3.2	464
Coarse aggregate	Depot	76.8	4,426
Cement	Depot	52.6	7,967
Boulder	Quarry B	80.1	1,253
Murram	Plant	32.1	3,974
Bitumen	Plant	2.1	3,719
Murram	Quarry A	89.2	2,480
Cement	Depot	72.3	4,989
Steel	Depot	14.2	9,067
Fine sand	Quarry B	35.3	4,453
Boulder	Quarry A	44.3	861
Steel	Quarry A	9.3	7,243
Fine sand	Depot	35.5	3,689
Fine sand	Plant	59.8	290
Bitumen	Quarry B	23.4	7,075
Coarse aggregate	Depot	15.5	1,998
Murram	Quarry B	83.2	7,920
Coarse aggregate	Quarry A	40.6	6,164
Steel	Plant	64.7	264
Steel	Quarry B	80.9	8,810

ANNEXURE VI — HYDRAULIC INVESTIGATION REPORT

Discharge estimation, HFL determination, afflux computation, scour depth and waterway adequacy.

Annexure VI — Hydraulic Computations — Sheet 1

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. LA-143, revision B.

Annexure VI — Hydraulic Computations — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 678	18.12	3.65	0.55	22	291.863
Item 301	21.47	4.83	1.95	6	219.965
Item 626	19.84	0.89	2.92	21	271.545
Item 714	24.15	2.66	0.73	6	247.837
Item 174	4.67	3.51	1.64	5	292.023
Item 141	22.60	0.74	1.83	14	344.813
Item 557	18.55	3.12	1.61	16	177.473
Item 606	12.75	0.57	1.34	13	251.902
Item 568	2.70	5.64	1.40	19	140.832
Item 165	33.90	1.01	0.86	6	352.996
Item 873	7.13	5.57	2.68	12	85.077
Item 746	23.71	3.83	1.71	15	148.341
Item 471	20.80	2.91	2.11	15	151.088
Item 614	39.47	0.93	2.32	14	127.877
Item 128	28.91	3.53	1.91	8	229.309
Item 280	30.37	1.78	0.51	9	236.799
Item 133	4.36	4.93	2.20	2	109.124
Item 278	25.28	0.75	1.34	20	280.690
Item 258	22.66	1.02	2.90	19	38.055
Item 677	10.24	3.45	2.61	6	30.106

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Annexure VI — Hydraulic Computations — Sheet 3

Design Computation

Shear = $64.477 \times 6.243 = 402.537$ mm (permissible 448.177) — safe

Settlement = $11.922 \times 0.732 = 8.730$ mm (permissible 15.058) — safe

Shear = $23.568 \times 1.037 = 24.439$ mm (permissible 43.511) — revise section

Moment = $28.349 \times 11.047 = 313.158$ kN (permissible 419.792) — safe

Deflection = $6.070 \times 6.061 = 36.794$ kNm (permissible 47.973) — safe

Deflection = $82.276 \times 9.415 = 774.639$ kPa (permissible 859.543) — revise section

Settlement = $9.590 \times 1.913 = 18.348$ kPa (permissible 27.391) — safe

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Annexure VI — Hydraulic Computations — Sheet 4

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. GA-971, revision B.

Annexure VI — Hydraulic Computations — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 675	32.98	0.48	2.18	24	200.401
Item 175	32.55	1.44	0.39	6	303.287
Item 899	25.70	1.83	2.76	22	65.670
Item 812	15.93	0.96	2.96	10	291.334
Item 615	22.52	3.36	0.69	17	152.808
Item 845	19.02	3.80	2.79	8	153.612
Item 235	21.06	1.90	2.67	23	397.155
Item 166	39.24	5.39	1.31	10	96.057
Item 281	8.14	4.60	1.11	23	93.296
Item 314	14.92	2.73	2.51	15	144.541
Item 310	15.10	3.01	0.33	1	274.808
Item 217	22.68	1.79	2.42	5	359.240
Item 121	37.92	2.68	1.38	11	347.089
Item 165	35.15	3.11	0.90	18	310.534
Item 494	35.74	2.72	0.39	7	29.201
Item 881	2.19	0.52	0.30	4	38.590
Item 756	35.93	3.22	2.43	9	391.464

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Annexure VI — Hydraulic Computations — Sheet 6

Design Computation

Settlement = $51.345 \times 6.412 = 329.221$ kN (permissible 486.556) — safe

Bearing pressure = $24.539 \times 4.752 = 116.613$ kNm (permissible 218.609) — safe

Moment = $68.656 \times 2.120 = 145.520$ kPa (permissible 194.932) — safe

Settlement = $4.024 \times 6.934 = 27.903$ kPa (permissible 41.985) — safe

Settlement = $75.510 \times 11.673 = 881.422$ kNm (permissible 1308.261) — safe

Shear = $88.922 \times 1.582 = 140.692$ mm (permissible 240.079) — safe

Deflection = $4.045 \times 3.809 = 15.406$ mm (permissible 19.810) — safe

Settlement = $45.315 \times 8.059 = 365.188$ kN (permissible 493.148) — safe

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Annexure VI — Hydraulic Computations — Sheet 7

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Reference: drawing no. HY-891, revision B.

Annexure VI — Hydraulic Computations — Sheet 8

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 463	37.55	2.08	0.26	4	257.480
Item 802	28.91	4.82	2.51	3	15.982
Item 540	38.60	3.59	1.80	19	58.263
Item 128	22.80	3.55	2.77	1	238.844
Item 655	25.95	4.95	2.53	10	91.613
Item 395	3.61	0.46	1.44	18	350.474
Item 582	21.62	3.64	0.44	16	170.520
Item 374	18.78	4.50	2.92	8	154.547
Item 125	38.59	4.68	2.70	6	335.621
Item 150	38.42	5.94	1.31	7	391.905
Item 723	32.08	0.39	0.41	4	331.645
Item 898	30.67	2.99	1.19	16	312.814
Item 187	20.34	5.06	1.23	21	107.461
Item 287	33.79	2.25	0.58	6	387.627
Item 339	27.04	2.09	0.83	22	231.090
Item 885	4.96	0.95	2.95	14	267.242
Item 746	38.51	3.54	1.81	21	114.235
Item 392	19.18	1.57	2.92	1	44.940
Item 348	3.82	2.54	0.45	23	41.724
Item 525	4.18	2.11	2.59	23	155.901
Item 898	25.49	1.72	2.70	8	134.323
Item 438	6.79	0.74	2.93	21	111.695

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Annexure VI — Hydraulic Computations — Sheet 9

Design Computation

Shear = $11.256 \times 11.854 = 133.425$ mm (permissible 206.623) — safe

Bearing pressure = $7.371 \times 1.970 = 14.519$ kPa (permissible 27.261) — safe

Settlement = $16.516 \times 6.677 = 110.284$ kN (permissible 167.525) — safe

Shear = $29.829 \times 4.986 = 148.733$ kN (permissible 217.661) — safe

Deflection = $59.777 \times 2.234 = 133.518$ mm (permissible 195.221) — safe

Deflection = $18.596 \times 0.617 = 11.475$ kN (permissible 19.087) — safe

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Annexure VI — Hydraulic Computations — Sheet 10

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. ST-419, revision C.

Annexure VI — Hydraulic Computations — Sheet 11

Traffic Volume Count

Hour	2-Wheeler	Car/Jeep	Bus	LCV	Truck	PCU
00:00	170	21	38	30	105	1250
21:00	161	115	9	40	47	1231
05:00	372	87	27	78	75	228
09:00	143	205	8	47	80	785
15:00	333	72	34	90	88	353
12:00	358	150	24	54	33	849
01:00	70	23	13	73	77	208
20:00	182	237	21	29	93	453
17:00	54	212	32	68	14	129
07:00	235	131	28	36	43	835
23:00	60	11	5	66	38	746
05:00	33	180	39	38	3	1104
17:00	385	202	17	35	38	478
18:00	158	35	16	14	47	546
15:00	365	203	39	74	44	223
01:00	189	112	28	9	96	853
00:00	103	16	25	60	40	224
23:00	193	182	38	24	101	435

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Annexure VI — Hydraulic Computations — Sheet 12

Design Computation

Deflection = $17.156 \times 9.336 = 160.171$ kNm (permissible 196.040) — safe

Shear = $36.962 \times 1.936 = 71.541$ kPa (permissible 88.067) — safe

Shear = $28.943 \times 0.431 = 12.468$ kPa (permissible 17.185) — safe

Settlement = $2.699 \times 1.333 = 3.598$ kNm (permissible 5.888) — safe

Settlement = $60.599 \times 1.180 = 71.535$ kPa (permissible 132.477) — safe

Deflection = $86.667 \times 3.855 = 334.059$ kPa (permissible 588.574) — safe

Bearing pressure = $48.821 \times 6.626 = 323.488$ kN (permissible 534.104) — safe

Bearing pressure = $72.849 \times 1.925 = 140.203$ kN (permissible 218.882) — safe

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Annexure VI — Hydraulic Computations — Sheet 13

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. ST-522, revision A.

Annexure VI — Hydraulic Computations — Sheet 14

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Boulder	Plant	25.9	6,888
Boulder	Quarry B	13.8	8,964
Steel	River bed	64.5	7,472
Fine sand	Quarry A	8.4	2,936
Steel	Quarry A	37.8	3,311
Fine sand	Quarry B	62.4	6,163
Cement	Quarry A	75.3	4,939
Bitumen	Quarry B	20.3	7,386
Fine sand	River bed	65.2	5,131
Murram	Depot	24.7	8,929
Murram	Quarry A	50.8	2,981
Boulder	Quarry A	35.4	4,760
Cement	Depot	63.9	5,809
Steel	River bed	37.4	6,381
Steel	River bed	15.5	7,807
Cement	Quarry B	21.4	3,537

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

Annexure VI — Hydraulic Computations — Sheet 15

Design Computation

Settlement = $6.213 \times 9.642 = 59.907$ mm (permissible 65.786) — safe

Moment = $44.096 \times 4.120 = 181.663$ kPa (permissible 291.692) — safe

Deflection = $7.859 \times 2.080 = 16.343$ kN (permissible 21.423) — safe

Bearing pressure = $23.222 \times 6.262 = 145.418$ kN (permissible 181.413) — safe

Deflection = $66.719 \times 2.616 = 174.570$ mm (permissible 289.383) — safe

Settlement = $67.630 \times 1.917 = 129.642$ kNm (permissible 192.042) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Annexure VI — Hydraulic Computations — Sheet 16

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. ST-937, revision A.

Annexure VI — Hydraulic Computations — Sheet 17

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Coarse aggregate	Quarry A	54.4	3,507
Fine sand	Quarry A	71.0	2,615
Coarse aggregate	Quarry B	85.7	6,652
Coarse aggregate	River bed	77.5	8,968
Bitumen	Quarry B	46.6	1,301
Murram	Depot	49.2	6,932
Murram	Quarry A	84.3	5,577
Cement	Depot	40.8	4,646
Murram	Quarry B	12.4	2,991
Coarse aggregate	Quarry A	36.0	1,940
Cement	Plant	18.1	5,679
Coarse aggregate	Quarry B	64.2	2,498
Murram	Quarry A	78.0	1,745
Bitumen	Plant	32.2	5,507
Bitumen	Plant	19.1	3,553
Cement	Quarry B	86.3	9,161
Cement	Plant	28.2	2,884
Coarse aggregate	Depot	77.3	1,460
Coarse aggregate	Plant	46.1	3,565
Cement	Depot	12.2	8,033
Boulder	Quarry B	46.5	1,329
Bitumen	Quarry A	18.9	906

Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

ANNEXURE VII — COPIES OF STATUTORY APPROVALS

Copies of the administrative sanction, waterway clearance, pollution control board consent and approval of river training works.

Annexure VII — Approval Correspondence — Sheet 1

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

The design conforms to the relevant IRC and IS codes in force on the date of this report. Load combinations have been considered in accordance with IRC:6 and the limit state method adopted throughout.

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Reference: drawing no. ST-894, revision B.

Annexure VII — Approval Correspondence — Sheet 2

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 738	21.10	3.92	2.87	24	164.247
Item 306	31.88	5.84	0.28	8	58.537
Item 321	24.70	5.08	2.88	12	98.215
Item 698	3.17	5.82	1.02	18	215.140
Item 621	3.67	0.34	1.88	18	126.133
Item 426	34.34	2.71	2.57	7	23.425
Item 356	9.50	3.86	2.63	20	207.810
Item 160	5.78	1.02	1.74	2	19.757
Item 323	16.06	4.75	2.97	3	341.883
Item 434	5.96	4.77	2.84	23	183.493
Item 176	3.54	0.85	0.87	3	384.507
Item 795	38.45	2.28	2.74	20	310.385
Item 798	28.52	2.28	1.52	13	284.689
Item 145	7.77	0.35	0.51	20	377.995
Item 804	20.11	3.62	2.48	2	278.195
Item 539	3.85	5.84	0.66	10	68.388
Item 200	4.54	5.25	2.66	22	367.575
Item 658	2.86	5.66	1.57	11	259.770
Item 660	29.37	0.47	0.51	7	176.419
Item 413	10.91	5.08	1.60	13	161.333
Item 423	28.59	3.24	2.03	24	62.608
Item 805	3.68	1.12	0.88	3	80.974
Item 431	18.54	1.54	1.60	1	210.447
Item 831	5.81	0.43	0.79	4	57.954
Item 851	7.10	5.80	2.92	24	217.017
Item 268	37.70	5.92	0.35	16	215.261

Material sources have been identified within an economic lead. Test results for coarse aggregate, fine aggregate and water from the identified sources are enclosed and satisfy the specification requirements.

Annexure VII — Approval Correspondence — Sheet 3

Design Computation

Settlement = $31.464 \times 3.076 = 96.770$ kNm (permissible 126.622) — safe

Settlement = $60.983 \times 5.477 = 333.978$ kPa (permissible 442.731) — safe

Deflection = $70.327 \times 3.593 = 252.705$ kPa (permissible 434.821) — safe

Moment = $37.002 \times 10.657 = 394.331$ mm (permissible 476.404) — safe

Settlement = $87.782 \times 6.992 = 613.742$ kPa (permissible 1006.473) — revise section

Bearing pressure = $64.114 \times 9.245 = 592.745$ mm (permissible 670.080) — revise section

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Annexure VII — Approval Correspondence — Sheet 4

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

Quantities have been computed from the approved General Arrangement Drawings and cross-checked against the detailed measurement sheets enclosed at the Annexure. Rates are as per the schedule in force with lead and lift adjustments applied.

Reference: drawing no. LA-799, revision C.

Annexure VII — Approval Correspondence — Sheet 5

Detailed Measurement Sheet

Item	L (m)	B (m)	D (m)	Nos	Quantity
Item 221	20.38	3.12	2.33	7	309.838
Item 357	38.30	1.79	0.48	16	260.388
Item 477	33.19	1.56	0.60	4	121.053
Item 664	10.37	4.47	1.78	5	331.269
Item 149	24.37	3.83	0.50	16	119.955
Item 818	33.80	4.55	1.55	6	32.488
Item 178	24.76	3.55	2.50	7	308.520
Item 208	12.25	4.19	0.89	15	348.282
Item 281	31.99	3.58	1.43	14	203.892
Item 382	10.23	0.87	2.16	5	221.148
Item 405	20.55	5.72	1.46	17	201.796
Item 133	37.96	1.13	2.01	21	146.760
Item 608	21.32	3.57	2.64	23	346.061
Item 781	2.61	2.23	2.28	14	284.371
Item 861	34.61	2.37	2.27	12	393.919
Item 346	24.45	3.22	2.79	19	204.440
Item 520	23.12	4.59	2.26	1	94.134
Item 821	35.97	2.98	0.65	5	233.595
Item 207	32.81	4.66	1.30	2	278.109
Item 525	24.48	3.11	1.31	7	23.553
Item 380	8.09	3.92	0.62	5	149.023

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

Annexure VII — Approval Correspondence — Sheet 6

Design Computation

Shear = $48.767 \times 7.607 = 370.986$ mm (permissible 501.431) — safe

Bearing pressure = $76.163 \times 7.862 = 598.774$ mm (permissible 775.738) — safe

Bearing pressure = $37.870 \times 11.306 = 428.168$ mm (permissible 561.980) — safe

Bearing pressure = $56.961 \times 3.580 = 203.910$ mm (permissible 373.679) — safe

Settlement = $62.415 \times 8.230 = 513.708$ kNm (permissible 938.756) — safe

Shear = $52.417 \times 9.699 = 508.374$ kN (permissible 791.165) — safe

Deflection = $18.998 \times 1.375 = 26.116$ kN (permissible 47.514) — safe

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

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Hydrological analysis was carried out using the discharge records of the nearest gauging station for the period of record available. The design discharge corresponds to a return period of 100 years as required for a structure of this class.

The alignment has been fixed after detailed reconnaissance survey and in consultation with the concerned district authorities. Ground control points were established at intervals of 500 m and connected to the nearest GTS benchmark.

The construction methodology envisages work in two seasons with the river training works taken up in the lean period. A detailed programme is placed at the Annexure.

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.

Reference: drawing no. ST-626, revision B.

Annexure VII — Approval Correspondence — Sheet 8

Lead Statement

Material	Source	Lead (km)	Rate (Rs)
Murram	Plant	60.7	787
Steel	Quarry B	86.2	9,103
Coarse aggregate	River bed	31.9	6,168
Steel	Depot	55.9	975
Coarse aggregate	Quarry A	60.2	5,969
Bitumen	Quarry A	42.9	237
Cement	Quarry A	49.7	4,485
Coarse aggregate	River bed	69.6	7,849
Cement	River bed	41.2	7,459
Cement	River bed	25.2	2,760
Cement	River bed	84.9	333
Bitumen	River bed	11.6	1,266
Murram	River bed	79.3	1,681
Boulder	Quarry A	64.5	6,414
Fine sand	Depot	64.4	7,026
Coarse aggregate	Depot	21.5	8,598
Bitumen	Quarry B	45.2	7,935
Coarse aggregate	River bed	78.8	2,984

Soil investigation was carried out by rotary drilling at the locations shown on the layout plan. Standard penetration tests were conducted at 1.5 m intervals and undisturbed samples collected in the cohesive strata for laboratory testing.

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Design Computation

Shear = $59.041 \times 0.678 = 40.008 \text{ kN}$ (permissible 62.459) — safe

Shear = $89.442 \times 0.584 = 52.192 \text{ kN}$ (permissible 97.322) — safe

Settlement = $61.314 \times 5.475 = 335.700 \text{ kN}$ (permissible 409.567) — safe

Settlement = $26.432 \times 1.223 = 32.336 \text{ kNm}$ (permissible 52.538) — safe

Bearing pressure = $51.078 \times 11.620 = 593.530 \text{ mm}$ (permissible 716.305) — safe

Consultations were held with the Gram Panchayat and the affected households. Minutes of the consultation meetings, along with the attendance register, are placed at the Annexure to this chapter.